

Neighborhood Revitalization and Residential Sorting

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Abstract

The HOPE VI Revitalization program sought to transform high-poverty neighborhoods into mixed-income communities through the demolition of public housing projects and the construction of new housing. We use longitudinal administrative data to investigate how the program affected both neighborhoods and individual residential outcomes. In line with the stated objectives, we find that the program reduced poverty rates in targeted neighborhoods and enabled subsidized renters to live in lower-poverty neighborhoods. The primary beneficiaries were not the original neighborhood residents, most of whom moved away. Instead, subsidized renters who moved into the neighborhoods after an award experienced the largest reductions in neighborhood poverty. The program reduced the stock of public housing in targeted neighborhoods but expanded access to housing vouchers in other, lower-poverty neighborhoods. Spillover effects on the poverty rates of other neighborhoods were small and dispersed throughout the surrounding cities. Our estimates imply that cities that revitalized half of their public housing stock reduced the average neighborhood poverty rate among all subsidized renters by 4.0 percentage points.

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1 Introduction

Low-income families in the United States tend to live in high-poverty neighborhoods, which research shows can adversely affect residents' health, education, behavior, and labor market outcomes (Chyn and Katz, 2021). One class of interventions, place-based policies, targets low-income places rather than low-income people. This place-based approach can improve equity and efficiency in the presence of market failures (Kline and Moretti, 2014), but risks displacing the very residents they aim to help. Understanding how such interventions affect residential mobility is therefore critical to designing policies that improve low-income neighborhoods and benefit low-income families.

This paper investigates how the HOPE VI Revitalization program, among the largest place-based policies in recent U.S. history, affected both neighborhoods and individual-level residential outcomes. The program allocated \$6.3 billion through 261 grants between 1993–2010 and sought to transform high-poverty neighborhoods into mixed-income communities by demolishing distressed public housing projects and constructing a mix of new subsidized and market-rate housing units. The neighborhood-level changes were intended to benefit subsidized renters by allowing them to live in lower-poverty neighborhoods. Yet, because the program could have affected residential sorting, a reduction in poverty rates in targeted neighborhoods need not benefit the original, subsidized renters in those neighborhoods.

Our analysis seeks to answer two key questions: i) Did the HOPE VI Revitalization program reduce poverty rates in targeted neighborhoods, and if so, through what mechanisms? ii) Did the program enable subsidized renters to live in lower-poverty neighborhoods?

Our empirical strategy compares neighborhoods that received an award to failed applicants—i.e., those that applied for but never received funding. The program targeted some of the most distressed public housing projects in the country. However, due to funding constraints, there were many similar projects that applied for but never received an award. We track people and places over time using individual-level administrative data on income, place of

residence, and subsidized housing participation. The data allow us to follow the original residents as they moved across neighborhoods and to observe everyone who moved into Revitalization or failed applicant neighborhoods after an award.

We begin by showing that the Revitalization program led to a large, persistent reduction in poverty rates in targeted neighborhoods. We estimate a stacked event-study regression, reweighting by the inverse propensity score, and find that the program reduced poverty rates in targeted neighborhoods by 8 percentage points 15 years after an award. Decomposing the neighborhood-level changes reveals that the program reduced poverty rates primarily by changing who moved into the neighborhood as opposed to by displacing the original residents or increasing the incomes of subsidized renters. The change in migration flows is attributable to a reduction in the stock of public housing units—which led fewer poor households to move in—and the construction of new market-rate units—which attracted higher-income households. In this way, the program achieved one of its primary stated objectives of creating more mixed-income neighborhoods.

Next, we investigate whether the poverty reductions in Revitalization neighborhoods led subsidized renters to live in lower-poverty neighborhoods. There are four groups of subsidized renters who could have been affected: i) the original Revitalization neighborhood residents, ii) subsidized renters who moved into Revitalization neighborhoods after an award, iii) subsidized renters who would have moved into a Revitalization neighborhood absent an award, and iv) residents of other neighborhoods potentially affected by spillovers. We estimate impacts on exposure to neighborhood poverty for each group.

Most of the original Revitalization neighborhood residents (group i) moved away and had limited exposure to the reduction in poverty rates in targeted neighborhoods. Within 5 years of the award, 74 percent of the original public housing project residents in Revitalization neighborhoods had moved to a new neighborhood. Because of these high rates of residential mobility, the reduction in poverty rates for Revitalization neighborhoods is five times as large as the reduction in exposure to neighborhood poverty experienced by the original residents.

Critics of HOPE VI argue that high out-migration rates reflect forced displacement (Goetz, 2013). In response, Congress commissioned the HOPE VI Panel Study (Popkin et al., 2009) and the HOPE VI Residents Tracking Study (Buron et al., 2002) to track the outcomes of the original residents. Both studies are descriptive and follow residents over time, without a control group to establish counterfactual outcomes. This matters because low-income renters in high-poverty neighborhoods are highly mobile even without redevelopment. We estimate that while some residents were displaced, three-fourths of those who moved would have left absent the intervention. The demolitions and subsequent rising housing prices also led the original residents without subsidized housing to move away at higher rates. However, we find no impacts on the migration rates of residents with access to Housing Choice Vouchers (hereafter, housing vouchers). This is consistent with housing vouchers diminishing displacement by limiting exposure to price increases.

The program enabled the subsidized renters who moved into Revitalization neighborhoods after an award (group ii) to live in lower poverty neighborhoods. This is not a mechanical result despite the observed reduction in poverty rates in Revitalization neighborhoods; the program could have attracted residents who, absent the intervention, would have lived in non-Revitalization neighborhoods with similar poverty rates to the post-Revitalization neighborhoods. We find that subsidized renters who move into Revitalization neighborhoods and failed applicant neighborhoods after an award move from neighborhoods with similar poverty rates. This suggests that the new subsidized renters in Revitalization and failed applicant neighborhoods had similar outside housing options. Using data on the characteristics of prior neighborhoods and individual incomes, we estimate that the program enabled the subsidized renters who moved into Revitalization neighborhoods after an award to live in neighborhoods with poverty rates that were 7.6 percentage points lower than the neighborhoods they would have lived in absent the award.

The program also could have affected the residents of neighborhoods that were not directly targeted. First, we consider subsidized renters who would have moved into the Revi-

talization neighborhoods absent the award (group iii). Because Revitalization neighborhoods were among the highest-poverty neighborhoods in their cities before the intervention, these individuals likely ended up in lower-poverty neighborhoods. Using origin neighborhoods of new residents as a proxy for the counterfactual neighborhood of these displaced residents, we estimate that the program reduced exposure to neighborhood poverty by 14 percentage points for the subsidized renters who would have moved in absent the award. Second, we consider residents of other neighborhoods affected by spillovers (group iv). We show that the program did not significantly affect poverty rates in nearby neighborhoods and, by analyzing migration flows in and out of the Revitalization and failed applicant neighborhoods, we rule out the possibility that the program significantly increased poverty rates in other neighborhoods via concentrated migration flows. We find the program increased exposure to neighborhood poverty by at most 0.3 percentage points for residents of neighborhoods impacted by spillovers.

Did the Revitalization program reduce exposure to neighborhood poverty for subsidized households, on average? We take a weighted-average of our group-specific estimates of the effect on exposure to poverty and find that cities that revitalized half of their public housing stock reduced the average neighborhood poverty rate among all subsidized renters in the city by 4.0 percentage points. Reductions in exposure to neighborhood poverty for the new neighborhood residents explain 68 percent of this aggregate effect. While the program did not change the city-wide poverty rate, the reductions in poverty were concentrated in neighborhoods with many subsidized renters while the increases in poverty from spillovers were dispersed across many neighborhoods with fewer subsidized renters. In this way, the program achieved its goal of enabling subsidized renters to live in lower poverty neighborhoods.

Our findings point to three lessons relevant for the trade-off between neighborhood revitalization and displacement. First, low-income households in distressed neighborhoods move frequently so the primary beneficiaries of neighborhood-level interventions are likely to be new residents. Unless there are effective ways to encourage the original residents to stay,

efforts to maximize a program’s impact may be most productive if focused on these newcomers. Second, the provision of tenant-based subsidized housing (i.e., housing vouchers) can help to mitigate the displacement effects generated by rising housing prices. Third, because of the high rates of residential mobility, efforts to minimize displacement effects will not necessarily conflict with the goal of changing the composition of neighborhoods.

The main contribution of this paper is to estimate the effect of a large, place-based intervention on both neighborhoods and individual residential outcomes. Prior research on HOPE VI uses repeated cross-sectional data to estimate the effect on neighborhoods (Zielenbach and Voith, 2010; Tach and Emory, 2017; Aliprantis and Hartley, 2015; Sandler, 2017; Bruhn, 2018; Blanco, 2023). Our individual-level panel data allow us to decompose the neighborhood-level changes into changes in individual outcomes, which illuminates the tension between neighborhood revitalization and displacement. Previous research estimating impacts on individuals focuses on the original residents of the HOPE VI public housing projects (Kingsley et al., 2003; Jacob, 2004; Chyn, 2018; Haltiwanger et al., 2020; Tach and Emory, 2017). We extend this work by considering impacts on all subsidized renters, which clarifies who benefits from place-based interventions. In the HOPE VI context this is especially important since most of the original residents moved away. Almagro et al. (2023) estimate a structural model and find that the HOPE VI program reduced the welfare of low-income, minority renters.¹ Whereas they structurally model neighborhood-level changes for all low-income households, we provide reduced-form estimates of the impact on exposure to neighborhood poverty for four distinct groups of subsidized renters—a key input in determining the overall welfare impacts of the program.

More generally, we contribute to research on urban renewal policies and the drivers, dynamics, and distributional impacts of neighborhood-level change.² Chyn and Katz (2021)

¹Almagro et al. (2023) study data from Chicago, which may not be representative of the broader program, since the Chicago Housing Authority was characterized by poor performance and weak management during this period (GAO, 1995).

²Papers include Baum-Snow and Marion (2009); Collins and Shester (2013); Busso et al. (2013); Harari et al. (2018); Rojas Ampuero (2022); Blanco and Neri (2023); Gechter and Tsivanidis (2023); Rossi-Hansberg et al. (2010); McKinnish et al. (2010); Guerrieri et al. (2013); Boustan et al. (2019); Diamond and McQuade

note that “a final frontier research area involves the estimation of the impact of place-based policies to improve low-income neighborhoods on the intended beneficiaries—the incumbent (preexisting) adult residents and their children.” Our findings confirm the importance of separately estimating impacts on the incumbent residents and on the targeted places, and additionally highlights the value of considering program impacts on other groups of beneficiaries; in the case of the Revitalization program, the new residents of targeted neighborhoods.

The rest of the paper is structured as follows. Section 2 provides background on the HOPE VI program. Section 3 describes the data. Section 4 presents the conceptual framework and empirical strategy. Section 5 presents our empirical results and Section 6 concludes.

2 The HOPE VI Revitalization Program

The HOPE VI program was created in response to a report by the National Commission on Severely Distressed Public Housing that raised concerns about living conditions in public housing projects (Green and Lane, 1992). While there were several distinct programs included under the broader HOPE VI umbrella, 94 percent of HOPE VI funding was allocated to the Revitalization program. The FY1999 HUD appropriations bill (Public Law 105-276) that authorized HOPE VI identifies four main goals of the Revitalization program:

(1) improving the living environment for public housing residents of severely distressed public housing projects through the demolition, rehabilitation, reconfiguration, or replacement of obsolete public housing projects (or portions thereof); (2) revitalizing sites (including remaining public housing dwelling units) on which such public housing projects are located and contributing to the improvement of the surrounding neighborhood; (3) providing housing that will avoid or decrease the concentration of very low-income families; and (4) building sustainable communities.³

In short, the program sought to transform distressed public housing projects into economically integrated neighborhoods in which families receiving subsidized housing assistance would live alongside higher-income neighbors who paid market prices for housing.

(2019); Couture et al. (2019); Couture and Handbury (2020); Qiang et al. (2020); Brummet and Reed (2021); Pennington (2021); Su (2022); Lee (2022); Asquith et al. (2023), and Kennedy and Wheeler (2023).

³42 United States Code §1437v(a).

The majority of the funds from the HOPE VI Revitalization grants were spent on the demolition of distressed public housing projects and the construction of new market-rate and public housing units. The grants also funded an expansion in vouchers to help offset displacement costs. The Revitalization program represented a large investment in neighborhoods. In total, the program awarded 261 grants totaling \$6.3 billion between 1993 and 2010 with a mean award size of \$24 million. On average, each grant leveraged an additional \$42 million from non-HOPE VI funding sources making the full size of the neighborhood-level intervention often exceeded the listed grant amount (Gress et al., 2016). To put these numbers in perspective, the mean Revitalization project had 360 occupied units in 1993.

Revitalization grants were awarded through a competitive process in which Public Housing Agencies (PHAs)—the city-level agencies authorized to administer HUD programs—could submit one application per year. Each year HUD scored applications based on four key factors including need, quality of the proposal, capacity to undertake development, and the amount of outside resources brought in (leveraging), and then allocated grants to the highest ranked applicants subject to funding constraints. The proposals were complex and evaluated by federal officials who lacked detailed knowledge of each site, which likely introduced noise into the scores, leading to different outcomes for otherwise similar applicants. Indeed, HUD’s Inspector General found inconsistencies in the scoring process and HOPE VI officials attributed errors to the short period allowed for grant review (GAO, 2003).

A challenge in interpreting HOPE VI research is that most studies conflate the two distinct arms of the HOPE VI program: Demolition grants, which funded only the removal of public housing, and Revitalization grants, which added funding to rebuild mixed-income communities after the demolitions. While each awarded a similar number of grants (285 Demolition vs. 261 Revitalization), Revitalization grants were on average 20 times larger.

3 Data

We combine survey and administrative data to track neighborhoods and individuals over time. The two key strengths of our data are that they include the near universe of Revitalization grants, and they contain annual data on income, place of residence, and participation in subsidized housing for nearly all individuals in the United States.

Applicants and Awardees. We use publicly available data to identify HOPE VI projects and the neighborhoods in which they were located. All applicants and awardees are publicly listed. We link these records to project-level summary files from HUD and assign each public housing project to one of three categories:

- i) *Revitalization*: projects that received a HOPE VI Revitalization grant,
- ii) *Failed Applicant*: projects that applied for but did not receive HOPE VI funding, and
- iii) *Non-Applicant*: projects that did not apply for or receive a HOPE VI grant.⁴

We limit our sample to large, non-senior public housing projects for which we can identify the geographic location. Table A.1 presents the impact of each sample restriction; after imposing all of the restrictions, we retain 88 percent of all Revitalization awards. We impose the same restrictions on non-Revitalization projects but additionally require that they be located at least one mile away from the nearest Revitalization site. This restriction is made because the Revitalization program could have directly affected nearby neighborhoods.⁵ Our final

⁴The HUD data identify projects with an alpha-numeric project ID and we assign a project ID to 99 percent of Revitalization grantees, and 98 percent of HOPE VI applicants. All the Revitalization grantees and most of the applicants that were not assigned a project ID were scattered sites, which we omit from the analysis anyways. We use data from HUDUSER's, "Picture of Subsidized Households" to identify all projects that existed in 1993, map the project names to the project IDs, and measure project-level characteristics such as the geographic location. We also identify all projects that were awarded a HOPE VI Demolition grant and we drop these projects from our sample.

⁵Our sample excludes the 39 demonstration grants awarded between 1993 and 1995. The demonstration grants differed in their objectives (creating economically integrated neighborhoods became an explicit goal only in 1996) and their approach (later awards placed a larger emphasis on leveraging additional sources of funding). Project-level data on applicants is missing for 2009 and 2010 but there were only 80 applicants in these two years compared to 778 applicants across all other years. Furthermore, many of these applicants likely applied in previous years and therefore still appear in our data.

sample contains 251 Revitalization, 166 failed applicant, and 5,783 non-applicant projects.⁶ Figure B.1 maps the location of each Revitalization and failed applicant project.

Neighborhoods. We use data from HUDUSER’s “Picture of Subsidized Households” to identify the Census block group in which each project is located.⁷ Block groups are defined by the U.S. Census Bureau and typically contain around 1,500 residents. To account for the fact that multiple projects might be located in a single block group and some projects span multiple block groups, we group together clusters of nearby public housing projects into neighborhoods defined by the connected set formed by projects and Census tracts.⁸ 90 percent of neighborhoods contain one or two block groups, and the two largest neighborhoods contain five block groups. For much of our analysis we collapse the data to these neighborhoods, of which there are 125 failed applicant and 204 Revitalization neighborhoods.⁹

Revitalization neighborhoods contain public and non-public housing units prior to an award. The average Revitalization neighborhood contained 1,059 households, 45 percent of whom lived in public housing in 1993. The percent of Revitalization neighborhoods living in public housing ranged from 29 to 60 percent at the 25th and 75th percentiles, respectively.

Individuals. We use linked survey and administrative data to measure the outcomes of individuals. We rely on four main data sources. First, we use administrative data from HUD, which allow us to identify which individuals receive subsidized housing and where these individuals live.¹⁰ These data cover a number of federal rental assistance programs but 97 percent of individuals in the Revitalization and failed applicant neighborhoods participate in the public housing, Section 8 vouchers, or project-based Section 8 programs. Second, we use administrative data from tax records, which provide annual measures of income and

⁶A single Revitalization grant or application could affect multiple projects. In most cases projects affected by a single grant are clustered within the same neighborhood.

⁷We use the definition of a block group based on the 2010 Census geography.

⁸We use tracts to form the connected sets as this allows us to group together two projects that are in distinct block groups but in the same tract.

⁹Our data include 308, 178, and 5,391 block groups that contain a Revitalization, a failed applicant, and a non-applicant public housing project, respectively.

¹⁰The HUD files we use are the Tenant Rental Assistance Certification Systems (TRACS) and Public and Indian Housing Information Center (PIC) files.

residential location for all filers. Third, we use administrative data from the U.S. Census Bureau’s Master Address File (MAF), which provide additional address information for individuals not in the tax records. Fourth, we use survey data from the Decennial Census Survey and the American Community Survey, which provide information on other variables including income, rent, and home ownership status. We link these datasets together to construct an individual-level panel that includes one observation per person per year between 1995 and 2018 for the near universe of people living in the United States.¹¹

We use adjusted gross income (AGI) to identify poor households. Specifically, we define individuals as poor if they reside in a household whose average AGI over the past five years is less than \$15,000.¹² The \$15,000 threshold is approximately equal to the poverty line for a three-person family in 2007, which is the middle of our sample period. While there is considerable mobility in and out of poverty from year to year, our measure captures a persistent form of income deprivation (Larrimore et al., 2020).¹³ Unless otherwise noted, we use the term “neighborhood poverty rate” to refer to the share of individuals who meet this AGI-based definition of poverty.

Our AGI-based measure of poverty closely tracks traditional survey-based measures. Figure B.2 presents a binscatter plot of the relationship between our AGI-defined poverty rates and decennial Census estimates at the block group level in 2000. The measures are well aligned: the fitted slope is 1.2 (s.e. = 0.002) with an R-squared of 0.69, indicating that our measure captures similar variation in neighborhood poverty to the traditional measure.

We combine the data on neighborhoods and individuals to create two analysis datasets. We construct a repeated cross-sectional dataset that summarizes the population that lives in each neighborhood in each year. We use these data to measure how neighborhoods change over time and to study the new residents that move into the sample neighborhoods after

¹¹While the coverage is quite comprehensive, address data are not available for every individual in every year. We make the assumption that an individual lives in the same address until we observe a new address. For children, we use the address data from the parent who most recently claimed them as a dependent.

¹²For children, we use the AGI from the parent who most recently claimed them as a dependent.

¹³Using a five-year average of AGI has the added advantage of providing a simple way to account for the fact that we do not have tax records for 1996 and 1997.

an award. Second, we construct a panel of original residents that follows the individuals who lived in the targeted neighborhoods in the year before an award. We use these data to measure the outcomes of the original residents and explore how a key group of intended beneficiaries were affected by the Revitalization program. The two datasets provide annual panels tracking the neighborhoods and original residents between 1995 (one year before the first award) and 2018 (eight years after the last award). In some cases we supplement these data with additional neighborhood-level information from the publicly available summary files from the 1990 and 2000 Decennial Census Surveys and the annual data from the 2009-2017 waves of the ACS, all of which are obtained through IPUMS (Manson et al., 2021).¹⁴

4 Conceptual Framework and Identification Strategy

4.1 Target Estimands

We aim to estimate the impact of the Revitalization program on: i) poverty rates in targeted neighborhoods, and ii) exposure to neighborhood poverty for subsidized renters.

To formalize the target estimands, we define the outcome space as the triplet $\{N_{it}^d, P_{it}^d, S_{it}^d\}$, where i indexes individuals, $t \in \{t', t''\}$ denotes the time period (pre- or post-treatment), and $d \in \{0, 1\}$ indicates the potential outcome defined by whether the local PHA received a Revitalization grant. N_{it}^d is the neighborhood of residence, P_{it}^d is an indicator for income below the poverty line, and S_{it}^d indicates subsidized housing status. To denote observed outcomes we drop the superscript d . For example, the observed neighborhood of residence for individual i at time t is N_{it} . To simplify notation, assume that each PHA applies for at most one Revitalization grant, and $D^d(n)$ indicates whether neighborhood n received a grant between t' and t'' . In the section below we discuss the empirical challenges that arise when a PHA receives multiple grants. We assume that baseline outcomes are unaffected by sub-

¹⁴Annual neighborhood-level statistics are from the 5-year ACS estimates, which are constructed by pooling data from the 5-year period centered on the year of interest. For example, we use data from the 2015-2019 ACS surveys to estimate neighborhood characteristics in 2017.

sequent treatment so that $Y_{it'}^1 = Y_{it'}^0 \equiv Y_{it'}$ for $Y \in \{N, P, S\}$. Let $\bar{P}_t^d(n) \equiv \mathbb{E}[P_{it}^d \mid N_{it}^d = n]$ denote the average poverty rate among individuals living in neighborhood n at time t .

Impact on Neighborhoods. Our first estimand is the average effect of the Revitalization program on neighborhood poverty rates in targeted neighborhoods:¹⁵

$$\mathbb{E}[\bar{P}_{t''}^1(n) - \bar{P}_{t''}^0(n) \mid D(n) = 1], \quad (1)$$

where $D(n) = 1$ denotes the set of neighborhoods that received a Revitalization grant.

To understand the mechanisms, we leverage our longitudinal individual-level data to decompose the changes in neighborhood poverty rates into changes in the size of subpopulations defined by the interaction of three binary variables: poverty status, subsidized housing receipt, and incumbent residency. The neighborhood poverty rate for neighborhood n at time t'' can be expressed as:

$$\bar{P}_{t''}^d(n) = \frac{\sum_{i:N_{it''}^d=n} \sum_{s=0}^1 \sum_{o=0}^1 \mathbb{1}\{P_{it''}^d = 1, S_{it''}^d = s, O_{it''}^d = o, \}}{\sum_{i:N_{it''}^d=n} \sum_{p=0}^1 \sum_{s=0}^1 \sum_{o=0}^1 \mathbb{1}\{P_{it''}^d = p, S_{it''}^d = s, O_{it''}^d = o\}}, \quad (2)$$

where $O_{it''}^d \equiv \mathbb{1}\{N_{it''}^d = N_{it''}^d\}$ is an indicator for being an original resident of the neighborhood. In our analysis of neighborhood-level impacts, we also seek to estimate the impact on the size of these subpopulations defined by $\sum_{i:N_{it''}^d=n} \mathbb{1}\{P_{it''}^d = p, S_{it''}^d = s, O_{it''}^d = o\}$ for all possible combinations of p , s , and o . This framework allows us to quantify the drivers of neighborhood change. For example, do poverty rates change because the original, low-income residents are displaced? Or because new, higher-income residents move in?

Impact on Individuals. Our second set of estimands allow us to examine whether the Revitalization program led subsidized renters to live in lower-poverty neighborhoods. Let:

$$\Delta_i \equiv \bar{P}_{t''}^1(N_{it''}^1) - \bar{P}_{t''}^0(N_{it''}^0) \quad (3)$$

¹⁵For all expectations, project-level outcomes are aggregated across sites using weights proportional to each project's baseline population.

denote the effect on individual i 's exposure to neighborhood poverty, which may reflect changes in both where they live and the poverty level of that location.

Given concerns about displacement, a central group of interest are the original residents—individuals who lived in Revitalization neighborhoods prior to the award (i.e., $D(N_{it'}) = 1$). For these residents, we estimate how the program affected their exposure to neighborhood poverty, as well as their likelihood of moving or losing access to subsidized housing. These outcomes help quantify the potential displacement costs.

We additionally assess whether the program reduced exposure to neighborhood poverty for distinct groups of individuals in subsidized housing after the award. To do so, we partition the population of subsidized renters in the post-period into four mutually exclusive and exhaustive groups:

1. *Original Residents.* Those who lived in targeted neighborhoods before the award and remained in subsidized housing after:

$$\mathcal{G}_1 = \{D(N_{it'}) = 1, S_{it''} = 1\}. \quad (4)$$

2. *New Residents.* Subsidized renters who moved into Revitalization neighborhoods after the award:

$$\mathcal{G}_2 = \{D(N_{it'}) = 0, D(N_{it''}^1) = 1, S_{it''} = 1\}. \quad (5)$$

3. *Displaced Residents.* Subsidized renters who, in the absence of the Revitalization program, would have moved into a treated neighborhood but instead reside elsewhere:

$$\mathcal{G}_3 = \{D(N_{it'}) = 0, D(N_{it''}^1) = 0, D(N_{it''}^0) = 1, S_{it''} = 1\}. \quad (6)$$

4. *Residents of Other Neighborhoods.* Subsidized renters who reside in other neighborhoods, potentially affected by spillover effects:¹⁶

$$\mathcal{G}_4 = \{D(N_{it'}) = 0, D(N_{it''}^1) = 0, D(N_{it''}^0) = 0, S_{it''} = 1\}. \quad (7)$$

The overall effect on subsidized renters' exposure to neighborhood poverty can be written as a weighted-average of the group-specific effects:

$$\sum_{g=1}^4 \omega_g \mathbb{E}[\Delta_i | \mathcal{G}_g], \quad (8)$$

¹⁶We implicitly assume that these individuals live in the same PHA that received a Revitalization award.

where ω_g is the share of individuals in group g . Equation (8) allows us to quantify both the average effect of the program on all subsidized renters as well as the the distributional impacts across different groups.

Even if the Revitalization program reduced poverty rates in targeted neighborhoods, it is not obvious that these neighborhood-level changes would enable subsidized renters to live in lower-poverty neighborhoods. The program could have displaced subsidized renters to other high-poverty neighborhoods, potentially increasing their exposure to neighborhood poverty as well as the exposure to poverty for other residents in those destination neighborhoods.

We make this point formally in Appendix C and interpret the Revitalization program through the lens of a model of residential sorting. Our model extends the framework of Schelling (1971, 2006) and Glaeser (2008) to include public housing and heterogeneous agents and shows how improvements in the Revitalization neighborhoods could either increase or decrease the exposure to neighborhood poverty for subsidized renters depending on how subsidized households value neighborhood amenities and the extent to which they are affected by increases in housing prices.

4.2 Empirical Design and Estimation Strategy

The ideal research design to identify the causal effect of HOPE VI on targeted neighborhoods would involve randomly assigning grants to a subset of distressed public housing projects. In such an experiment, comparing post-treatment poverty rates in neighborhoods that received a grant to those that did not would identify the causal effect on neighborhood-level poverty. This same experiment would also identify the causal effect of HOPE VI on the outcomes of original residents by comparing the post-treatment neighborhood characteristics of original residents in treated and control sites. In practice, grants were not randomly assigned. But we approximate this ideal experiment by comparing the recipients of Revitalization grants to projects that applied for but did not receive funding.

As motivation, Figure B.3(A) presents a binned scatter plot of the change in neighborhood poverty rates between 1980 and 1990 against the average neighborhood poverty rate in 1980.

Prior to the award, changes in neighborhood poverty rates were unrelated to treatment status. Figure B.3(B) repeats the analysis for changes in poverty rates between 1990 and 2017. Revitalization sites experienced substantially larger reductions in poverty than both failed applicants and non-applicants with similar baseline poverty rates. These patterns provide preliminary evidence that HOPE VI reduced poverty in targeted neighborhoods and highlight the importance of comparing treated sites to similar untreated projects.

Failed applicants provide a natural comparison group for three reasons. First, demand far exceeded supply: only 29% of applicants in a given year received funding, and 34% never received a grant (see Figure B.4). Second, with average awards over \$20 million, PHAs likely could not have replaced Revitalization funding to carry out planned projects. Third, the act of applying signals that the PHA viewed the project as a strong candidate for revitalization, making failed applicants more comparable to awardees—both on observables and, plausibly, unobservables—than non-applicants. Consistent with this, Table A.2 shows that while Revitalization projects were somewhat more disadvantaged, differences from failed applicants are meaningfully smaller than those from non-applicants.

While there are observable differences between the failed applicant and Revitalization neighborhoods on average, there exist failed applicant neighborhoods that are observably similar to Revitalization neighborhoods. Figure B.5(A) shows substantial overlap in 1990 poverty rates and Panels B-D show similar overlap for other pre-award characteristics. Given the overlap, we view inverse propensity score weighting, as a promising way to adjust for observable differences between the two groups.

We compare Revitalization awards from a given grant year to all failed applicants. To facilitate this comparison we create a stacked dataset that contains one observation for each Revitalization neighborhood and 15 observations for each failed applicant neighborhood, one for each of the 15 possible grant years between 1996 and 2010 (in this way, each failed applicant neighborhood appears in the event-specific panel for all grant years). Within each grant year, we estimate a propensity score by estimating a logistic regression of receipt of a

Revitalization award on the population of the neighborhood, the share of the neighborhood that was Black non-Hispanic, the share below the poverty line, the median home value, and the size of the public housing project. Our approach allows for changes in the selection process over time and aims to balance the observable characteristics of the failed applicant and Revitalization neighborhoods within each grant year. When conducting the analysis at the neighborhood-level, we multiply the inverse propensity score weights by the neighborhood population in the year before the award. We normalize the weights for the failed applicant group such that within each grant year the sum of the weights across failed applicant neighborhoods is equal to the sum of the weights across the Revitalization neighborhoods.¹⁷

Weighting the data by the inverse estimated propensity score eliminates observable pre-Revitalization differences between the Revitalization and failed applicant neighborhoods. Figure B.6 presents balance tests in which we regress a neighborhood characteristic measured prior to the award on an indicator for receipt of a Revitalization grant. Panel A weights by population and illustrates that there are some differences between the Revitalization and failed applicant neighborhoods. Panel B uses inverse propensity score weights and shows that these weights eliminate observable differences between the two groups.¹⁸

To estimate each of the key estimands, we rely on the comparison between the Revitalization and failed applicant neighborhoods. Therefore, the following conditional mean independence assumption is central to our empirical analysis.

Assumption 1. Conditional on observable characteristics defined prior to the award, $X_{it'}$, Revitalization grants are as good as randomly assigned among neighborhoods that applied for a Revitalization grant. This assumption implies conditional mean independence for sub-populations who would live in the target neighborhoods. Formally, let $C(n)$ be an indicator

¹⁷For results based on individual-level data, we multiply the inverse propensity score weights by the baseline population in the neighborhood divided by the number of individuals in the sample for that analysis. Thus, the sum of these weights across individuals within a project is equal to the weights used in the neighborhood-level analysis. This weighting scheme makes our results comparable across specifications since all analyses, whether based on neighborhood- or individual-level data, weight by the exact same measure: baseline population in the neighborhood.

¹⁸Figure B.7 present analogous results for the contiguous Census block groups. Weighting by the inverse estimated propensity score eliminates observable pre-award differences for these nearby census blocks.

for whether n is a failed applicant neighborhood with $D^1(n) = n$ if $C(n) = 1$. Then,

$$\begin{aligned} \mathbb{E}[\bar{P}_{t''}^d(N_{it}^d) \mid D(N_{it'}) = j, D(N_{it''}^1) = k, D(N_{it''}^0) = l, S_{it''}^0 = v, S_{it''}^1 = u, X_{it'}] \\ = \mathbb{E}[\bar{P}_{t''}^d(N_{it}^d) \mid C(N_{it'}) = j, C(N_{it''}^1) = k, C(N_{it''}^0) = l, S_{it''}^0 = v, S_{it''}^1 = u, X_{it'}], \end{aligned} \quad (9)$$

for $d \in \{0, 1\}$; $t \in \{t', t''\}$; $v \in \{0, 1\}$; $u \in \{0, 1\}$; and at least one of j , k , or l is equal to one (i.e., i would live in the target neighborhood at some point under treatment or control).

We often have data on pre-treatment outcomes and we operationalize the comparison between Revitalization neighborhoods and failed applicant neighborhoods using a stacked difference-in-differences estimator. To address concerns related to the staggered timing of treatment, we follow Cengiz et al. (2019) and stack the data, creating a panel dataset that has a clean “event” for each Revitalization grant year. For each potential grant year between 1996-2010, the panel includes all failed applicants and the awardees from that year.¹⁹ We then stack the datasets from each award year and estimate the following specification,

$$y_{nt} = \sum_{j \neq -1} \beta^j D_{nt}^j + \lambda_{gt} + \delta_{ng} + \epsilon_{nt} \quad (10)$$

where δ_{ng} is a neighborhood by grant year fixed effect, λ_{gt} is a fixed effect for grant year by year relative to award, and D_{nt}^j treatment indicator equal to one if neighborhood n received a Revitalization grant and year t is j years after the award.²⁰ We estimate this specification via weighted least squares using the inverse propensity score as weights and cluster standard errors at the level of the neighborhood. The main coefficients of interest are the β^j , which trace out treatment effects over time relative to the year before the award.

The key identifying assumption is the “parallel trends” assumption which corresponds to a specific case of Assumption 1: $\mathbb{E}[\bar{P}_{t''}^d(n) \mid D(n) = 1, \bar{P}_{t''}^d(n)] = \mathbb{E}[\bar{P}_{t''}^d(n) \mid C(n) =$

¹⁹This specification effectively compares the Revitalization awardees in a given year to all failed applicants. The propensity score is estimated within grant year to adjust for observable differences between these two groups. Standard errors are clustered at the level of the neighborhood to account for the fact that we reuse the failed applicants as controls across grants years.

²⁰We use the neighborhood by grant year fixed effect because we stack the data and a panel of the failed applicant neighborhoods appears in the data for each grant year.

$1, \bar{P}_{it'}^d(n)]$ for $d \in \{0, 1\}$. We assess the plausibility of this assumption by examining trends in the estimated pre-treatment effects and by exploring the sensitivity of the results to using alternative comparison groups, estimators, and data sources.

In cases where we lack annual measures of the outcome variable prior to and after the award, we estimate the following specification,

$$y_n = \alpha + \beta D_n + \lambda_g + \epsilon_n \quad (11)$$

where y is the outcome variable (typically measured as an average between the 10 to 15 years after the award) for neighborhood n , λ_g is a fixed effect for the grant year, and D is an indicator equal to one if the neighborhood received a Revitalization grant. We estimate this specification via weighted least squares using the inverse propensity score as weights and cluster standard errors at the level of the neighborhood. When possible, we control for pre-treatment covariates to increase precision. Equation 11 identifies the causal impact under Assumption 1 when $X_{it'}$ includes only one pre-treatment measure of the outcome or does not include any pre-treatment outcome measures.

Under Assumption 1, the comparison to failed applicant neighborhoods identifies the causal effect on treated neighborhoods and all original residents. While this assumption is necessary, it is not sufficient to identify impacts for the subpopulations defined in Section 4.1. Estimating effects for these groups requires additional assumptions and distinct estimators. However, because all analyses rely on Assumption 1, we start by presenting neighborhood-level effects and assessing the validity of Assumption 1 in that context. We then proceed to analyze individual-level outcomes, introducing the relevant estimators and additional identifying assumptions in the sections that follow.

An implicit assumption in our empirical analyses is that potential outcomes are not a function of the treatment status of other projects. This assumption could be violated if, for example, the Revitalization program reduced poverty in surrounding neighborhoods, thereby

indirectly benefiting nearby failed applicant sites and biasing our estimates. Several facts help to alleviate this concern. First, 71 percent of failed applicant neighborhoods are located in a PHA where no project received a Revitalization grant. Second, we drop from the analysis any failed applicant neighborhoods located within one mile of a Revitalization site. Third, 40 percent of Revitalization neighborhoods are the only such project within their PHA and another 40 percent are located in PHAs with 2 or 3 sites. Even in the PHAs with multiple sites, 85 percent of project pairs are located more than one mile apart, and 98 percent are separated by more than half a mile. Lastly, we find that only 3 percent of individuals who ever live in a Revitalization or failed applicant neighborhood subsequently appear in another neighborhood in our sample. Together, these patterns suggest that both physical proximity and residential mobility between treatment and control neighborhoods are limited, reducing the risk of bias due to spatial spillovers.

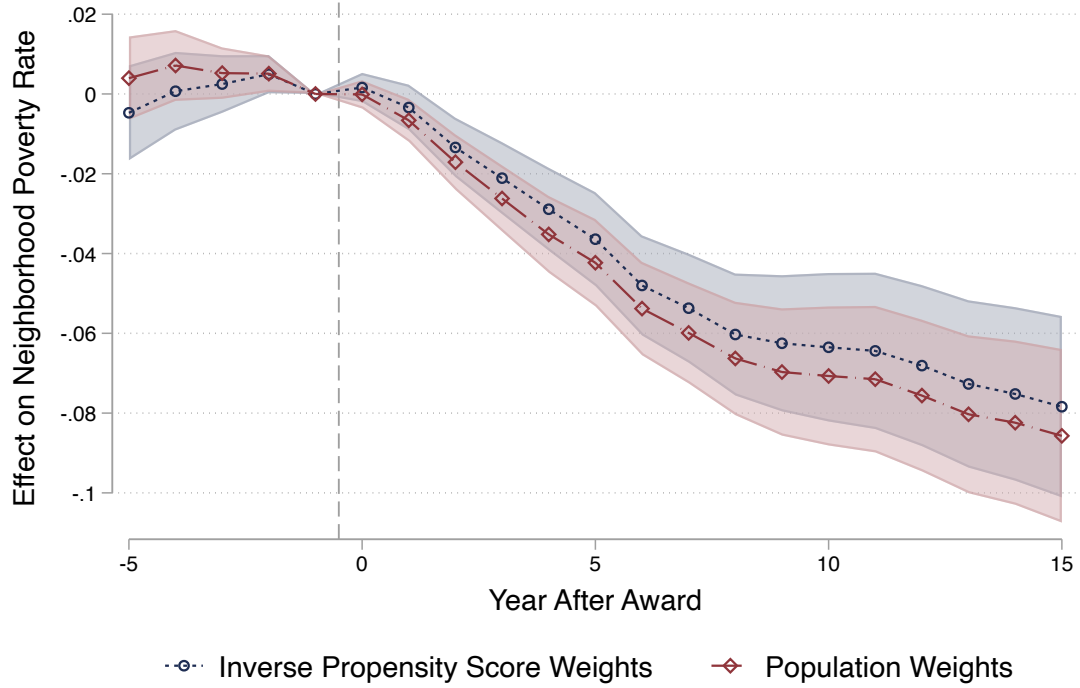
5 Effect of the HOPE VI Revitalization Program

5.1 Effect on Neighborhoods

We first examine how the Revitalization program affected total population in order to establish the timeline of the intervention. Figure B.8 presents results from equation 10 where the outcome variable is the log of total population. The Revitalization program led to a 26 log point reduction in population five years after the award. But population quickly recovered over the next five years. The results suggest the following timeline: (*short-run*) in the five years following the award people moved out and housing units were demolished; (*medium-run*) in the next five years people moved back into the neighborhood as housing units were rebuilt; and (*long-run*) around ten years after the award construction was largely complete and the population fully recovered.²¹

²¹Figure B.9 presents the effect on the physical housing stock. These results suggest a similar timeline, with a reduction in total housing units in the first five years and the construction of new housing units taking place over the subsequent five years. These estimates are less precise as the housing stock variables are measured from public data in 1990, 2000, and annual data are only available starting in 2007.

Figure 1: Effect on Neighborhood Poverty



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The two series present estimates from specifications that weight by the inverse propensity score and the baseline population. The outcome is the proportion of current residents who are poor. The shaded regions denote the 95 percent confidence intervals.

The Revitalization program led to a large, persistent reduction in neighborhood poverty rates. The blue circles in Figure 1 show that the program led to an 8 percentage point reduction in the share of residents who were poor 15 years after an award. We find similar reductions in poverty rates among households with and without children (see Figure B.10). These results indicate that the changes were not driven by an influx of childless adults, which is a concern often raised in the gentrification literature (McKinnish et al., 2010). We also calculate the proportion of residents whose AGI falls within narrow ranges 10 to 15 years after the awards and estimate equation 11. We find that the program led to a reduction in the proportion of families in all income categories below \$15,000. Conversely, the program led to an increase in moderate-income households, with the largest effects for households with an AGI between \$20,000 and \$100,000 (see Figure B.11).

The red diamonds in Figure 1 show similar results when using population weights instead of inverse propensity score weights. The insensitivity of the estimates is a consequence of the fact that the pre-award differences in observable characteristics are modest and the failed applicant neighborhoods were relatively stable throughout the study period while revitalized neighborhoods changed dramatically. Figure B.12 illustrates this by plotting the average within-county poverty rank for Revitalization and failed applicant neighborhoods from 1995 to 2018.²² In 1995—one year before the first HOPE VI grant—Revitalization neighborhoods ranked in the 95th percentile of poverty within their counties. By 2018—eight years after the final award—they had fallen to the 87th percentile. Failed applicant neighborhoods were consistently ranked between the 89th and 91st percentiles.

Table A.3 demonstrates the robustness of the estimated effect on neighborhood poverty rates across alternative estimators, comparison groups, and data sources. We regress the change in block group poverty rate between 1990 and 2017 on an indicator for receiving a Revitalization grant and a vector of baseline covariates.²³ The outcome variable and covariates are measured using publicly available data from the 2017 ACS or the 1990 Decennial Census. Column 1 of Panel A presents estimates using the failed applicant comparison group and shows that we find similar effects using the alternative estimator and data: the Revitalization program led to a 7.7 percentage point reduction in neighborhood poverty rates 7-21 years after a Revitalization.²⁴ Columns 2 and 3 compare the Revitalization sites to block groups that contain projects that never applied for HOPE VI funding and block groups that do not contain public housing projects, respectively. Both specifications show similarly large impacts on neighborhood poverty rates. Columns 4 and 5 present estimates with these same two comparison groups but limit the sample to block groups between one and five miles of a

²²We use within-county ranks to net out city-wide fluctuations in poverty rates.

²³Dube et al. (2023) describe this estimator, which, like the stacked event-study estimator, accounts for issues that arise from staggered treatment adoption and heterogeneous effects.

²⁴As discussed in Section 3, we classify households with an AGI below \$15,000 as poor. Figure B.13 presents estimates from equation 10 where the outcome is the survey-based measure of neighborhood poverty rates. We impute missing values between Census surveys by assuming poverty rates do not change unless we have data that indicates otherwise. The survey-based poverty measure yields similar results: the program led to a 11 percentage point reduction in poverty rates 15 years after the award.

Revitalization site; the specifications also include fixed effects for the closest HOPE VI site. These estimates also show large impacts on poverty rates when comparing HOPE VI sites to similar, nearby neighborhoods. The specifications in panel B do not control for baseline differences and the estimates are substantially larger. Taken together, Table A.3 shows that we find similar effects using alternative comparison groups, provided that we control for observable pre-award differences between Revitalization and comparison neighborhoods.

While not directly comparable, our estimates of the impact on neighborhood poverty rates are qualitatively similar to those in Tach and Emory (2017). Tach and Emory (2017) pool together Demolition and Revitalization grants and estimate the combined effect of the two programs on the change in poverty rates between 1990 and 2010. They find that the program led to a 11 percentage point reduction in neighborhood poverty rates and point towards the potential displacement of the original, low-income public housing residents as an explanation. Our data allow us to directly assess the drivers of neighborhood change.

Using the decomposition in equation 2, we show that poverty rates fell in Revitalization neighborhoods primarily because fewer poor, subsidized households and more non-poor, unsubsidized households moved in after awards. Table 1 presents estimates from equation 11 where the outcome variable is the average number of people (in groups defined by poverty, tenure, and subsidy status) in the neighborhood 10–15 years after the award.²⁵ To increase precision, the specifications control for the number of poor and non-poor individuals in the neighborhood in the year before the award. Columns 1 and 2 indicate that the Revitalization program decreased the number of poor residents by 269 and increased the number of non-poor residents by 156, on average.²⁶ Columns 3 through 6 illustrate that both the reduction in poor residents and the increase in non-poor residents is driven by a change in the composition of *new residents*, as opposed to differences in out-migration for original residents. Specifically,

²⁵We use equation 11 for this analysis because we estimate the impact on the size and characteristics of the population that moved in after, and this population is not defined before the award.

²⁶Figure B.14 shows the same finding using equation 10 to estimate the effect on log population by poverty status. In the long-run, the program led to a 10 log point reduction in the number of poor residents and a 20 log point increase in the number of non-poor residents.

Table 1: Effect on Size of Sub-Populations

	By Income		By Income and Tenure				By Income, Tenure, and Subsidized Housing							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Revitalization	-269 (74) [0.000]	156 (65) [0.017]	-74 (15) [0.000]	-195 (71) [0.006]	-42 (10) [0.000]	199 (65) [0.002]	-43 (11) [0.000]	-31 (10) [0.001]	-168 (53) [0.002]	-27 (41) [0.520]	-20 (7) [0.008]	-23 (9) [0.012]	-2 (33) [0.958]	201 (54) [0.000]
Mean	1580	1100	285	1290	224	879	108	177	696	595	52	172	335	544
Poor	yes	no	yes	yes	no	no	yes	yes	yes	yes	no	no	no	no
Original resident			yes	no	yes	no	yes	yes	no	no	yes	yes	no	no
Subsidized housing							yes	no	yes	no	yes	no	yes	no
Observations	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000

Notes: Each column presents estimates from a separate regression in which an outcome variable is regressed on an indicator for the Revitalization award. The outcome variable differs across columns and is the size of the population defined by poverty, tenure, and subsidized housing status. Regressions are estimated via weighted-least squares using the inverse propensity score. All regressions control for the year of first award as well as the number of poor and non-poor households who lived in the neighborhood in the year before the award. Standard errors are clustered at the neighborhood and are presented in parentheses and p-values are presented in brackets.

the program reduced the number of new, poor residents by 195 and increased the number of new, non-poor residents by 199. Columns 7 through 14 further break these results out by subsidized housing status. The decline in the population of poor households is driven by fewer new residents with subsidized housing (column 9) and, to a lesser extent, a decline in the number of poor, original residents in subsidized housing. The growth in the population of non-poor individuals is exclusively driven by new, non-poor residents without subsidized housing (column 14).

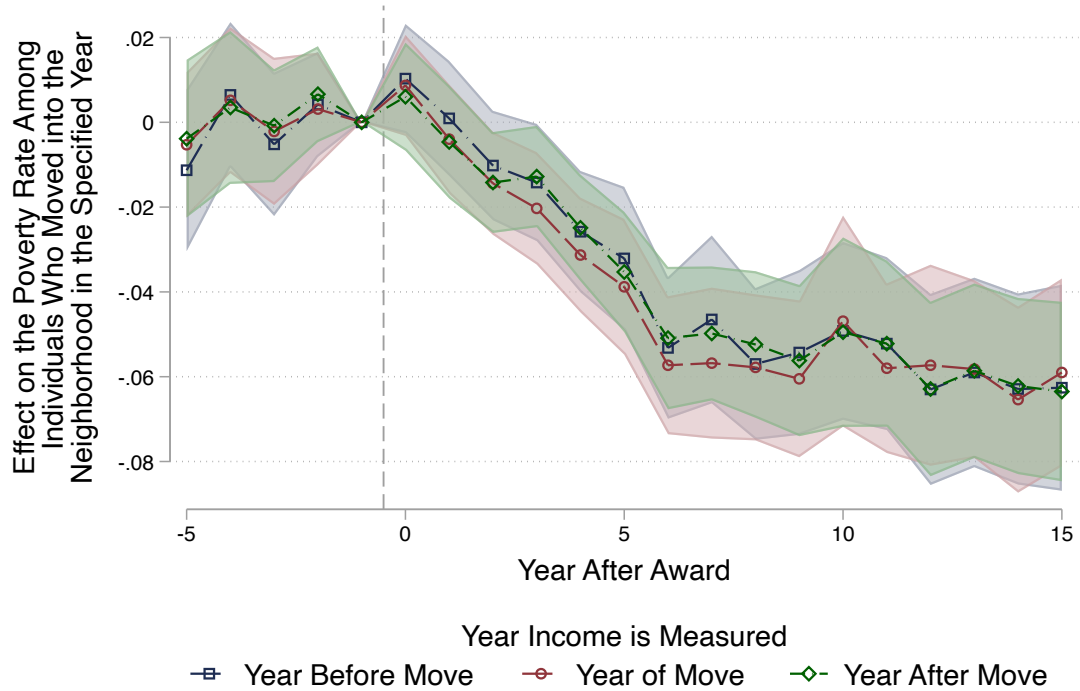
The program reduced the poverty rate in targeted neighborhoods, but did so at the cost of reducing access to these neighborhood for subsidized households. Columns 7 through 14 of Table 1 imply that the program reduced the size of the population with subsidized housing by 20 percent and increased the size of the population without subsidized housing by 8 percent.²⁷ Figure B.15 presents estimates from equation 10 and shows a long-run decline in the subsidized population of a similar magnitude. However, a reduction in the number of subsidized renters in the Revitalization neighborhoods need not imply a reduction in subsidized units in the broader housing market. Indeed, Section 5.3 presents evidence that at the city-level, the program reduced the stock of public housing but this was offset by an increase in the provision of housing vouchers.

The program reduced poverty rates by attracting residents who would have had higher incomes regardless of where they lived as opposed to increasing the income of new residents. For each neighborhood and year, we calculate the poverty rate for the population of individuals who move into the neighborhood in that year using their income in the previous, current, or subsequent year.²⁸ Figure 2 presents estimates from equation 10, using these three measures of poverty as the outcome variable. The red circles represent estimates when measuring the income of the new neighborhood residents in the year of the move and show

²⁷We arrive at these calculations by taking the sum of the treatment effects divided by the sample mean size of the populations: $-0.20 = (-43 - 168 - 20 - 2) / (108 + 696 + 52 + 335)$ and $0.08 = (-31 - 27 - 23 + 201) / (177 + 595 + 172 + 544)$.

²⁸In contrast to our main measure of poverty, these outcomes are based on the individual's income in a single year, as opposed to the five-year average.

Figure 2: Effect on the Poverty Rate of New Residents



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. We limit the sample to the population of individuals who move in during a the year specified by the horizontal axis. The three series represent estimates from three separate regressions in which we measure the poverty rate of this population using their income in the previous, current, or subsequent year. The shaded regions denote the 95 percent confidence intervals.

that the poverty rate of the flow of households moving into the Revitalization neighborhoods drops by 6 percentage points in the long run. The blue squares and green diamonds represent estimates using the poverty rate measured using individual income in the year before the move or in the year after the move, respectively; both estimates are indistinguishable from those based on the contemporaneous measure of income. The higher-income residents who moved into the revitalized sites were already higher-income before moving into these neighborhoods.

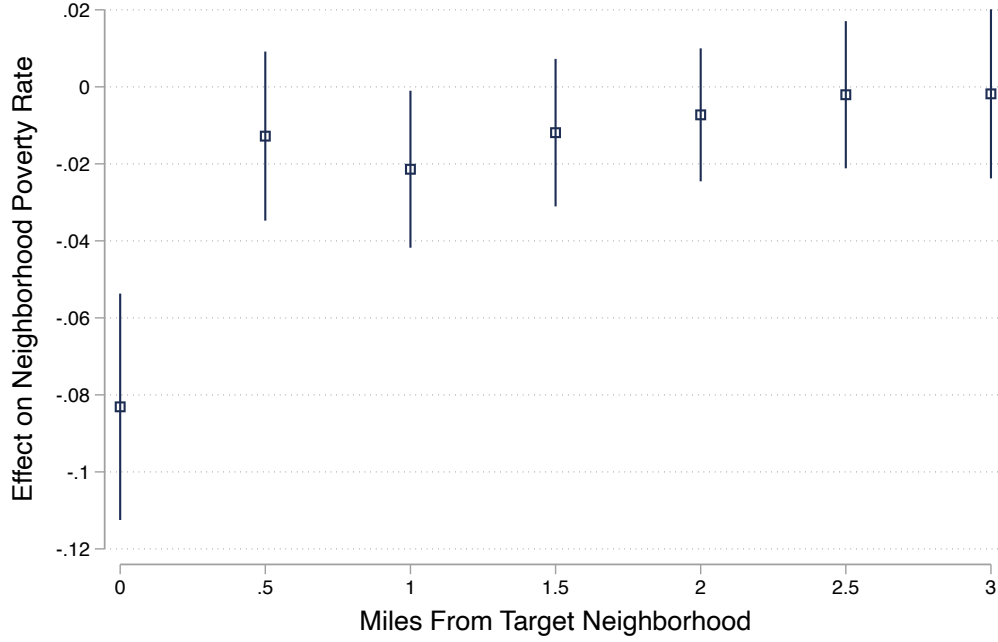
Figure B.16 presents results showing how the Revitalization program affected the racial and ethnic composition of the neighborhoods. Many HOPE VI projects were predominantly Black non-Hispanic, which explains why the Black non-Hispanic population share in Revi-

talization neighborhoods declined in the short-term as the public projects were demolished; the share of neighborhood residents who were White non-Hispanic responded in the opposite way: increasing immediately following a Revitalization grant and remaining persistently higher up to 10 years after a demolition. The share of Black non-Hispanic residents had largely recovered by 15 years after a Revitalization; we fail to reject that there is no difference between Revitalization and applicant neighborhoods in terms of the proportion of residents who were White non-Hispanic or Black non-Hispanic 15 years post-Revitalization. We find no evidence that the Revitalization program affected the proportion of neighborhood residents who were Hispanic in any post-event year.

The Revitalization program led to a substantial increase in housing costs, but had no impact on rates of home ownership. We measure housing-related variables using responses to the ACS in the 10 to 15 years after the award.²⁹ Since we do not have ACS data in the pre-award period, we use equation 11 to estimate the effect of the program the outcomes from the ACS. Columns 1 to 9 of Table A.4 show the effect on rent, home values, and monthly mortgage payments. For each outcome we present results from a baseline model, which controls for pre-award neighborhood characteristics, a modified specification that also controls for observable housing unit characteristics (building type, year built, persons per room), which is our preferred specification, and a specification that limits the sample to units that were built prior to the award. Across all outcomes and specifications we find positive effects on housing prices. The robustness of the price effects to controlling for housing characteristics and limiting the sample to pre-existing housing units, suggests that the program made the neighborhoods more appealing places to live rather than just improving the physical characteristics of the housing stock. Columns 10 and 11 show that the program had no impact on home ownership rates.

²⁹When measuring housing prices, we exclude households who receive subsidized housing. Respondents to the ACS often report the rent they pay (the tenant payment) as opposed to their contract rent. Because HOPE VI reduces the share of units that are subsidized, using all responses to the ACS would lead us to overestimate the effect on prices. Indeed, when we use all respondents to the ACS, we find increases in rent that are 43 percent larger than the estimates in Table A.4.

Figure 3: Effect on Neighborhood Poverty Rates in Nearby Block Groups



Notes: The sample includes target neighborhoods as well as all block groups within three miles of the Revitalization or failed applicant neighborhoods. We estimate a regression where the outcomes variable is the change in neighborhood poverty rate between on year before and fifteen years after the award. The independent variables in the specification include the interaction between the Revitalization indicator and the distance category (grouped into .5 mile increments) and the block group poverty rate in the year before the award. The regression is estimated via WLS, weighting by the inverse propensity score weights. Standard errors are clustered at level of the targeted neighborhoods and the vertical spikes denotes the 95 percent confidence intervals.

We also investigate whether the program had spillover effects on the surrounding neighborhoods. To do so, we measure the change in poverty rates in all block groups that are within a 3 mile radius of the target neighborhood and group the surrounding block groups into half mile bins based on the distance between their centroid and the centroid of the focal neighborhood. We then regress the change in poverty rate between the year before to fifteen years after the award on the interaction between the Revitalization indicator and the distance category. Figure 3 presents the estimates and shows that there is limited evidence of spatial spillover effects on nearby neighborhoods.³⁰ These results show that the effects were

³⁰Figure B.17 presents similar results, which use equation 10 to estimate the impact on poverty rates in contiguous Census block groups.

quite local and were concentrated in the neighborhoods directly targeted by the intervention, fading away quickly as distance from the neighborhoods increases.

5.2 Effect on Individual-Level Residential Outcomes

We next investigate whether the reduction in poverty rates in the Revitalization neighborhoods led subsidized renters to live in lower poverty neighborhoods.

5.2.1 Original Residents

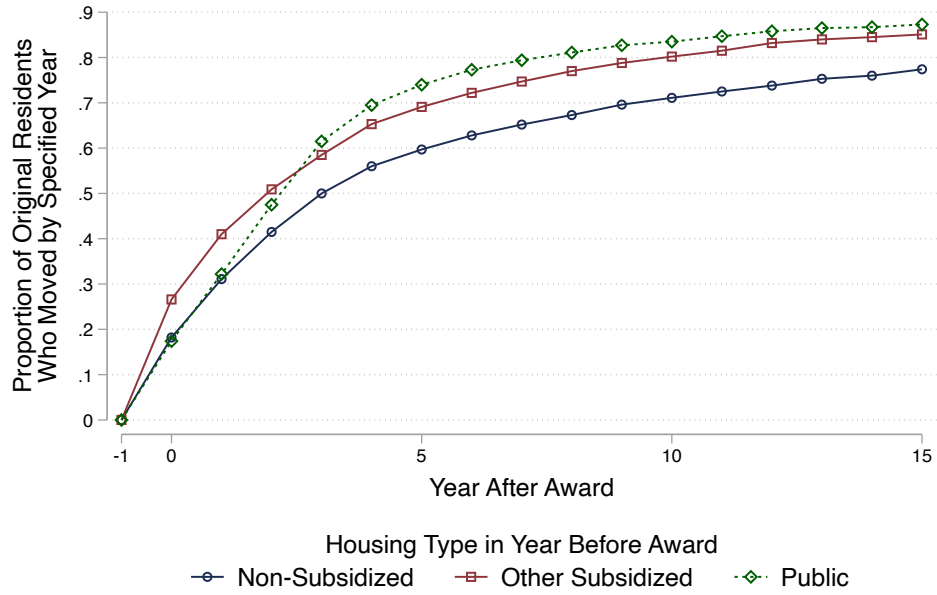
We begin by documenting the migration responses of the original residents. Figure 4(A) plots the proportion of original residents of the Revitalization neighborhoods—i.e., residents of the neighborhood in the year before the award—who had moved away 0-15 years after an award, broken out by housing subsidy type in the year before the award. Within five years of an award, over 60 percent of all original residents had moved away, and high migration rates are common to both households with and without subsidized housing and even those in public housing. These high migration rates overstate displacement effects, as there were similarly high rates of out-migration in the failed applicant neighborhoods (see Figure B.18).

To directly assess displacement, Figure 4(B) presents the effects on out-migration based on equation 10. Displacement effects for the original public housing residents peak at 18 percentage points five years after the award, but then fade to 6 percentage points 15 years after the award as most of the original residents of the failed applicant sites had also moved away by that time. Original residents without access to subsidized housing were also 5 percentage points more likely to have moved within five years after an award and this difference persists 15 years after the award. One explanation is that rising local housing costs—documented in Table A.4—may have priced these residents out of the neighborhood. Consistent with this hypothesis, we find no displacement effect for the original residents who had access to other forms of subsidized housing which would have limited exposure to price increases.

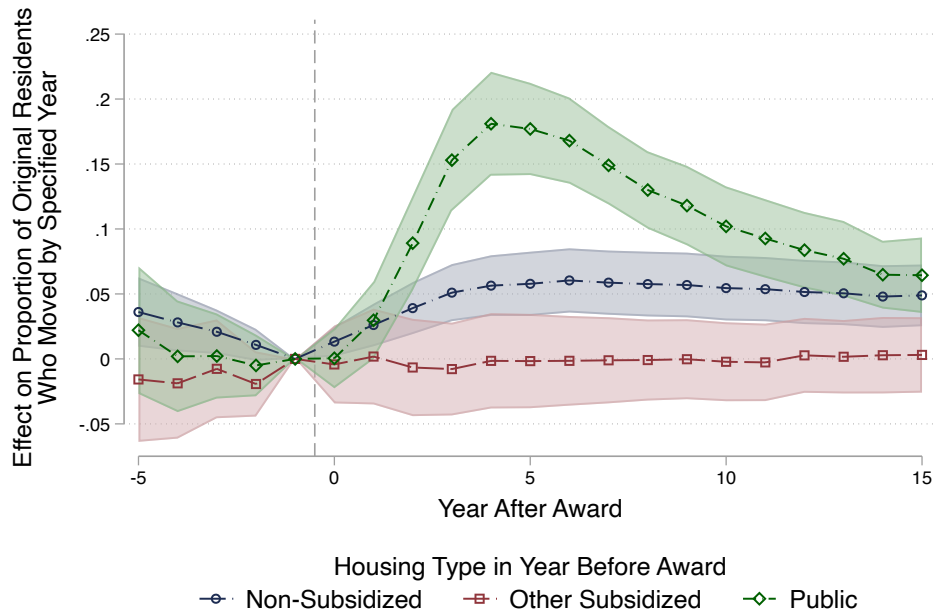
Because of these high migration rates, the effect of the Revitalization program on exposure to neighborhood poverty for original residents was significantly smaller than the effect on the targeted neighborhoods. The red triangles in Figure 5 presents the effect on exposure

Figure 4: Residential Mobility of the Original Residents

(A) Rates of Residential Mobility in Revitalization Neighborhoods



(B) Effect of the Revitalization Program



Notes: Panel A plots the proportion of the original residents of the Revitalization neighborhoods who moved to a different Census block group by a given year after the award. Panel B plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the proportion of the original residents who moved. The shaded regions denote the 95 percent confidence intervals. All results are presented separately by original housing type (i.e., non-subsidized, other subsidized, or public housing).

to neighborhood poverty for original residents—i.e., all residents of the neighborhood in the year before the award. The Revitalization program led to a 1.5 percentage point decrease in the poverty rate of the neighborhoods in which the original residents lived 15 years after the award. For comparison, the blue circles reproduces the estimates from Figure 1 of the effect of the program on poverty rates in targeted neighborhoods. The reduction in poverty rates in targeted neighborhoods is 5.4 times as large as the reduction in exposure to neighborhood poverty rates for the original residents. Figure B.19 illustrates how reductions in neighborhood poverty differ for groups of original residents defined by their housing subsidy in the year before the award. Residents of non-subsidized, public, and other subsidized housing experience similar reductions in exposure to neighborhood poverty.³¹

Consistent with Chyn (2018) and Haltiwanger et al. (2020), we find no evidence that the program affected the income of the adult original residents. Figure B.21 presents estimates from equation 10 where the outcome variable is the poverty rate of the original residents. For all time periods, the estimates are economically small and statistically insignificant.

Assumption 1 identifies the effect of the program on original residents. While this is a population of interest given concerns related to displacement, the decomposition in equation 8 focuses on individuals in subsidized housing after the revitalization. To recover effects for this group—i.e., the subset of the original residents who are in subsidized housing in the post-award period, defined in equation 4—we require the following additional assumption.

Assumption 2. The average potential outcomes of original residents who have subsidized housing in the post-period under the control regime are equal to those with subsidized housing under the treatment regime. Formally,

$$\mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) \mid C(N_{it'}) = 1, S_{it''}^0 = 1] = \mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) \mid C(N_{it'}) = 1, S_{it''}^1 = 1]. \quad (12)$$

³¹Figure B.20 presents estimates for the original residents without subsidized housing broken out by baseline poverty status. We find no evidence of worsening neighborhood conditions for poor residents without subsidized housing.

This assumption holds if the Revitalization program does not affect which original residents appear in subsidized housing in the post-treatment period—i.e., if $S_{it''}^0 = S_{it''}^1$.

Two pieces of evidence support the plausibility of Assumption 2. First, Figure B.22(A) shows that the program did not significantly effect the probability that the original residents appeared in subsidized housing 10 to 15 years after the award.³² Second, we find no difference between the average pre-award income of the original residents of the Revitalization neighborhoods who have subsidized housing after the award and the original residents of the failed applicant neighborhoods who have subsidized housing after the award—i.e., $\mathbb{E}[P_{it'} \mid D(N_{it'}) = 1, S_{it''} = 1] = \mathbb{E}[P_{it'} \mid C(N_{it'}) = 1, S_{it''} = 1]$. Specifically, we find an insignificant difference in average income of \$45 (standard error=\$396).

Assumptions 1 and 2 imply the following identification result,

$$\mathbb{E}[\bar{P}_{t''}(N_{it''}) \mid D(N_{it'}) = 1, S_{it''} = 1] - \mathbb{E}[\bar{P}_{t''}(N_{it''}) \mid C(N_{it'}) = 1, S_{it''} = 1] = \mathbb{E}[\Delta_i \mid \mathcal{G}_1]. \quad (13)$$

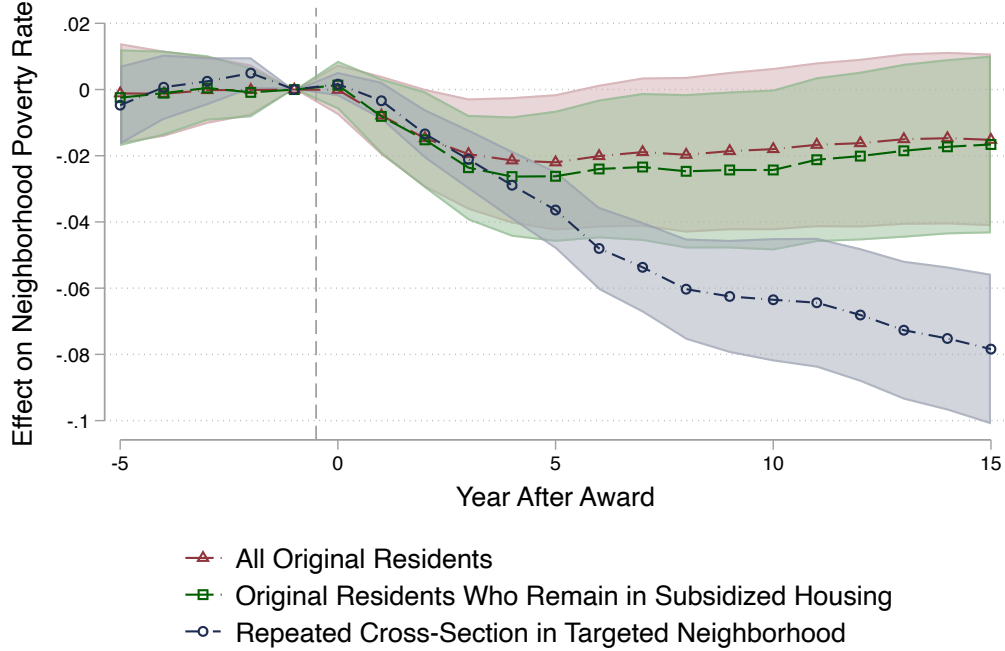
For notational simplicity, we omit explicit conditioning on $X_{it'}$. But for all identification results expectations should be interpreted as being first taken conditional on $X_{it'}$, and then averaged over the distribution of $X_{it'}$. See Appendix D for proof.

The green squares in Figure 5 presents the empirical analog of equation 13, where we re-estimate our main event-study specification on the subsample of original residents who appear in subsidized housing ten years after the award. The results imply that the Revitalization program reduced neighborhood poverty exposure for the original residents by 1.7 percentage points 15 years after the award: $\hat{\mathbb{E}}[\Delta_i \mid \mathcal{G}_1] = -0.017$.

To summarize, most original residents moved away within a few years of the Revitalization awards and therefore were not exposed to the changed neighborhood conditions. The high rates of out-migration were not driven by Revitalization-induced displacement. Instead, they

³²Figure B.22(B) presents results for the original residents who had subsidized housing before the award. Here we see that the probability of living in public housing declined, but this reduction was fully offset by a corresponding increase in voucher receipt.

Figure 5: Effect on Exposure to Neighborhood Poverty for Original Residents



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the poverty rate of the original or current neighborhood. The red triangles present the estimated effect on the poverty rates of the neighborhoods that the original residents live in, tracking these individuals even if they move to different neighborhoods. The green squares repeat the analysis but limit the sample to original residents who were in subsidized housing ten years after the award. The blue circles present estimates of the effect on the poverty rates in the targeted neighborhoods. The shaded regions denote the 95 percent confidence intervals.

are largely a shared feature of low-income populations in high-poverty neighborhoods. Thus, the likely beneficiaries are the residents who move in post-award rather than the incumbent residents. It is important to emphasize that the Revitalization program did displace some original residents. While we find evidence that these families ended up living in slightly lower-poverty neighborhoods, these forced moves also could have imposed disruption costs.³³

5.2.2 New Residents

A reduction in poverty rates in revitalized neighborhoods does not, by itself, imply a reduction in exposure to neighborhood poverty for the new residents. This is because the program

³³We also note that Chyn (2018) and Haltiwanger et al. (2020) find that the children who lived public housing projects demolished under HOPE VI earned more in adulthood as a result of the program.

could have attracted individuals who would have chosen to live in different, lower-poverty neighborhoods absent the intervention.

Using the framework in Section 4, we can formally decompose the treatment effect for new residents into two components:

$$\mathbb{E}[\Delta_i \mid \mathcal{G}_2] = \underbrace{\mathbb{E}[\bar{P}_{t''}^1(n) - \bar{P}_{t''}^0(n) \mid D(n) = 1]}_{\text{Effect on Neighborhoods}} + \underbrace{\mathbb{E}[\bar{P}_{t''}^0(N_{it''}^1) - \bar{P}_{t''}^0(N_{it''}^0) \mid \mathcal{G}_2]}_{\text{Sorting Effect}}. \quad (14)$$

This expression shows that effect on new residents' exposure to poverty depends not only on how the program affected poverty rates in targeted neighborhoods, but also where these individuals would have lived absent the intervention. If they would have lived in lower-poverty areas absent the program, then the sorting effect could offset the neighborhood-level poverty reduction.

To assess the importance of the sorting effect, we compare subsidized renters who move into the HOPE VI and failed applicant neighborhoods after an award and estimate the difference between the poverty rates in their *origin neighborhoods*—i.e., the neighborhoods where they lived prior to moving. We use these origin neighborhoods as proxies for where individuals would have lived absent the intervention. If the newcomers in the revitalized neighborhoods tend to come from lower-poverty areas than those entering failed applicant neighborhoods, it suggests the program attracted individuals who would have otherwise lived in different, lower-poverty neighborhoods. Assumption 3 formalizes this intuition.

Assumption 3. There is a linear relationship between the poverty rate in the origin neighborhood and non-treated counterfactual,

$$\bar{P}_{t'}(N_{it'}) = \alpha + \theta \bar{P}_{t''}^0(N_{it''}^0) \text{ with } \theta > 0. \quad (15)$$

Assumptions 1 and 3 yield the following identification result:

$$\begin{aligned}
& \mathbb{E}[\bar{P}_{t'}(N_{it}) \mid D(N_{it'}) = 0, D(N_{it''}) = 1, S_{it''} = 1] \\
& \quad - \mathbb{E}[\bar{P}_{t'}(N_{it}) \mid C(N_{it'}) = 0, C(N_{it''}) = 1, S_{it''} = 1] \\
& = \theta \mathbb{E}[\bar{P}_{t''}^0(N_{it''}^0) - \bar{P}_{t''}^0(N_{it''}^1) \mid \mathcal{G}_2],
\end{aligned} \tag{16}$$

implying that the difference in average origin-neighborhood poverty rates is proportional to the sorting effect.³⁴ See Appendix D for proof.

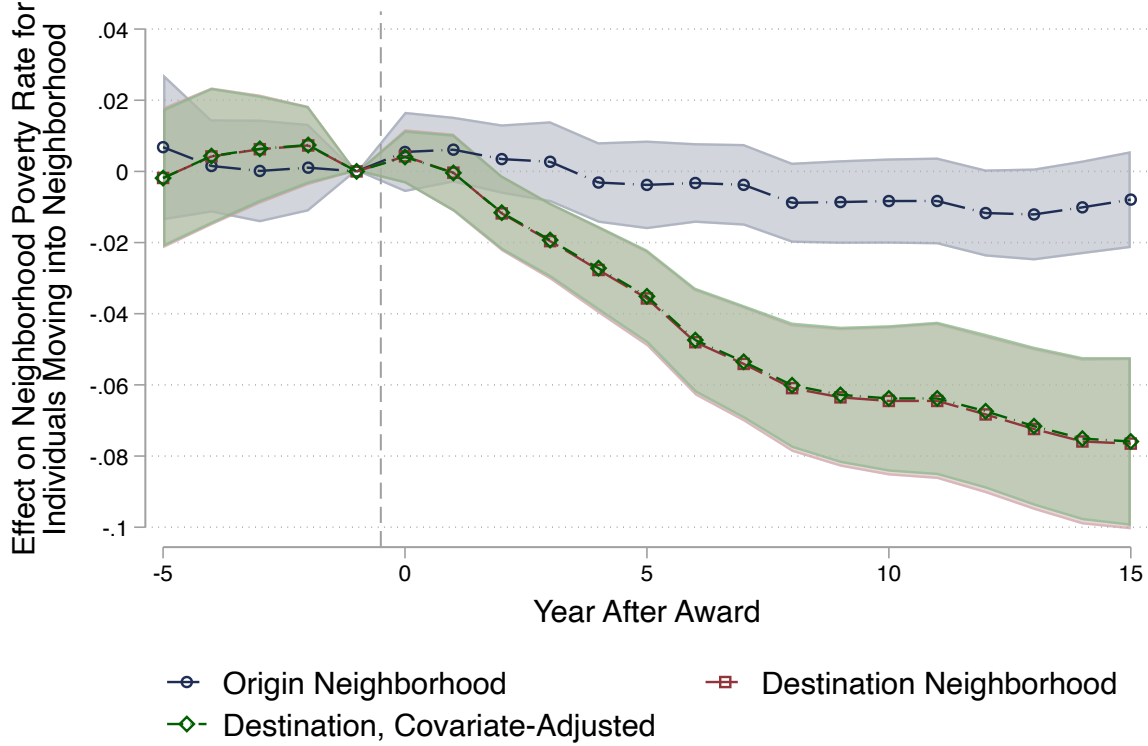
To estimate this difference, we construct a sample of subsidized households who move into a failed applicant or Revitalization neighborhood between 5 years before an award and 15 years after an award and estimate equation 10 on individual-level panel data where the outcome is the poverty rate in the *origin* neighborhood (i.e., the neighborhood the individual lived in before moving to the Revitalization or failed applicant neighborhood).

We find that subsidized renters who moved into the revitalized neighborhoods came from neighborhoods with similar poverty rates to the subsidized renters who moved into failed applicant neighborhoods. The blue circles in Figure 6 present the estimates and show that e that the new residents of the revitalized neighborhoods moved from neighborhoods that had quantitatively similar and never statistically different poverty rates: origin neighborhood poverty rates are 0.8 percentage points lower 15 years after the award. For comparison, the red squares in Figure 6 present the estimated effects on the poverty rate of the destination neighborhoods (these results simply replicate the findings in Figure 1), and show that families who move into Revitalization neighborhoods after the award move into lower poverty neighborhoods than those who move into failed applicant neighborhoods. These results suggest that the sorting effects are small relative to the effect on neighborhood poverty.

To further quantify the importance of sorting, we estimate the effect on the destination

³⁴There is a slight abuse of notation in our use of t' , which refers to the pre-award period in the conceptual framework. In the empirical analysis, we measure neighborhood poverty based on the location the individual lived in just prior to entering a targeted neighborhood. This may not correspond exactly to the pre-award neighborhood if individuals moved multiple times before entering the targeted neighborhood.

Figure 6: Effect on Exposure to Neighborhood Poverty for New Subsidized Renters



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10 but estimated on individual-level panel data. The sample includes subsidized renters who move into a Revitalization or failed applicants neighborhood in the year specified by the horizontal axis. For the blue circles, the outcome is the neighborhood poverty rate in the origin neighborhood. For the red squares and green diamonds, the outcome is the poverty rate in the destination neighborhood. The specification underlying the green triangles additionally controls for poverty rate in the origin neighborhood, the income of the individual before the move, and a quadratic in age. The shaded regions denote the 95 percent confidence intervals.

neighborhood's poverty rate using a modified version of equation 10, which controls for the poverty rate in the origin neighborhood, the income of the individual before the move, and a quadratic in age. The green diamonds in Figure 6 display these estimates. The results are almost identical to the unadjusted estimates (the red squares): 15 years after the award the adjusted estimate is just 0.06 percentage points (or 1 percent) smaller. The adjusted estimates represent the causal effect of HOPE VI on exposure to neighborhood poverty for the new residents if the covariates fully capture the differences in the outside housing options for the subsidized families who move into the HOPE VI and failed applicant neighborhoods.

We view the estimates in Figure 6 as providing strong support for the idea that the reduction in poverty in targeted neighborhoods translated into a reduction in exposure to poverty for the new subsidized households. In other words, the estimates from Figure 6 suggest that the program reduced the new residents' exposure to neighborhood poverty by 7.6 percentage points 15 years after the award: $\hat{\mathbb{E}}[\Delta_i | \mathcal{G}_2] = -0.076$.

The new residents of Revitalization neighborhoods could also differ on unobserved dimensions. We assess this concern by controlling for a richer set of characteristics. Panel A of Table A.6 presents estimates of a regression of an outcome on the Revitalization indicator and the interaction between award year and year of move, estimated on a sample of individuals who move in after the award. The outcomes in columns 1 and 2 are the poverty rate in the prior and current neighborhood, respectively. These columns simply replicate the findings in Figure 6, but summarize the average difference instead of presenting differences by award year. Columns 3-5 add an increasingly rich set of covariates to the model: column 3 adds the baseline covariates included in the green series Figure 6; column 4 controls for contemporaneous income; and column 5 adds 14 additional characteristics from the HUD data, including race, ethnicity, sex, disability status, and income amounts from various sources.

Figure B.23 presents the effect of the Revitalization program on out-migration. We estimate equation 10 where the outcome variable is the proportion of current residents who move to a different neighborhood in the following year. For subsidized renters, out-migration rates spike four years after the award, but, in the long-run, there is a reduction in out-migration rates: subsidized renters living in Revitalization neighborhoods 10 to 15 years after the award are 3 percentage points less likely to move compared to those in failed applicant neighborhoods. For non-subsidized households we find much smaller displacement effects in the short-run and no evidence of a difference in out-migration rates in the long-run.

5.2.3 Residents Who Would Have Moved in Absent the Award

The Revitalization program reduced the subsidized housing population in targeted neighborhoods by approximately 20 percent in the long-run (see Table 1 and Figure B.15). How-

ever, as we show in Section 5.3, the overall citywide supply of subsidized housing remained unchanged due to an offsetting expansion in housing vouchers. This implies that some subsidized renters who might otherwise have moved into Revitalization neighborhoods instead used vouchers to locate elsewhere.

While we cannot directly identify these individuals, we approximate their counterfactual outcomes using the origin neighborhood poverty rate among those who moved into failed applicant neighborhoods after the award. We require the following assumption.

Assumption 4. This assumption has two components. First, for individuals who would have moved into a Revitalization neighborhood absent the award, their counterfactual destination poverty rate is equal, on average, to the poverty rate in their origin neighborhood. Second, among individuals who would have moved into a subsidized revitalized unit absent the program, the poverty rate of the origin neighborhood is unrelated to whether they would have moved in under treatment. Formally,

$$\begin{aligned} \mathbb{E}[\bar{P}_{it''}^1(N_{it''}^1) \mid D(N_{it'}) = 0, D(N_{it''}^1) = 0, D(N_{it''}^0) = 1, S_{it''}^1 = 1] \\ = \mathbb{E}[\bar{P}_{it''}^0(N_{it'}) \mid D(N_{it'}) = 0, D(N_{it''}^1) = j, D(N_{it''}^0) = 1, S_{it''}^k = 1], \end{aligned} \quad (17)$$

for $j \in \{0, 1\}$ and $k \in \{0, 1\}$. This assumption holds if, absent the intervention, displaced individuals would have remained in their origin neighborhoods (which are unaffected by spillovers) and are, on average, similar to those who would have moved into the Revitalization neighborhoods regardless of treatment.

Under Assumptions 1 and 4, we obtain the following identification result:

$$\mathbb{E}[\bar{P}_{it''}(N_{it'}) - \bar{P}_{it''}(N_{it''}) \mid C(N_{it'}) = 0, C(N_{it''}) = 1, S_{it''} = 1] = \mathbb{E}[\Delta_i \mid \mathcal{G}_3], \quad (18)$$

where the left-hand side is directly observable in the data and corresponds to the difference in average neighborhood poverty rates between the origin and destination neighborhoods for subsidized renters who move into the failed applicant neighborhoods after the award. See

Appendix D for proof.

To estimate this quantity, we identify all subsidized households who moved into failed applicant neighborhoods 10–15 years after the award. For these individuals, we compute the difference between the poverty rate in their destination (failed applicant) neighborhood and their origin neighborhood. On average, the poverty rate in the destination neighborhood is 14 percentage points higher than in their origin neighborhood. Under Assumptions 1 and 4, this is, on average, the change in poverty rate for the displaced subsidized renters who would have moved into a Revitalization neighborhood absent the intervention: $\hat{\mathbb{E}}[\Delta_i \mid \mathcal{G}_3] = -0.14$.

While Assumption 4 is strong, the magnitude of this estimate is supported by the extreme poverty of the Revitalization neighborhoods prior to treatment. In 1990, 86 percent of these neighborhoods were in the top decile of the county-level poverty distribution. The 25th and 50th percentiles of the Revitalization neighborhood distribution corresponded to the 95th and 98th percentiles of the broader county distribution, respectively. For households diverted away from these neighborhoods, it would have been difficult to relocate elsewhere without reducing exposure to poverty.

5.2.4 Spillovers

If the program shaped the spatial distribution of poverty by affecting where poor adults lived (as opposed to changing the income of individuals), then the reduction in poverty rates in Revitalization neighborhoods must be offset by an increase in poverty rates in other neighborhoods. If poor households were displaced into a small number of neighborhoods with many subsidized renters, then the increase in exposure to poverty for these incumbent residents could offset the gains experienced by families in the Revitalization neighborhoods. Our goal is to estimate an upper bound for the “migration spillover” effect—i.e., the effect attributable to individuals who would have lived in the Revitalization neighborhoods absent the award but instead live elsewhere. Our approach relies on the following assumption.

Assumption 5. This assumption has two components. First, for neighborhoods that are not directly targeted by the program, the treatment effect on poverty rates is attributable

to migration of the displaced individuals—i.e., individuals who would have lived in targeted neighborhoods absent the award. Formally, for all neighborhoods n such that $D^1(n) = 0$:

$$\begin{aligned}\bar{P}_{t''}^1(n) &= (1 - \lambda_n)\bar{P}_{t''}^0(n) + \lambda_n \mathbb{E}[P_{it''}^1 \mid N_{it''}^1 = n, D(N_{it''}^0) = 1] \\ \sum_i \mathbb{1}\{N_{it''}^1 = n\} &= \sum_i \mathbb{1}\{N_{it''}^0 = n\} = \sum_i \mathbb{1}\{N_{it''} = n\},\end{aligned}\tag{19}$$

where $\lambda_n = \frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it''}^0) = 1\}}{\sum_i \mathbb{1}\{N_{it''}^1 = n\}}$ is the share of neighborhood n 's residents at time t'' who are displaced from treated neighborhoods. Second, individuals who would have moved in after, instead stay in their origin neighborhoods,

$$N_{it''}^1 = N_{it'} \text{ for } i : D^1(N_{it''}^1) = 1, D(N_{it''}^0) = 1.\tag{20}$$

Assumption 5 allows us to empirically identify which neighborhoods may have been impacted by spillovers and implies that the spillover effect can be bounded as follows,

$$\begin{aligned}\bar{P}_{t''}^1(n) - \bar{P}_{t''}^0(n) &\leq \frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it'}^1) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}} \\ &\quad + \frac{\sum_i \mathbb{1}\{N_{it'} = n, D(N_{it''}^0) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}},\end{aligned}\tag{21}$$

for all $D^1(n) = 0$. See Appendix D for proof.

The numerator of the first term on the right-hand-side of equation 21 is the number of individuals who lived in a Revitalization neighborhood before an award and who moved to neighborhood n after the award. This quantity is observable in the Revitalization sample. The numerator of the second term is the number of individuals who lived in neighborhood n before the award and would have subsequently moved into a Revitalization neighborhood absent the award. This quantity is observable in the failed applicant sample. This is an upper bound because i) the population sizes are weakly larger than the number of poor displaced residents, and ii) individuals who were directly displaced by the program could have replaced other poor households in n .

To estimate an upper bound for the migration spillover effects, we estimate the number of poor households who were displaced using Table 1 and then allocate these displaced households in a way that is both consistent with observed migration flows and maximizes the potential increase in average neighborhood poverty rates for subsidized renters.

We estimate the two components in equation 21 separately. In both cases we implement the following procedure for each target neighborhood:

1. Estimate the size of the displaced population. See details below.
2. Estimate the maximum size of the population of poor individuals displaced into each neighborhood. See details below.
3. Rank destination neighborhoods by the number of subsidized renters.
4. Calculate a counterfactual poverty rate in the destination neighborhood with the most subsidized renters by removing the displaced poor individuals from this neighborhood and assuming that they are replaced by non-poor individuals. If the size of the displaced population exceeds the number of people who moved to this neighborhood, then remove the remaining displaced households from the second-ranked neighborhood, and so on.
5. Summarize the spillover effects from the displaced original residents by taking the weighted-average of the increase in neighborhood poverty rates among the set of destination neighborhoods that had a non-zero change in poverty, weighting by the size of the subsidized population in the destination neighborhood.

This methodology yields an estimate of the spillover migration effects for each target neighborhood. See Table A.7 for a formal description of each step. We summarize these effects by then taking a weighted-average across projects, weighting by the baseline population in the target neighborhood.

We start by discussing the results for the displaced original residents, i.e. $\frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it'}) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}}$.

For each Revitalization neighborhood we implement steps 1 through 5 described above. In step 1 we estimate the size of the displaced population as being equal to the number of poor, original residents who remain in the neighborhood 10 years after the award multiplied by 0.35, which is based on the estimates from Column 3 of Table 1.³⁵ In step 2, we

³⁵Column 3 of Table 1 shows that the Revitalization program reduced the number of poor, original residents who remained in the Revitalization neighborhoods by 26 percent (0.26=74/285). Thus, the size of this population would be 1.35 times larger (1.35=1/(1-0.26)) absent the program, and the size of the displaced population is $0.35 \times p = 1.35 \times p - p$, where p is the number of remaining, poor, original residents.

identify all poor, original Revitalization residents who lived in a different neighborhood 10 years after the award and count the number of people who moved to each of the destination neighborhoods. Migration patterns of the original residents were dispersed, as 97 percent of the migration flows between Revitalization and destination neighborhoods consisted of fewer than 10 people. In step three, we find that the original residents tended to move to neighborhoods with fewer subsidized renters: the average Revitalization neighborhood contains 5.6 times more subsidized renters than the average destination neighborhood. Step 4 provides an estimate of how much lower poverty rates would be in the destination neighborhoods absent the displaced households and step 5 takes an average across neighborhoods affected by the given Revitalization neighborhoods. Taking the weighted average across all Revitalization projects implies that migration spillover effects from the displaced original residents could have increased poverty rates by at most 0.30 percentage points for subsidized renters.

We follow an analogous approach to estimate the spillover effects attributable to households who would have moved in after the award, i.e., $\frac{\sum_i \mathbb{1}\{N_{it'}=n, D(N_{it''}^0)=1\}}{\sum_i \mathbb{1}\{N_{it''}=n\}}$. We have no way of identifying who would have moved into the Revitalization neighborhoods absent the award and we therefore study the migration flows into the failed applicant neighborhoods. For each failed applicant neighborhood we proceed as follows. In step 1, we estimate the size of the displaced population as being equal to the number of new, poor residents who lived in the neighborhood ten years after the award multiplied by 0.15, which is based on the estimates from Column 4 of Table 1 (0.15=195/1,290). In step 2, we identify all poor, new residents and count the number of people who moved from each origin neighborhood to the failed applicant neighborhood. Again, migration patterns are dispersed: 95 percent of the migration flows between origin and failed applicant neighborhoods consisted of fewer than 10 people. The third step shows that new residents came from neighborhoods with far fewer subsidized renters: the average failed applicant neighborhood contains 31 times more subsidized renters compared to the average origin neighborhood. After implementing the remaining steps we find that the migration spillover effects from the displacement of households who would have

moved in absent the award could have increased poverty rates by at most 0.19 percentage points for subsidized renters.

Column 6 of Table 1 also shows that the program increased the number of non-poor households who moved in after the award by 23 percent ($0.23=199/879$). Poverty rates could have risen in the neighborhoods from which these households moved if they were replaced by poor households. Following an analogous methodology, we find that the displacement of non-poor households who would have moved in absent the award could have increased poverty rates by 0.27 percentage points in the neighborhoods that these households would have lived in absent the award. Column 5 of Table 1 shows that the program reduced the number of non-poor original residents, suggesting that spillover effects from this group would actually reduce exposure to neighborhood poverty.

Taken together, the results suggest that households who were displaced by the Revitalization program were dispersed across many neighborhoods and that these neighborhoods had fewer subsidized households than the Revitalization neighborhoods. Because of this, the increases in poverty rates in other neighborhoods throughout the city due to program spillover were small in magnitude and had little effect on poverty exposure for subsidized renters, particularly when compared to the reduction in poverty rates in Revitalization neighborhoods. Summing the displacement effects discussed above yields an estimate of $\mathbb{E}[\Delta_i|\mathcal{G}_4] = -0.0076 = -0.0030 - 0.0019 - 0.0027$.

5.3 Aggregate Effects

We estimate the aggregate effect of the Revitalization program on the exposure to neighborhood poverty for all subsidized renters as the weighted average of the group-specific effects presented in Sections 5.2.1–5.2.4. The weights are a function of the share of subsidized units in the PHA that are targeted by the Revitalization program. See Appendix E for details.

Figure 7(A) presents the results from the decomposition described in equation 8 in the case where a PHA revitalizes half of their subsidized housing units.³⁶ The height of the

³⁶Among PHAs that received Revitalization funding, the average percent of units affected by HOPE VI

first four bars represents the group-specific estimate of $\omega_g \mathbb{E}[\Delta_i | \mathcal{G}_g]$ and the bar on the right presets the cumulative effect. Our estimates imply that PHAs that revitalized half of their subsidized units reduced the average neighborhood poverty rate for all subsidized renters by 4.0 percentage points.³⁷ Reductions in exposure to poverty for new, subsidized residents of the revitalized neighborhoods explain 68 percent of this effect.

As a complimentary analysis, we also show that subsidized households experienced a larger reduction in exposure to neighborhood poverty in PHAs that revitalized more public housing. We construct a PHA-level dataset and estimate the following specification:³⁸

$$y_h = \alpha + \beta D_h + \lambda S_h + \phi X_h + \delta D_h \times S_h + \theta S_h \times X_h + \epsilon_h \quad (22)$$

where h denotes the PHA, D_h is equal to one if any public housing project received a Revitalization grant and is zero if any project applied for Revitalization funding but no project received HOPE VI funding, S_h is the share of subsidized units in 1997 that were targeted by HOPE VI (awarded or applied for), and X_h is the average poverty rate of the HOPE VI and failed applicant sites in 1990.³⁹ The sample includes PHAs that received Revitalization funding or applied for but did not receive any HOPE VI funding. The main coefficient of interest is δ , which captures the effect of revitalizing a larger share of subsidized units. Equation 22 compares PHAs that were awarded HOPE VI funding to PHAs that applied to revitalize a similar share of units but were not awarded funding.

funding was 22 percent and the 5th and 95th percentiles were 4 percent and 48 percent, respectively. In other words, there were some PHAs that revitalized close to half of their subsidized housing units.

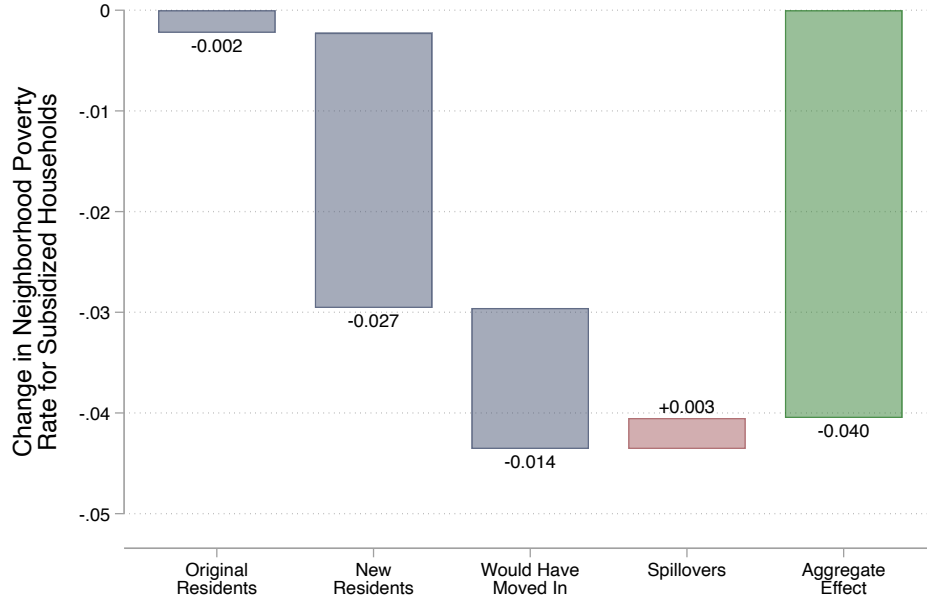
³⁷These estimates do not include effects from the spatial spillovers described in Figure B.17. If subsidized households were more likely to live nearby the Revitalization neighborhoods relative to other neighborhoods in the city then these spatial spillovers would also reduce exposure to poverty. In this case, our estimate understate the effect of the program.

³⁸This PHA-level analysis uses data from HUD User’s Picture of Subsidized Households and we focus on changes between 1997 (which is the earliest year in which all the necessary variables are available) and 2018.

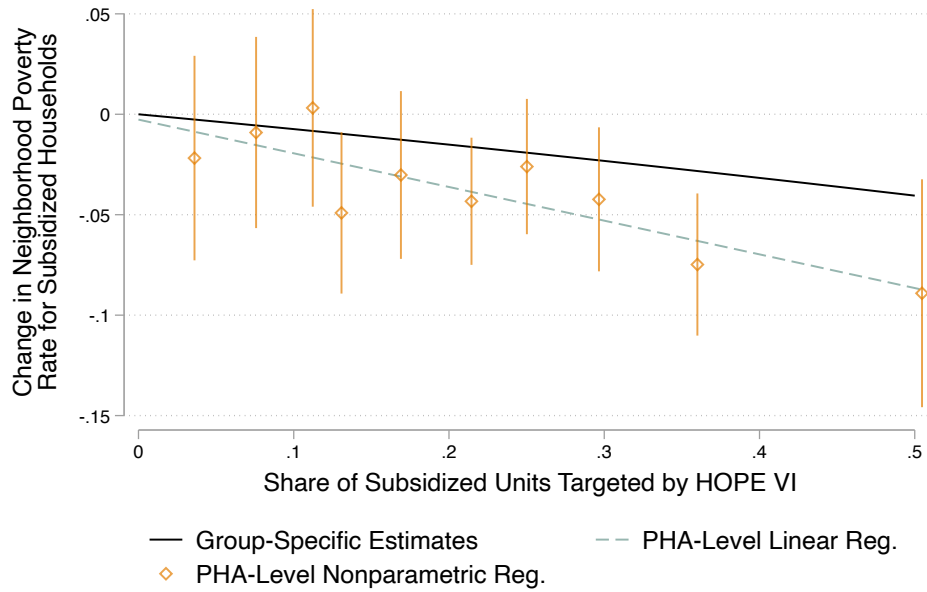
³⁹We control for baseline poverty rates, since this is a key potential confounder. We estimate that the program reduced poverty rates by 7.7 percentage points when regressing block group poverty rate in 2017 on the Revitalization indicator and a vector of 15 baseline covariates (see column 1 of Table A.3). If we estimate the same regression but only control for baseline poverty rates the estimate is 6.9 percentage points. If we include no covariates, the estimate drops to 3.5 percentage points. Following Feigenberg et al. (2023) we include the interaction between S_h and X_h .

Figure 7: Analysis of Public Housing Authorities

(A) Aggregation of Group-Specific Estimates



(B) Comparison to PHA-Level Regression



Notes: The height of first five bars in Panel A represents estimates of $\omega_g \mathbb{E}[\Delta_i | \mathcal{G}_g]$ from equation 8 in the case where the PHA revitalized half of the subsidized housing units. The bar on the right presents the aggregate effect. Panel B compares the aggregate effects based on the group-specific estimates to those obtained from the PHA-level regression described in equation 22. The figure presents the estimated effects as a function of the share of subsidized units that were targeted by HOPE VI.

Table A.8 presents the coefficient estimates from equation 22. The outcome in column 1 is the change in the average census tract poverty rate for subsidized households between 1997 and 2018. The estimates imply that revitalizing half of the subsidized units led to an 8.4 percentage point reduction in exposure to poverty ($0.084 = 0.168/2$). We estimate a more flexible specification by replacing the linear interaction between D_h and S_h with the interaction between D_h and indicators for whether the PHA was in different deciles of S_h . Figure 7(B) shows that the estimates from this less parametric PHA-level regression (the diamonds), equation 22 (the dashed line), and the aggregated estimates from Section 5.2 (the solid line) all suggest similar aggregate associations between the Revitalization program and exposure to neighborhood poverty.⁴⁰ Returning to Table A.8, the outcome in column 2 is the component of changes in exposure to poverty (outcome in column 1) attributable to changes in poverty rates in revitalized sites.⁴¹ The estimates suggest that reductions in neighborhood poverty in revitalized sites explain 60 percent of the total reduction in neighborhood poverty ($0.60 = 0.101/0.168$), similar to our findings based on equation 8 (68 percent compared to 60 percent).

The outcome variables in columns 2 to 5 of Table A.8 are the change in the share of subsidized units in public housing, total public housing stock, share of subsidized renters with households incomes below \$10,000, and the change in county-level poverty rates. These estimates suggest that HOPE VI shifted subsidized units from public to voucher housing but did not reduce the overall supply of subsidized housing. There is no evidence that the program displaced the neediest households entirely out of subsidized housing—an important concern, given that voucher recipients frequently encounter difficulties securing landlords willing to accept them Ellen (2020). Furthermore, there is no evidence that Revitalizations

⁴⁰The diamond markers in Figure 7(B) plot the estimated coefficient on the interaction between D_h and deciles of S_h against the average value of S_h for that decile.

⁴¹We can write the average poverty rate for subsidized households in a given PHA at time t as, $p_t = \omega_t p_t^r + (1 - \omega_t) p_t^n$, where ω_t is the share of units in revitalized neighborhoods and p^r and p^n denote the average poverty rates in revitalized and other neighborhoods, respectively. Then we can write the change as, $p_{t+1} - p_t = \omega_t(p_{t+1}^r - p_t^r) + p_{t+1}^r(\omega_{t+1} - \omega_t) + (1 - \omega_{t+1})p_{t+1}^n - (1 - \omega_t)p_t^n$. The outcome in column 1 of Table A.8 is $p_{t+1} - p_t$ and the outcome in column 2 is $\omega_t(p_{t+1}^r - p_t^r)$.

reduced the overall poverty rates in the city, which is consistent with the idea that the program altered where poor households lived but did not affect the incomes of individual adults.

We also use equation 22 to estimate impacts on exposure to poverty for all other households in the county. Specifically, using publicly available data, we estimate the change in exposure to neighborhood poverty between 1990 and 2017 for different groups of households based on their incomes. The results are presented in Table A.9. We find no evidence of impacts on the broader population. For example, column 1 implies that revitalizing half of the subsidized units is associated with an 1 percentage point reduction in exposure to poverty ($0.012 = 0.024/2$). We find similarly small and statistically insignificant impacts for all eight income groups. These analyses have limited statistical power to detect effects on the broader population, which is expected given that subsidized households represent a relatively small share of the overall population.

6 Conclusion

Addressing the harms from residential segregation is a central goal of housing policy. Yet place-based efforts to improve distressed neighborhoods often raise concerns about displacement of the very households they are intended to help. In this paper, we use individual-level administrative data to study how neighborhood-level transformation affects individual residential outcomes in the context of a large, place-based intervention: the HOPE VI Revitalization program.

We find that, as intended, the Revitalization program led to a large, persistent reduction in poverty rates in targeted neighborhoods, which enabled subsidized renters to live in lower-poverty neighborhoods. Fifteen years after a Revitalization award, the program reduced poverty rates by 8 percentage points in targeted neighborhoods by reducing the stock of public housing and attracting higher-income individuals to move into new, market-rate units.

To understand how the Revitalization program affected subsidized renters, we separately estimate the impact of the program on exposure to neighborhood poverty for four mutually exclusive and exhaustive groups of subsidized renters. The new residents of Revitalization neighborhoods experienced the largest reductions in exposure to neighborhood poverty while the original neighborhood residents experience poverty reductions that are less than one-fifth the size, largely due to high rates of residential mobility. Our estimates imply that cities that revitalized half of their public housing stock reduced the average neighborhood poverty rate among all subsidized renters by 4.0 percentage points.

The program required a large upfront investment but it produced a durable change in neighborhood conditions. In combination with existing estimates of households' willingness-to-pay for neighborhood amenities from Galiani et al. (2015), our estimates suggest that the present discounted value of cumulative benefits to subsidized renters from the reduction in exposure to neighborhood poverty exceed the program costs when considered over a time horizon of at least 14 years. See Appendix F for details. This calculation only considers the direct financial costs and benefits from a reduction in exposure to neighborhood poverty for subsidized renters. While these are key inputs to any cost-benefit analysis, the program could have imposed other costs and produced other benefits not captured by our analysis. For example, rising house prices could have adversely impacted low-income households without access to subsidized rent. Nonetheless the estimates suggest that the program generated gains that were large even when compared to the large upfront costs.

Our results suggest that place-based policies can be a useful tool for reducing exposure to neighborhood poverty for subsidized renters. However, high rates of residential mobility imply that place-based policies may be an ineffective way to target specific individuals (e.g., the original neighborhood residents) while still being an effective way to target broader populations of interest (e.g., subsidized renters). The provision of housing vouchers to the original neighborhood residents offers one promising approach for attenuating price-driven displacement that might occur after the implementation of a place-based policy.

While reducing residential segregation is a central objective of housing policymakers, the Revitalization program also sought to create living conditions that residents would value and benefit from. We present evidence that the program reduced rates of out-migration and increased housing prices, suggesting that the revitalized neighborhoods became more appealing places to live. Future work should aim to more completely quantify the improvements in neighborhood conditions and detail the welfare gains and losses for different sub-populations. For example, determining whether the program improved outcomes for children who grew up in revitalized neighborhoods or estimating impacts on poor households who lack subsidized housing are both promising avenues for future research.

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Appendix A Additional Tables

Table A.1: Sample Restrictions

	Non-Applicant		Failed Applicant		Revitalization	
	count	percent	count	percent	count	percent
Full sample	16,300	100%	244	100%	286	100%
Located in state	15,932	98%	242	99%	286	100%
Not in indian housing	13,233	81%	242	99%	286	100%
In picture of subsized housing	13,233	81%	233	95%	272	95%
Not scattered site or vacant	12,618	77%	220	90%	271	95%
At least 25 units in 1993	8,913	55%	210	86%	266	93%
Not senior housing	6,946	43%	204	84%	262	92%
Non-missing location	6,084	37%	198	81%	251	88%
At least one mile from Revitalization	5,783	35%	166	68%	251	88%

Notes: The column headers define the sample (the three samples are mutually exclusive). Each row presents the count of projects that remain after imposing the sample restriction as well as the percent of the full sample.

Source: Authors' calculations based on project-level summary files from HUD and the publicly available list of awardees of and applicants to HOPE VI funding.

Table A.2: Baseline Characteristics

	Non-Applicant		Failed Applicant		Revitalization	
	mean	s.d.	mean	s.d.	mean	s.d.
A. Project						
Occupied units	430	960	293	200	667	756
Percent minority	61.6	38.9	77.7	33.0	77.9	35.6
Percent majority income earned	21.1	15.2	19.9	11.2	14.1	10.5
Average household size	2.3	1.1	2.7	1.0	2.4	1.1
Percent 62 or older	26.4	21.4	15.0	11.8	13.1	11.7
Percent with disability	12.9	11.1	10.9	7.5	10.0	8.9
Percent single parent	35.5	25.0	47.4	23.4	40.3	27.2
Percent with female head	69.0	25.7	75.4	24.6	71.4	31.5
Percent crowded housing	6.8	8.5	7.4	6.8	8.0	7.1
Average rent	273	145	248	123	189	130
Average income (thousands)	13.1	6.9	11.6	5.3	8.9	5.9
Observations	5,783		166		251	
B. Census block group						
Percent Black	31.9	33.8	48.3	36.2	72.0	32.0
Median rent	606	260	501	207	451	222
Median home value (thousands)	119	103	111	113	109	107
Percent with public assistance	19.2	13.0	31.3	16.8	38.2	17.2
Percent unemployed	12.1	8.2	19.4	12.1	24.0	14.1
Percent below poverty	30.8	17.2	45.0	19.0	55.0	20.6
Number of people (thousands)	1.5	0.6	1.6	0.6	1.9	0.8
Observations	5,391		178		308	

Notes: The columns define one of three mutually exclusive samples including non-HOPE VI, failed applicant, and Revitalization neighborhoods. Each row presents statistics for a different variable. In panel A the variables are measured at the project level in 1993. In panel B the variables are measured at the block group level in 1990. In panels A and B we weight by the population of the project and block group, respectively.

Source: Authors' calculations based on project-level summary files from HUD and summary files from the 1990 Decennial Census.

Table A.3: Effect on Neighborhood Poverty Using Alternative Estimator, Data, and Comparison Groups

Change in Block Group Poverty Rate Between 1990 and 2017

	(1)	(2)	(3)	(4)	(5)
A. With Covariates					
Revitalization	-0.077 (0.016) [0.000]	-0.118 (0.008) [0.000]	-0.093 (0.005) [0.000]	-0.101 (0.012) [0.000]	-0.066 (0.006) [0.000]
Covariates	baseline	baseline	baseline	baseline+fe	baseline+fe
Comparison group	failed applicants	non-applicants	non-public housing	non-applicants near Revitalization	non-public housing near Revitalization
Observations	485	5,694	209,685	1,130	25,512
B. Without Covariates					
Revitalization	-0.132 (0.020) [0.000]	-0.202 (0.008) [0.000]	-0.222 (0.005) [0.000]	-0.191 (0.012) [0.000]	-0.228 (0.006) [0.000]
Covariates	none	none	none	fe	fe
Comparison group	failed applicants	non-applicants	non-public housing	non-applicants near Revitalization	non-public housing near Revitalization
Observations	485	5,694	209,722	1,130	25,522

Notes: Each column presents estimates from a separate regression in which the change in Census block group poverty rate between 1990 and 2017 is regressed on an indicator for receiving a Revitalization grant. The comparison groups in columns 1-5 are block groups that contain failed applicant projects, projects that did not apply for HOPE VI funding, no public housing projects, projects that did not apply for HOPE VI funding and are located within 1-5 miles of a Revitalization site, and no public housing projects and are located within 1-5 miles of a Revitalization site. The vector of baseline covariates include the following characteristics of the block group measured in 1990: poverty rate, share of residents that are White, share of residents that are Black, median rent, median home value, share of households receiving public assistance income, share of residents with a college degree, unemployment rate, share of residents that are new renters or owners, share of units that are owner occupied, share of housing building before 1960, total population, total housing, and share of units vacant. Columns 4 and 5 include fixed effects (fe) for the closest Revitalization site. Each regression is estimated via weighted least squares, using block group population in 1990 as the weights. Standard errors are presented in parentheses and p-values in brackets.

Source: Authors' calculations based on summary files of Decennial Census survey and the ACS.

Table A.4: Effect on Housing Prices and Home Ownership

	Log Rent			Log Home Value			Log Monthly Mortgage			Owner	Renter
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Revitalization	0.249 (0.062) [0.000]	0.192 (0.061) [0.002]	0.136 (0.044) [0.002]	0.246 (0.074) [0.001]	0.136 (0.067) [0.042]	0.161 (0.069) [0.021]	0.125 (0.052) [0.018]	0.044 (0.048) [0.357]	0.101 (0.044) [0.021]	-0.014 (0.023) [0.537]	0.014 (0.023) [0.537]
Housing Characteristics Exclude New Housing		X	X		X	X		X	X		
Mean	6.560	6.560	6.730	11.700	11.700	11.700	6.760	6.760	6.690	0.348	0.652
Standard deviation	0.062	0.061	0.044	0.074	0.067	0.069	0.052	0.048	0.044	0.023	0.023
Observations	30,000	31,000	12,000	33,000	32,000	23,000	18,000	18,000	12,000	2,000	2,000

Notes: Each column presents estimates from a separate regression in which an outcome variable is regressed on an indicator for the Revitalization award. Regressions are estimated via weighted-least squares using the inverse propensity score. All regressions control for the year of first award and the calendar year as well as the vacancy rate, median home value, neighborhood poverty rate, and share of population that is Black in the year before the award. When indicated some regressions further control for housing characteristics (building type, year built, persons per room) and limit to house units built before the award. In columns 1-9 the unit of observation is the housing unit. In columns 10 and 11 the data are collapsed to the project level. Standard errors are clustered at the neighborhood and are presented in parentheses and p-values are presented in brackets. We also report the mean and standard deviations for the failed applicant neighborhoods.

Source: Authors' calculations based on linked survey and administrative data.

Table A.5: Effect on Housing Prices and Home Ownership in Surrounding Neighborhoods

	Log Rent			Log Home Value			Log Monthly Mortgage			Owner	Renter
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Revitalization	0.031 (0.032) [0.320]	0.053 (0.025) [0.037]	0.034 (0.021) [0.095]	0.141 (0.048) [0.004]	0.143 (0.044) [0.001]	0.154 (0.045) [0.001]	0.108 (0.038) [0.005]	0.125 (0.032) [0.000]	0.135 (0.033) [0.000]	-0.033 (0.022) [0.135]	0.033 (0.022) [0.135]
Housing Characteristics		X	X		X	X		X	X		
Exclude New Housing			X			X			X		
Mean	6.810	6.810	6.890	12.000	12.000	12.000	6.890	6.890	6.850	0.525	0.475
Standard deviation	0.032	0.025	0.021	0.048	0.044	0.045	0.038	0.032	0.033	0.022	0.022
Observations	157,000	157,000	73,000	217,000	212,000	157,000	125,000	125,000	86,000	2,000	2,000

Notes: Each column presents estimates from a separate regression in which an outcome variable is regressed on an indicator for the Revitalization award. Regressions are estimated via weighted-least squares using the inverse propensity score. All regressions control for the year of first award and the calendar year as well as the vacancy rate, median home value, neighborhood poverty rate, and share of population that is Black in the year before the award. When indicated some regressions further control for housing characteristics (building type, year built, persons per room) and limit to house units built before the award. In columns 1-9 the unit of observation is the housing unit. In columns 10 and 11 the data are collapsed to the project level. Standard errors are clustered at the neighborhood and are presented in parentheses and p-values are presented in brackets. We also report the mean and standard deviations for the failed applicant neighborhoods.

Source: Authors' calculations based on linked survey and administrative data.

Table A.6: Effect on Exposure to Poverty for New Subsidized Renters, Additional Covariates

	Poverty Rate In Prior Neighborhood	Poverty Rate In Current Neighborhood			
	(1)	(2)	(3)	(4)	(5)
A. Moved In After Award					
Revitalization	-0.009 (0.007) [0.203]	-0.072 (0.015) [0.000]	-0.069 (0.014) [0.000]	-0.068 (0.014) [0.000]	-0.068 (0.014) [0.000]
R-squared	0.016	0.089	0.154	0.159	0.181
Observations (thousands)	1,082	1,082	1,082	1,082	1,082
B. Moved In Before Award					
Revitalization	-0.002 (0.010) [0.874]	-0.003 (0.019) [0.888]	-0.001 (0.015) [0.935]	-0.002 (0.016) [0.894]	-0.001 (0.015) [0.931]
R-squared	0.102	0.187	0.364	0.347	0.377
Observations (thousands)	1,362	1,362	1,256	1,075	1,256

Notes: The sample in panels A and B includes the set of individuals who move into a Revitalization or failed applicant neighborhood after or before the award, respectively. Each column presents an estimate from a separate regression of an outcome variable on the Revitalization indicator, controlling for the interaction between the award year and years since the award. The outcome variable in column 1 is the poverty rate of the neighborhood that the individual lived in prior to moving into the target neighborhood. The outcome variable in columns 1-5 is the poverty rate of target neighborhood. Columns 3-5 add additional covariates. Column 3 adds controls for the poverty rate in the prior neighborhood, own income in the year before moving into the target neighborhood, and a quadratic in age. Column 4 adds a control for income in the current year. Column 5 adds controls for a vector of characteristics from the HUD administrative data, including ethnicity, race, sex, disability status, household head disability status, household size, type of subsidized housing, a flag for hardship, hard to house, separates measures for the amount of income from various sources (welfare, wages, pension, other, and unknown).

Table A.7: Algorithm for Constructing the Spillovers Estimator

Step	Description	Formal Notation
1	Estimate the size of the displaced population.	Let Γ_0 denote the size of the displaced population.
2	Estimate the maximum size of the population of poor individuals displaced into each neighborhood.	Let $\Gamma(n)$ denote the maximum size of the population of poor individuals displaced into neighborhood n .
3	Rank destination neighborhoods by the number of subsidized renters.	Let $s(n) = \sum_i \mathbb{1}\{N_{it''} = n, S_{it''} = 1\}$ be the number of subsidized renters in n and let $r(\cdot)$ be a ranking function that ranks neighborhoods by $s(n)$ in descending order such that $r(j) < r(k)$ if and only if $s(j) > s(k)$ for all neighborhoods j and k .
4	Calculate a counterfactual poverty rate in the destination neighborhood with the most subsidized renters by removing the displaced poor individuals from this neighborhood and assuming that they are replaced by non-poor individuals. If the size of the displaced population exceeds the number of people who moved to this neighborhood, then remove the remaining displaced households from the second-ranked neighborhood, and so on.	Iterate sequentially for $k = \{1, 2, 3, \dots\}$. Select neighborhood j such that $r(j) = k$ and define $d_j = \min\{\Gamma(j), \tilde{\Gamma}_k\}$, where $\tilde{\Gamma}_k = \Gamma_{k-1}$. Define $\delta_j = [\sum_i \mathbb{1}\{N_{it''} = j, P_{it''} = 1\} + d_j] / \sum_i \mathbb{1}\{N_{it''} = j\}$. Stop iterating when $d_j = 0$.
5	Summarize the spillover effects from the displaced original residents by taking the weighted-average of the increase in neighborhood poverty rates among the set of destination neighborhoods that had a non-zero change in poverty, weighting by the size of the subsidized population in the destination neighborhood.	Let $\mathcal{J} = \{j : d_j > 0\}$ denote the set of neighborhoods that had a non-zero change in poverty rate and let $\pi(j) = \sum_i \mathbb{1}\{N_{it''} = j, S_{it''} = 1\} / \sum_i \mathbb{1}\{N_{it''} \in \mathcal{J}, S_{it''} = 1\}$. Then we summarize the spillover effects as $\sum_{j \in \mathcal{J}} \pi(j) \delta_j$.

Notes: This table provides a step-by-step guide to the construction of the spillovers estimator, corresponding to the discussion in 5.2.4. We implement this procedure separately for each target neighborhood but do not index by target neighborhood to simplify exposition.

Table A.8: Analysis of Public Housing Authorities

Change in Outcome from 1997 to 2018						
	Census Tract Poverty Rate	Poverty Rate in Revitalized Sites	Share Subsidized Units in Public Housing	Log Subsidized Units	Share Subsidized Renters Very Low-Income	County Poverty Rate
	(1)	(2)	(3)	(4)	(5)	(6)
Revitalization	-0.003 (0.021) [0.900]	0.006 (0.010) [0.541]	0.035 (0.054) [0.516]	-0.020 (0.087) [0.816]	0.015 (0.039) [0.701]	-0.005 (0.010) [0.613]
Units Targeted	0.158 (0.077) [0.043]	0.250 (0.037) [0.000]	0.280 (0.197) [0.157]	-0.505 (0.318) [0.114]	0.002 (0.144) [0.987]	-0.014 (0.036) [0.703]
Revitalization $\times$ Units Targeted	-0.168 (0.074) [0.024]	-0.101 (0.035) [0.004]	-0.790 (0.184) [0.000]	0.360 (0.297) [0.227]	-0.194 (0.140) [0.167]	0.048 (0.033) [0.156]
Baseline Poverty	-0.193 (0.058) [0.001]	0.007 (0.027) [0.789]	-0.013 (0.146) [0.932]	-0.294 (0.237) [0.215]	-0.072 (0.107) [0.504]	-0.059 (0.027) [0.029]
Baseline Poverty $\times$ Units Targeted	-0.594 (0.189) [0.002]	-0.604 (0.090) [0.000]	-0.456 (0.476) [0.339]	0.919 (0.770) [0.234]	-0.016 (0.365) [0.965]	-0.044 (0.086) [0.610]
Observations	193	193	195	195	143	195

Notes: Each column presents estimates from a separate regression in which an outcome is regressed on an indicator for receiving Revitalization funding, the average poverty rate of targeted neighborhoods in 1990, the share of subsidized units targeted by HOPE VI, and the interaction between the share of subsidized units and the other two variables. The outcome variables in columns 1-6 include the change between 1997 and 2018 in the: average census tract poverty rate of subsidized households, the component of column 1 attributable to changes in poverty rates in the revitalized neighborhoods, share of subsidized units in public housing, total number of subsidized units, the share of subsidized renters whose household income is below 10,000 dollars, and county-level poverty rates. Each observation represents a PHA and the sample includes Revitalization PHAs (any public housing project received a Revitalization grant) and Failed Applicant PHAs (any public housing project applied for Revitalization funding but no project received HOPE VI funding). Each regression is estimated via weighted least squares, using the total number of subsidized units targeted by HOPE VI as the weights. Standard errors are presented in parentheses and p-values are presented in brackets.

Source: Authors' calculations based the 1997 and 2018 vintages of HUD User's Picture of Subsidized Households.

Table A.9: Analysis of Public Housing Authorities, All Households

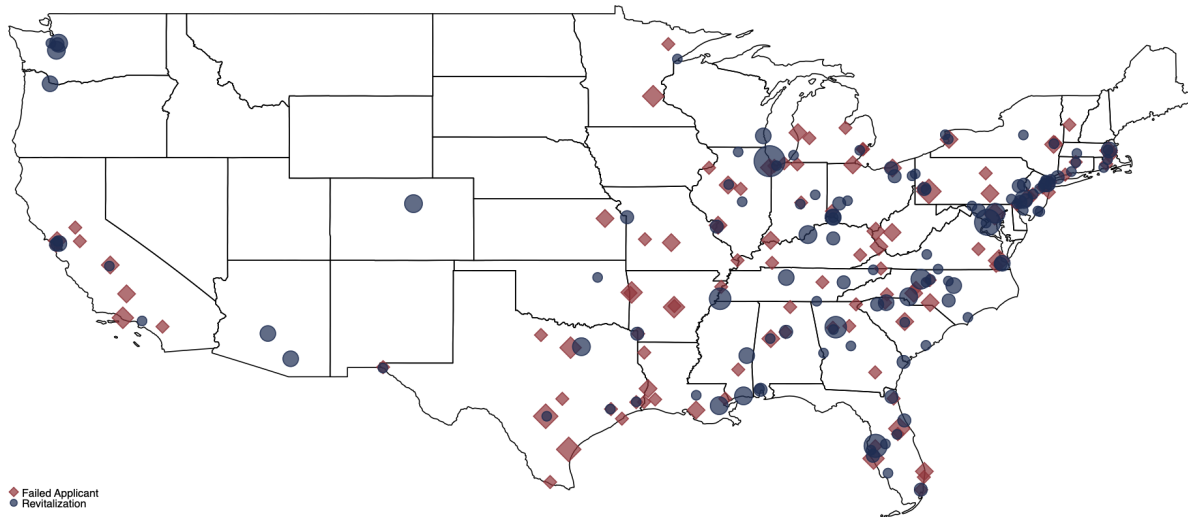
Change in Average Census Tract Poverty Rate Between 1990 and 2017, for Households in Specified Income Category								
	0-10k (1)	10k-20k (2)	20k-30k (3)	30k-40k (4)	40k-50k (5)	50k-75k (6)	75k-100k (7)	100k+ (8)
Revitalization	-0.003 (0.011) [0.820]	0.001 (0.009) [0.937]	0.001 (0.007) [0.844]	-0.002 (0.007) [0.823]	0.001 (0.006) [0.900]	0.002 (0.006) [0.728]	0.002 (0.005) [0.726]	0.003 (0.005) [0.520]
Units Targeted	0.039 (0.053) [0.457]	-0.003 (0.050) [0.952]	-0.033 (0.043) [0.442]	-0.033 (0.042) [0.439]	-0.035 (0.038) [0.352]	-0.015 (0.036) [0.684]	-0.008 (0.032) [0.797]	0.022 (0.026) [0.406]
Revitalization $\times$ Units Targeted	0.024 (0.036) [0.497]	0.028 (0.028) [0.317]	0.020 (0.022) [0.368]	0.022 (0.021) [0.286]	0.016 (0.019) [0.404]	0.010 (0.018) [0.574]	0.011 (0.017) [0.529]	-0.008 (0.016) [0.619]
Observations	195	195	195	195	195	195	195	195

Notes: Each column presents estimates from a separate regression. The outcome variable is the change in the average census tract poverty rate between 1990 and 2017 for households who live in the county and have a household income in the range defined by the column header. In each regression, the this outcome is regressed on an indicator for receiving Revitalization funding, the average census tract poverty rate for the population defined by the column header in 1990, the share of subsidized units targeted by HOPE VI, and the interaction between the share of subsidized units and the other two variables. Each observation represents a PHA and the sample includes Revitalization PHAs (any public housing project received a Revitalization grant) and Failed Applicant PHAs (any public housing project applied for Revitalization funding but no project received HOPE VI funding). Each regression is estimated via weighted least squares, using the total number of subsidized units targeted by HOPE VI as the weights. Standard errors are presented in parentheses and p-values are presented in brackets.

Source: Authors' calculations based the 1997 and 2018 vintages of HUD User's Picture of Subsidized Households.

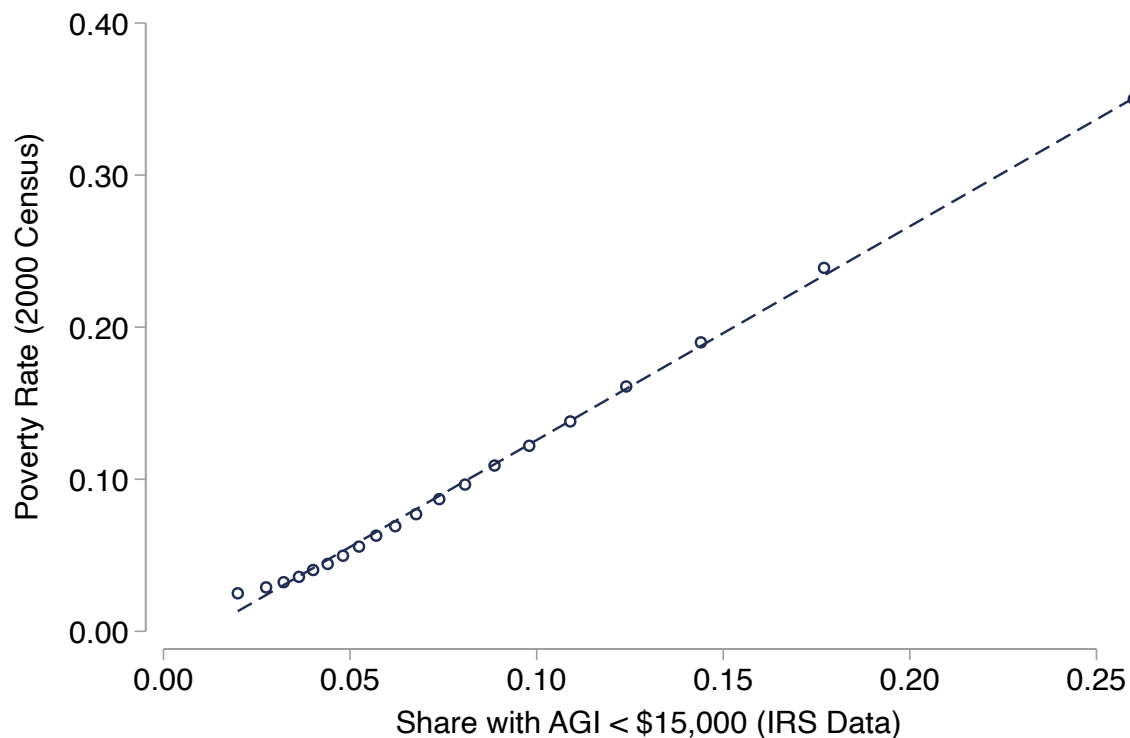
Appendix B Additional Figures

Figure B.1: Geographic Distribution of Applicants and Awardees



Notes: The blue circles and red diamonds represent the location of HOPE VI Revitalization and failed applicant sites, respectively. The size of the markers is proportional to the number of units in the project. Source: Authors' calculations based on project-level summary files from HUD.

Figure B.2: Alignment of IRS and Census Income Measures Across Neighborhoods

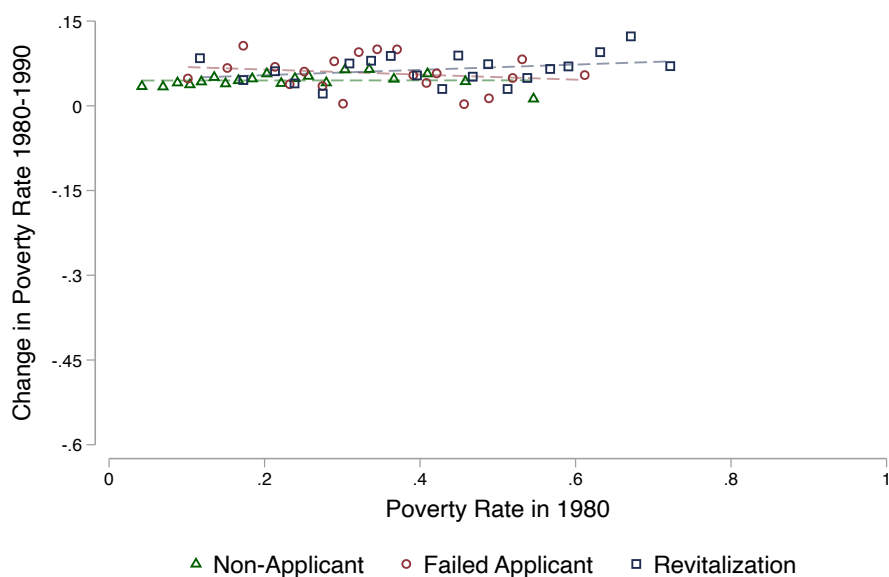


Notes: We measure household poverty using responses from the 2000 Decennial Census and identify whether individuals had adjusted gross income (AGI) below \$15,000 using IRS data. We compute the average of each measure at the census block group level based on residents living there in 2000. Block groups are grouped into ventiles based on the share of residents with AGI below \$15,000, and we plot the average poverty rate and AGI-based measure by ventile.

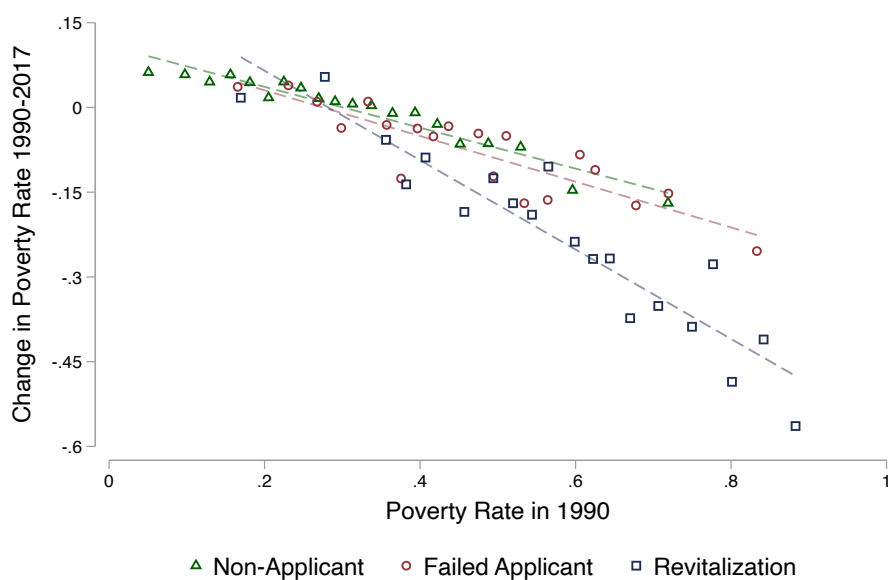
Source: Authors' calculations based on linked administrative data.

Figure B.3: Changes in Neighborhood Poverty Rates Near Public Housing Projects

(A) Before Revitalization Program



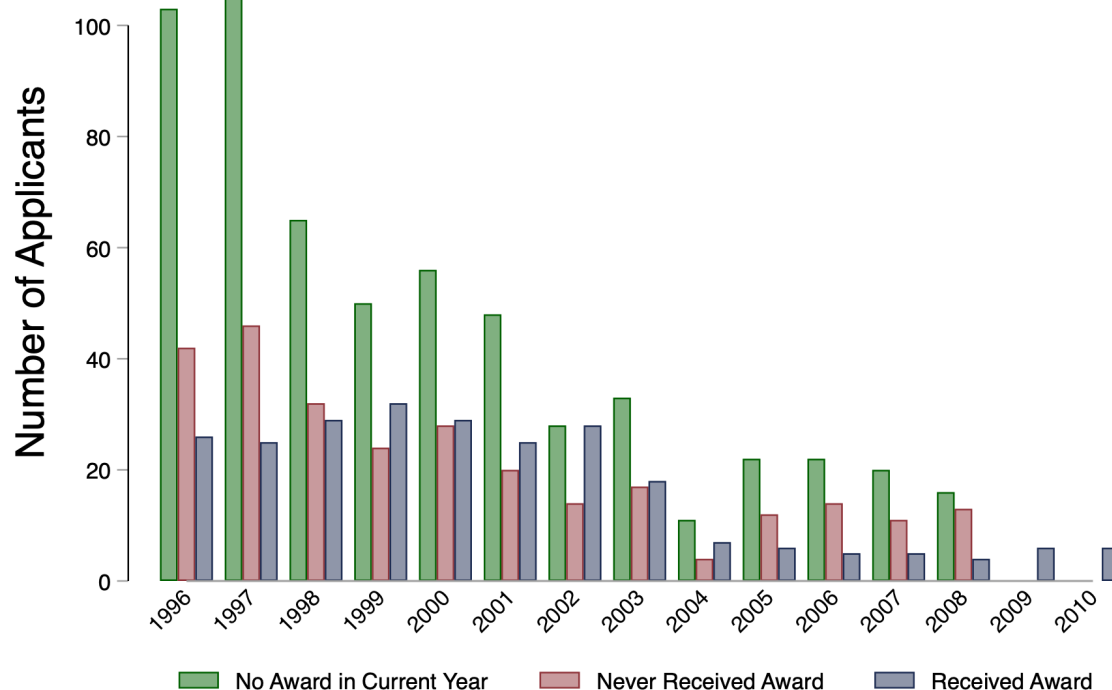
(B) After Revitalization Program



Notes: Each figure plots the change in neighborhood poverty rates against the baseline poverty rate. Within the non-applicant, failed applicant, and Revitalization samples, projects are grouped into population-weighted ventiles based on the baseline poverty rate. Each point represents a population-weighted average of observations in a given ventile. In Panel A the baseline year is 1980 and the change is measured between 1980 and 1990. In Panel B the baseline year is 1990 and the change is measured between 1990 and 2017.

Source: Authors' calculations based on summary files of Decennial Census survey and the ACS.

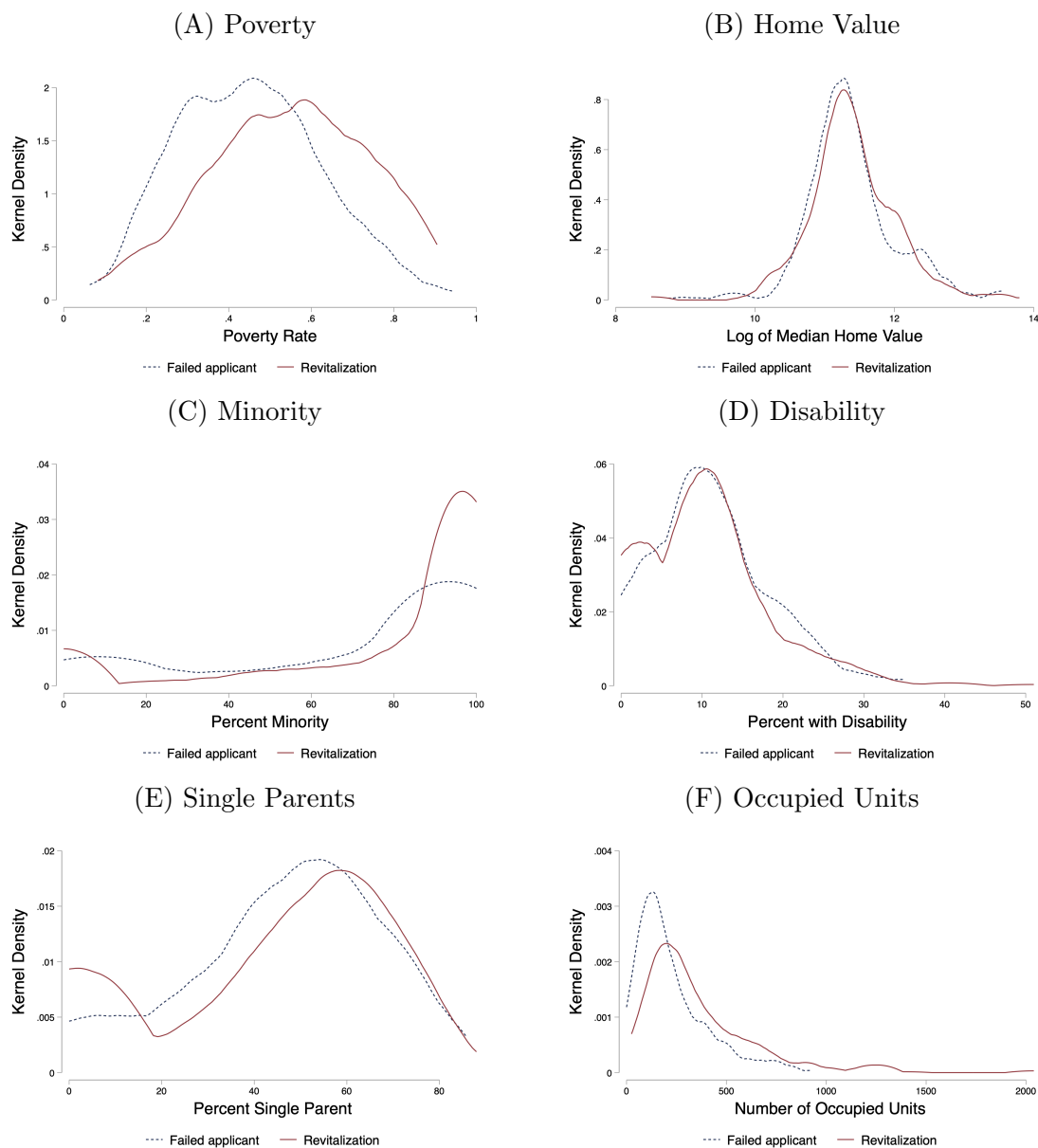
Figure B.4: Number of Awardees and Failed Applicants by Year



Notes: The figure presents the number of public housing projects per grant year for projects that applied for and: (1) did not receive funding in that year, (2) did not receive funding in any year, and (3) received a HOPE VI Revitalization grant in that year. If a project received more than one HOPE VI grant, then we report the earliest year. Projects not awarded a grant in one year could apply in a subsequent year. We do not observe application data for 2009 and 2010.

Source: Authors' calculations based on the publicly available list of awardees of and applicants to HOPE VI funding.

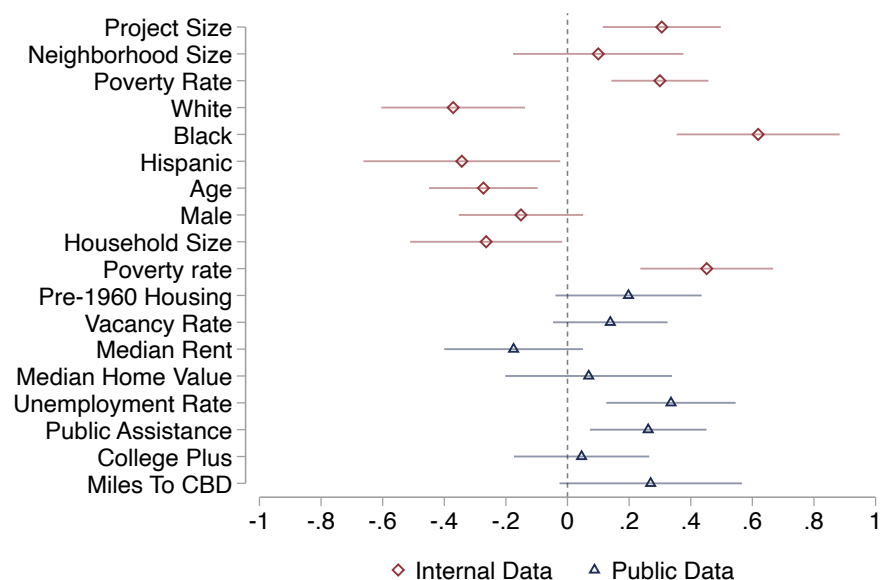
Figure B.5: Distribution of Baseline Characteristics



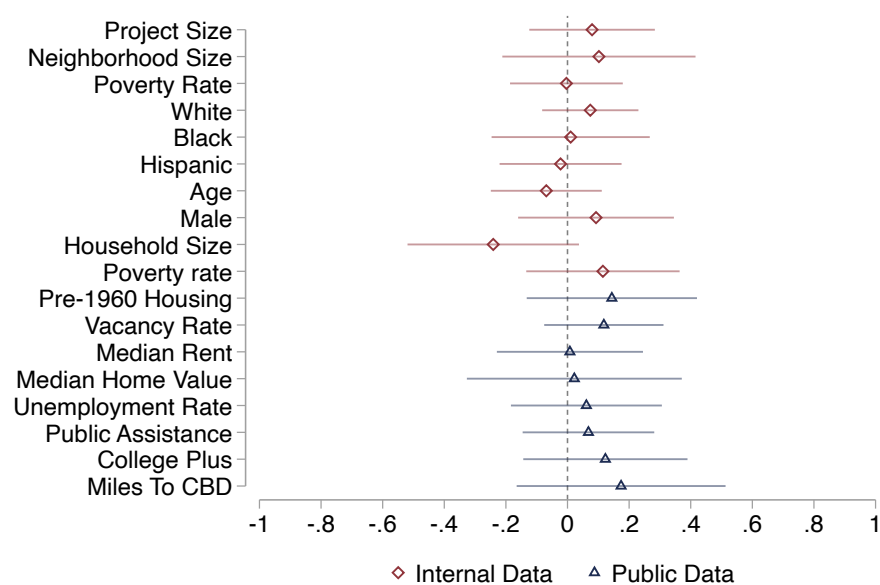
Notes: Each figure presents a kernel density plots of characteristics measured prior to the award. Panels A and B summarize characteristics of the block group and are based on data from the 1990 Decennial Census. Panels C-F summarize characteristics of the projects as measured in the 1993 vintage of HUD User's Picture of Subsidized Households. Distributions are presented separately by treatment status. Source: Authors' calculations based on summary files of 1990 Decennial Census survey and project-level summary files from HUD.

Figure B.6: Balance of Baseline Characteristics

(A) Population Weights



(B) Inverse Propensity Score Weights

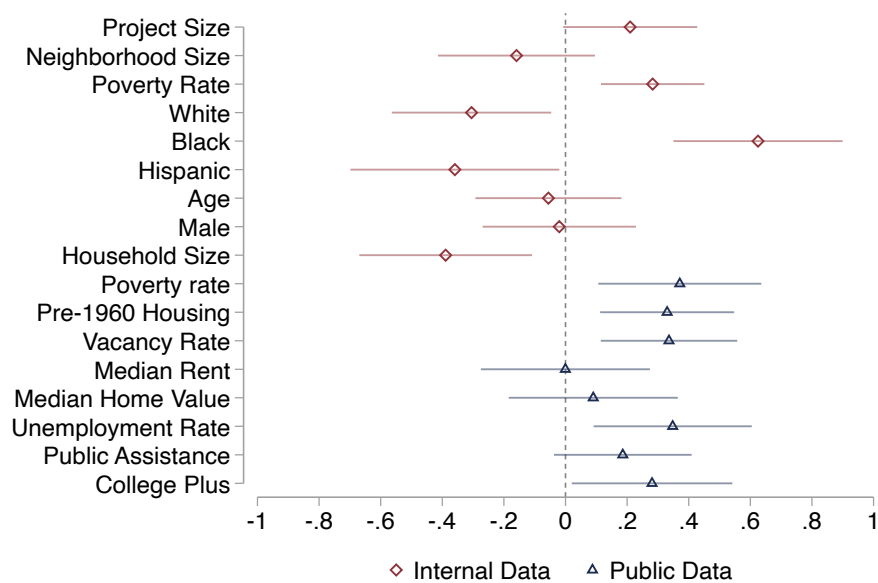


Notes: Each point plots the coefficient from a separate regression of a pre-award neighborhood characteristic (standardized by mean and standard deviation) on an indicator for whether the neighborhood received a revitalization award. The sample includes Revitalization and failed applicant neighborhoods. Panel A weights by neighborhood population; Panel B uses inverse propensity score weights. Red diamonds represent estimates using internal data; blue triangles use publicly available data. Standard errors are clustered at the neighborhood level, and 95 percent confidence intervals are shown with horizontal bars.

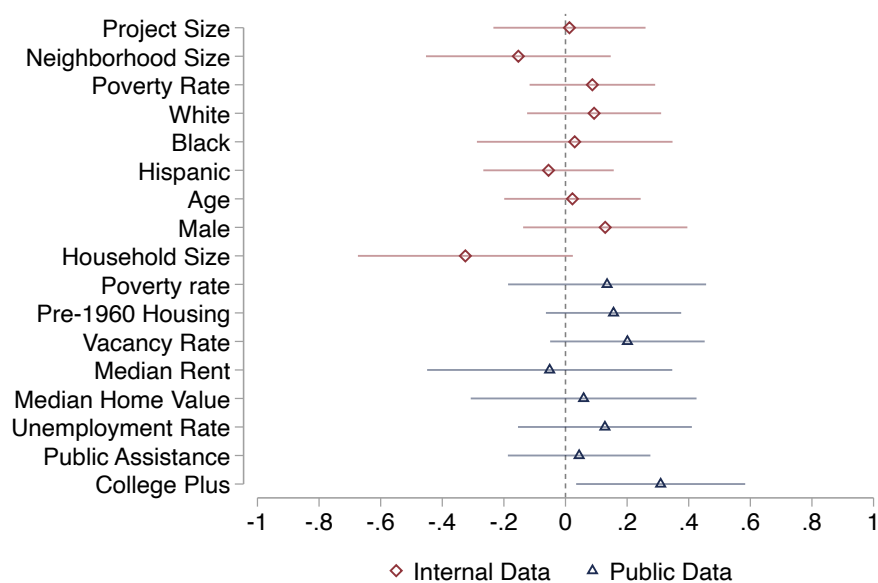
Source: Authors' calculations based on linked administrative data.

Figure B.7: Balance of Baseline Characteristics in Surrounding Neighborhoods

(A) Population Weights



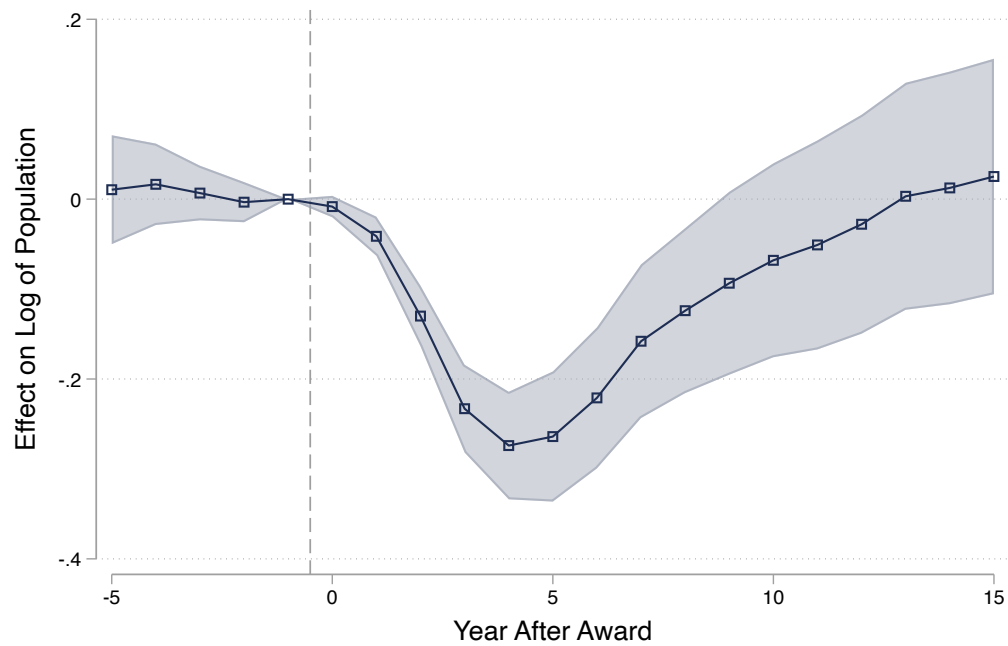
(B) Inverse Propensity Score Weights



Notes: Each point plots the coefficient from a separate regression of a pre-award neighborhood characteristic (standardized by mean and standard deviation) on an indicator for whether the neighborhood received a revitalization award. The sample includes the contiguous Census block groups to the Revitalization and failed applicant neighborhoods. Panel A weights by neighborhood population; Panel B uses inverse propensity score weights. Red diamonds represent estimates using internal data; blue triangles use publicly available data. Standard errors are clustered at the neighborhood level, and 95 percent confidence intervals are shown with horizontal bars.

Source: Authors' calculations based on linked administrative data.

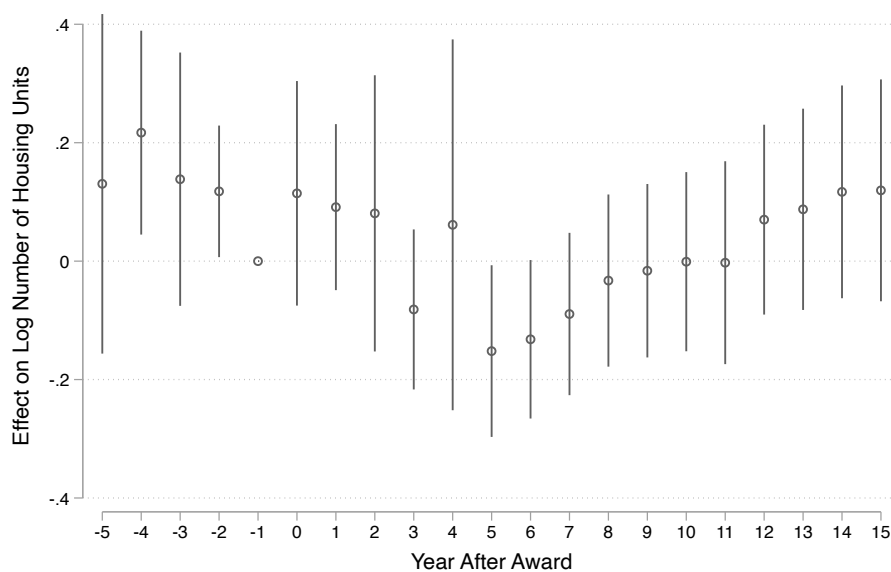
Figure B.8: Effect on Population of Neighborhood



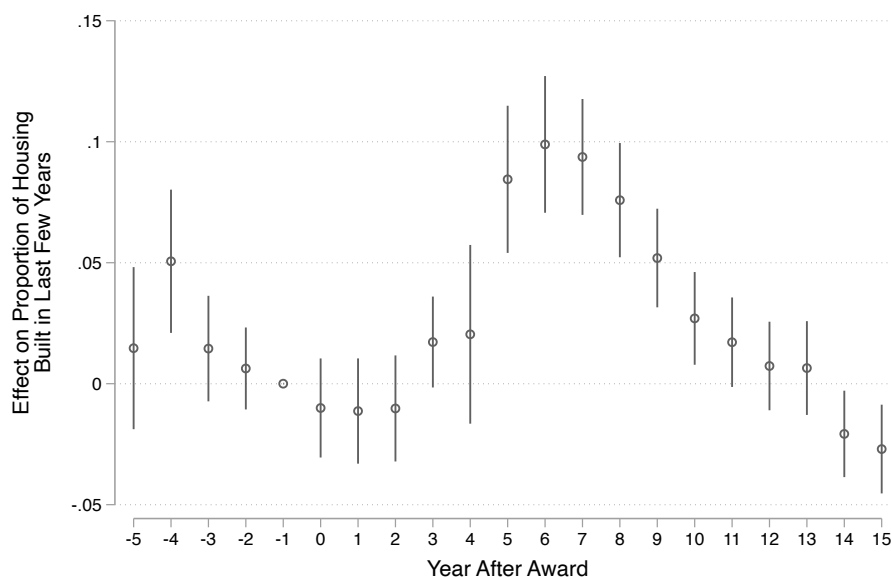
Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome variable is the log of the total number of people living in the neighborhood. The shaded regions denote the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure B.9: Effect on Housing Stock

(A) Total Housing



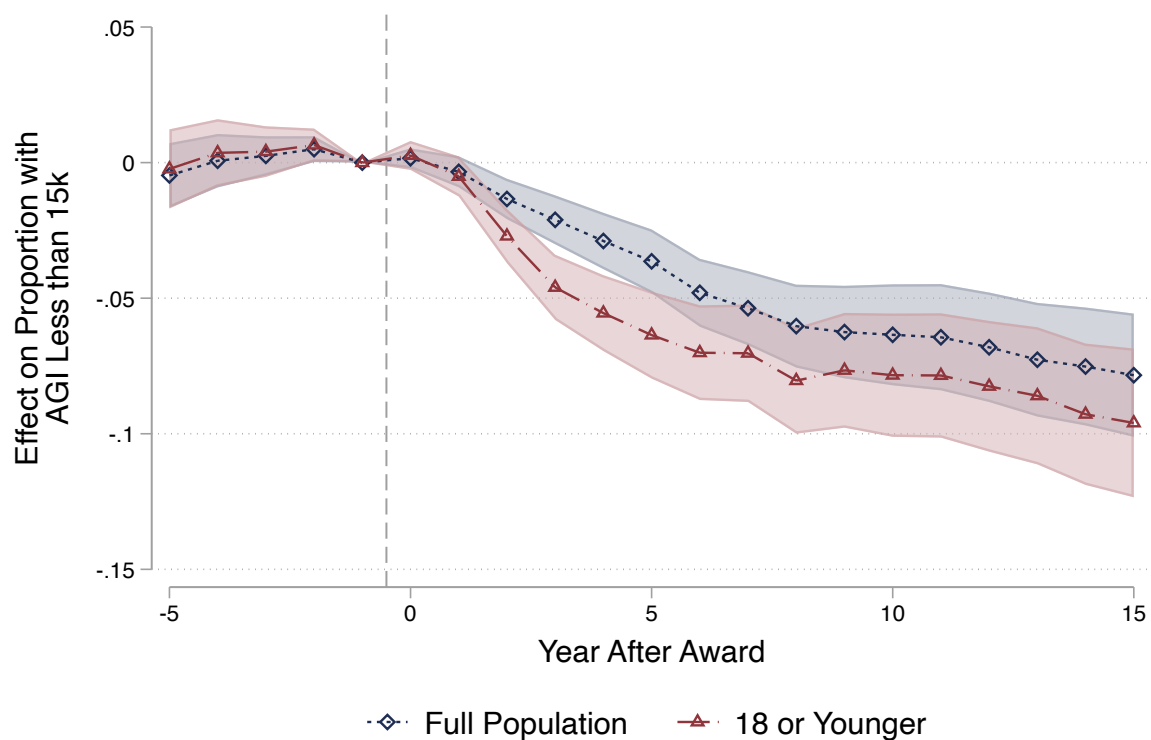
(B) New Housing



Notes: This figures plots estimates from the stacked difference-in-differences specification described in equation 10. The outcomes are the total housing stock and the proportion of housing built within the past few years. The vertical bars denote the 95 percent confidence intervals.

Source: Authors' calculations based on summary files of Decennial Census surveys and the ACS.

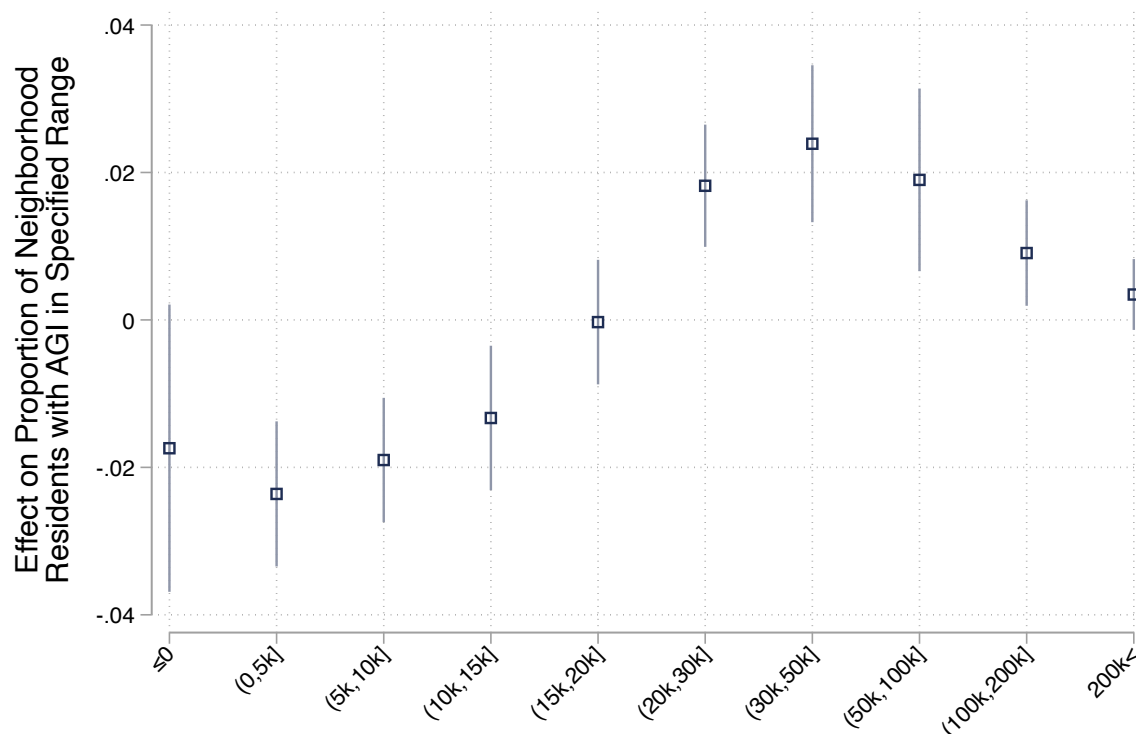
Figure B.10: Effect on Neighborhood Poverty, Families with Children



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the poverty rate for all individuals in the neighborhood (diamond markers) or the poverty rate for those 18 and younger (triangle markers). The shaded regions denote the 95 percent confidence intervals.

Source: Authors' calculations based on linked administrative data.

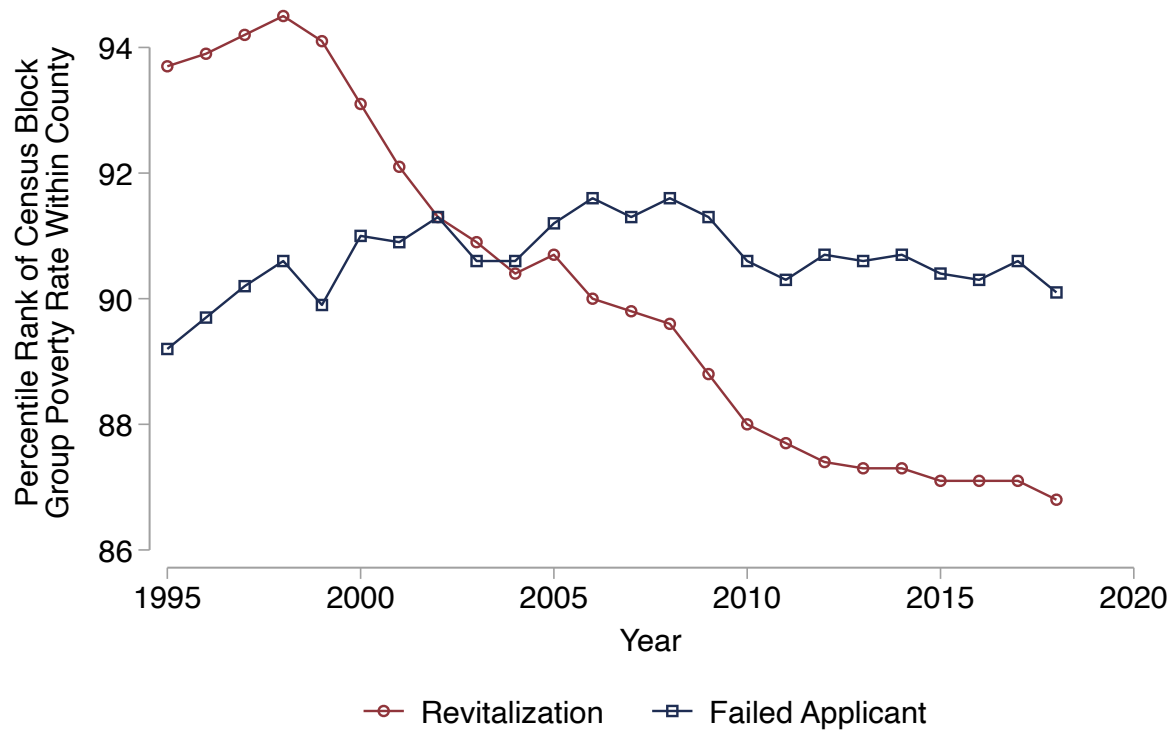
Figure B.11: Effect on Neighborhood Income Distribution



Notes: This figure plots estimates from the equation 11. Each point represents an estimate from a separate regression in which the outcome variable is the proportion of individuals living in the neighborhood 10 to 15 years after the award whose AGI falls within the range defined by the horizontal axis. The vertical bars denote the 95 percent confidence intervals.

Source: Authors' calculations based on linked administrative data.

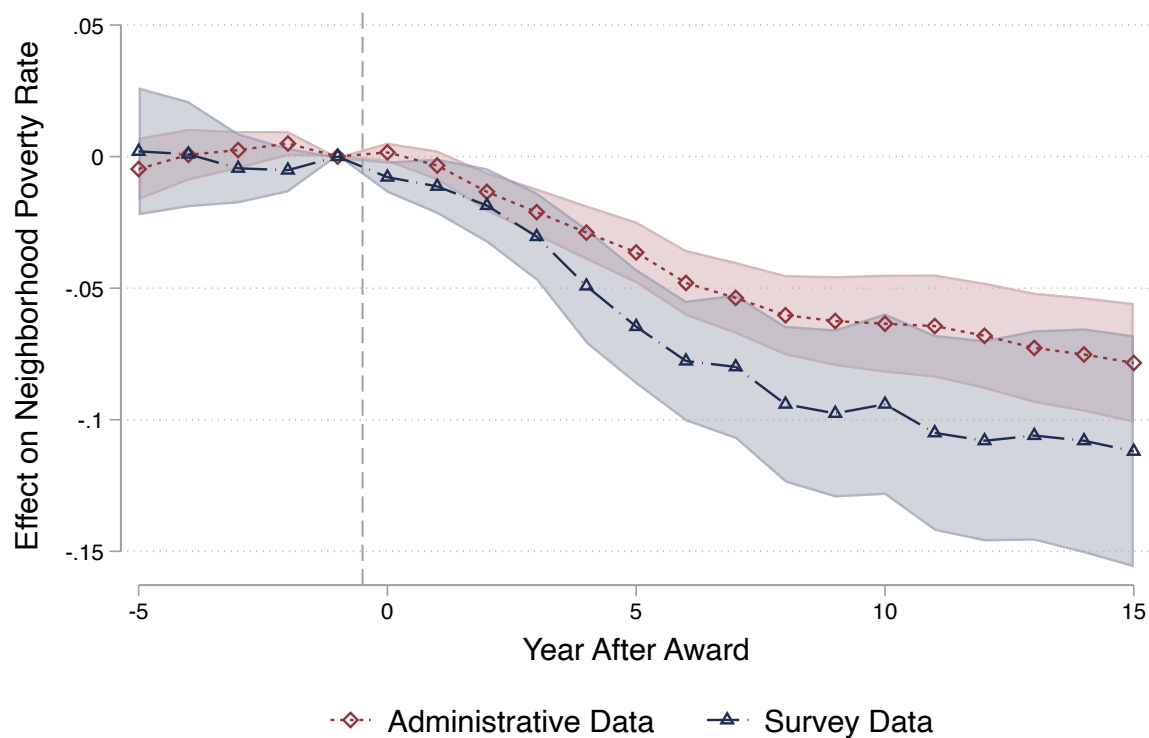
Figure B.12: Average Within-County Rank of Poverty Rate by Year and Treatment Status



Notes: For each census block group and year, we compute its percentile rank relative to all other block groups in the same county and year. Red circles show the average percentile rank for Revitalization neighborhoods; blue circles show the corresponding average for failed applicant neighborhoods.

Source: Authors' calculations based on linked administrative data.

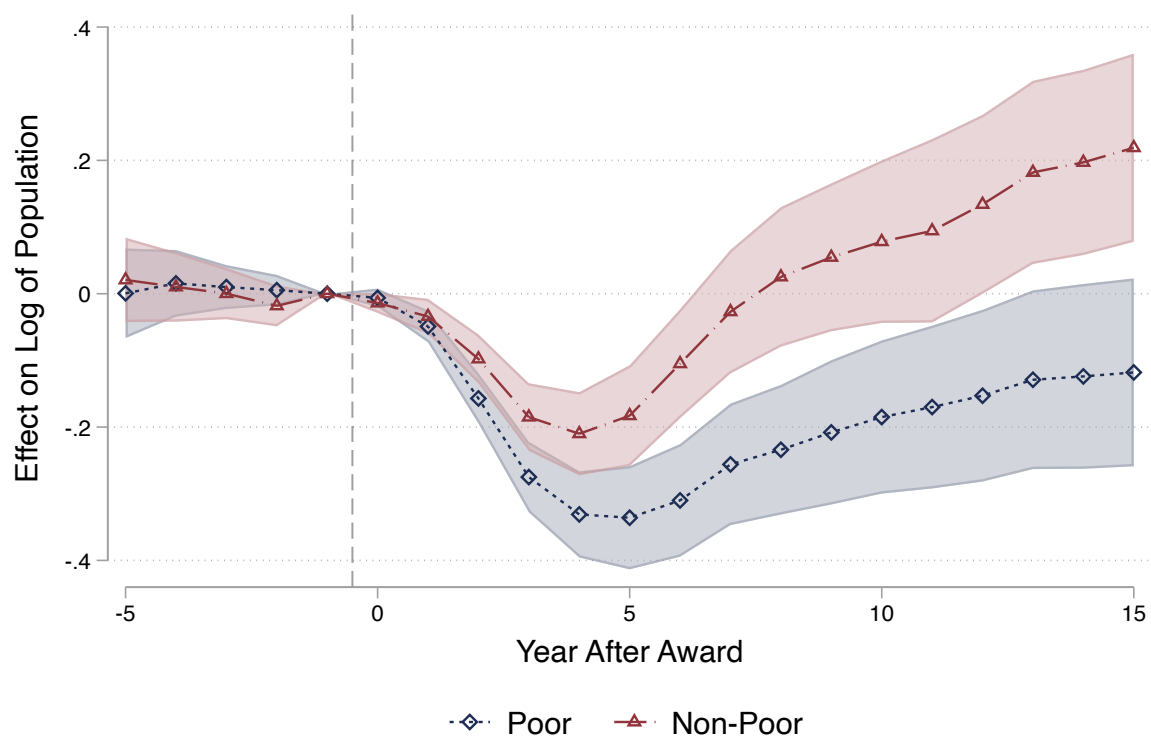
Figure B.13: Effect on Alternative Measures of Poverty Rates



Notes: This figure plots estimates from the specification described in equation 10. The outcome variable is either the share of the population with an AGI below \$15,000 (based on the administrative data) or the share of the population whose income falls below the poverty line (based on survey data). The shaded region denotes the 95 percent confidence intervals.

Source: Authors' calculations based on linked administrative data.

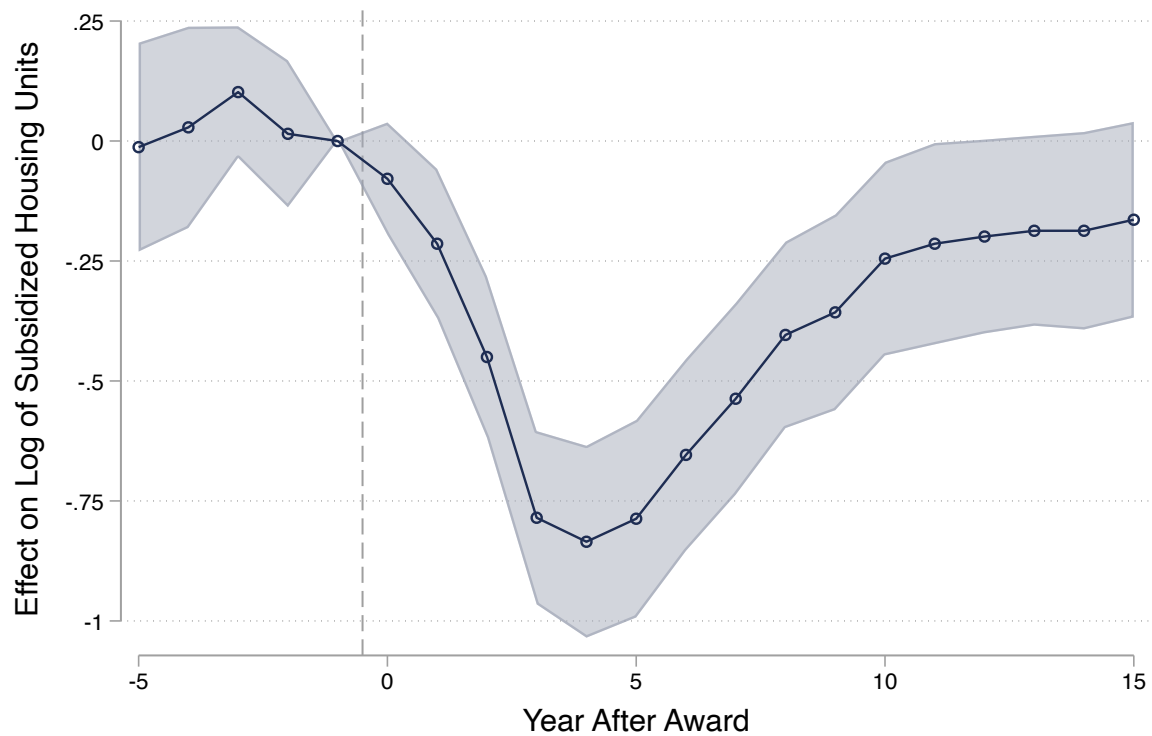
Figure B.14: Effect on Population by Poverty Status



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the log number of poor or non-poor individuals in the neighborhood. The shaded regions denote the 95 percent confidence intervals.

Source: Authors' calculations based on linked administrative data.

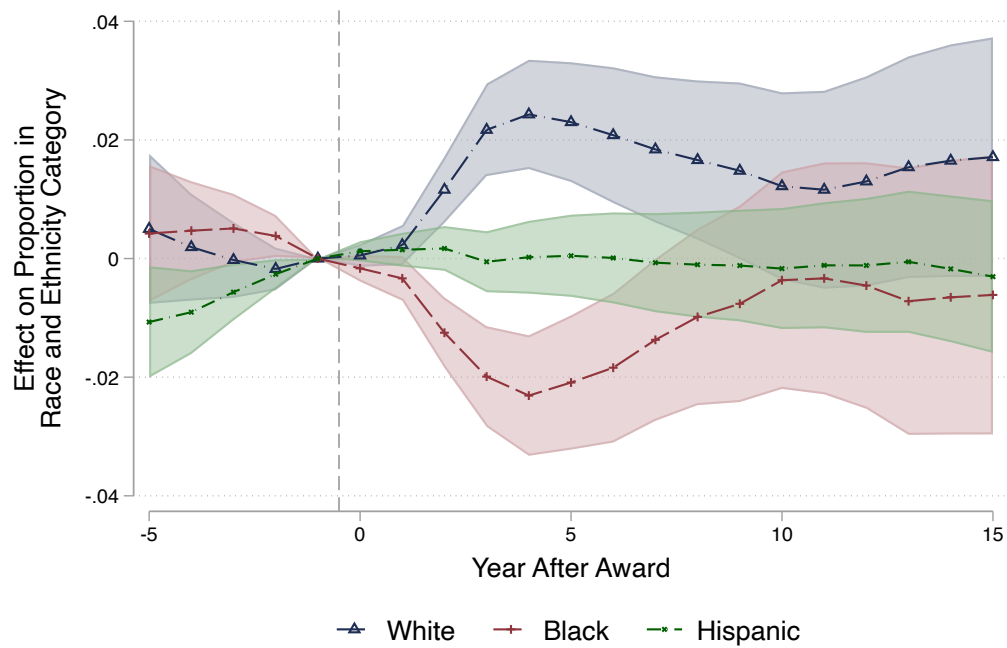
Figure B.15: Effect on Number of Subsidized Housing Units



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the log of the total number of subsidized housing units in the neighborhood. The vertical bars denote the 95 percent confidence intervals.

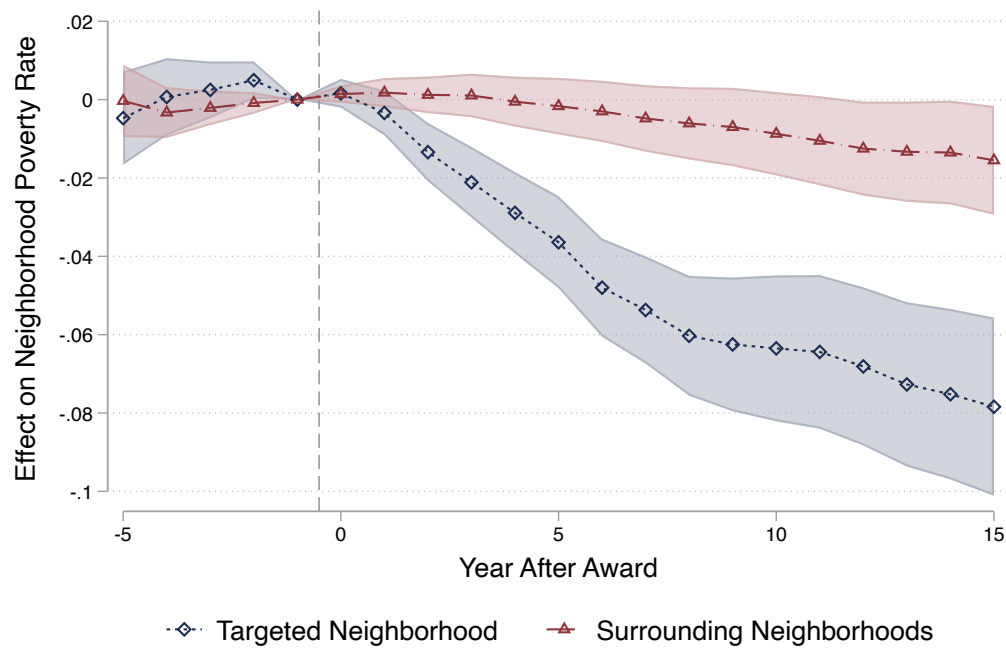
Source: Authors' calculations based on linked administrative data.

Figure B.16: Effect on Racial and Ethnic Composition of Neighborhood



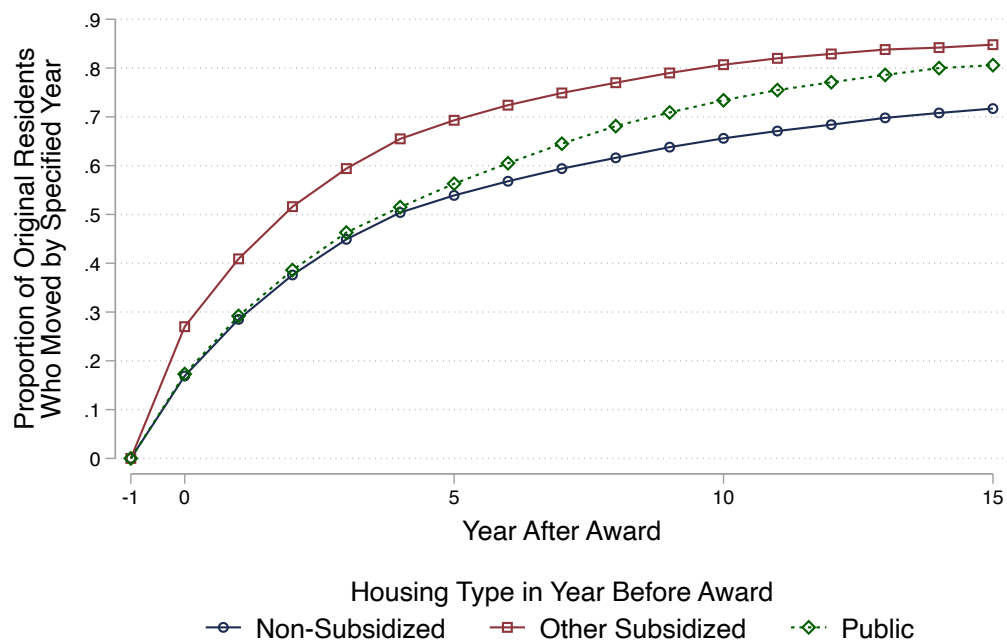
Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the proportion of the population who is White non-Hispanic, Black non-Hispanic, or Hispanic. The shaded regions denote the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

Figure B.17: Effect on Poverty Rates in Surrounding Neighborhoods



Notes: This figure plots estimates from the specification described in equation 10. The outcome variable is the poverty rate in the target neighborhoods or the poverty rate in the contiguous Census block groups. The shaded region denotes the 95 percent confidence intervals.
Source: Authors' calculations based on linked administrative data.

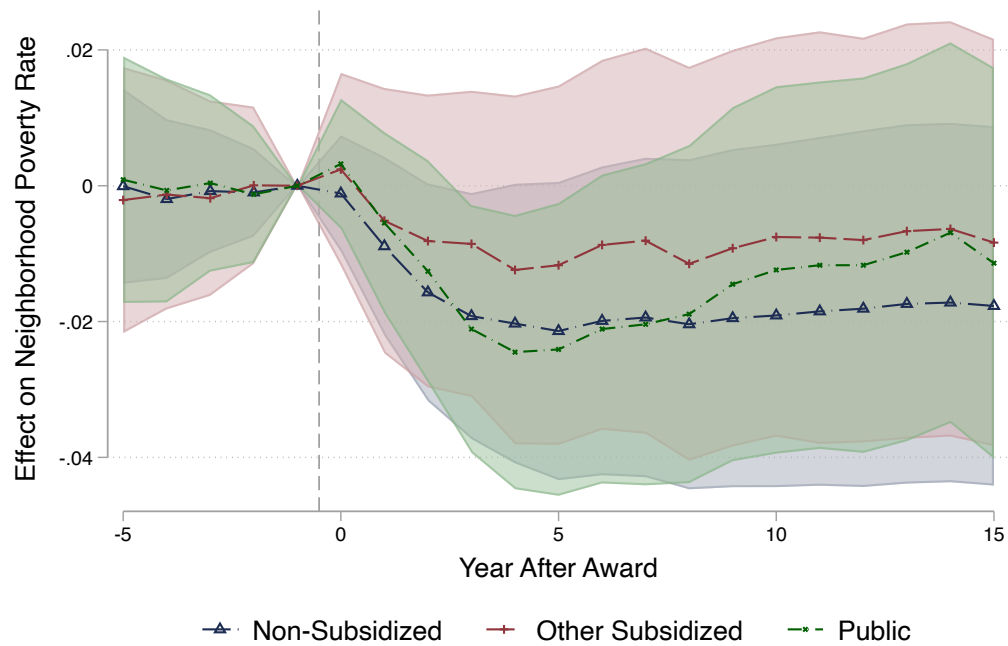
Figure B.18: Residential Mobility of the Original Residents, Failed Applicants



Notes: This figures plots the proportion of the original residents of the failed applicant neighborhoods who moved to a different Census block group by a given year after the award.

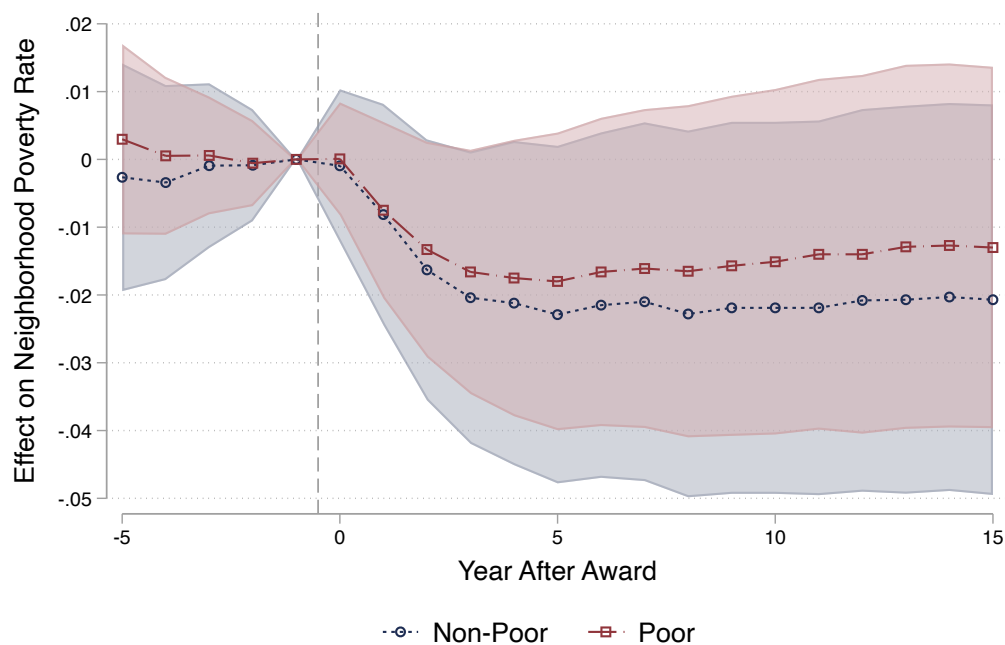
Source: Authors' calculations based on linked administrative data.

Figure B.19: Effect on Exposure to Poverty for Original Residents, By Subsidy Status



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the poverty rate of the current neighborhood, and results are estimated separately by subsidized housing status. The shaded regions denote the 95 percent confidence intervals. Source: Authors' calculations based on linked administrative data.

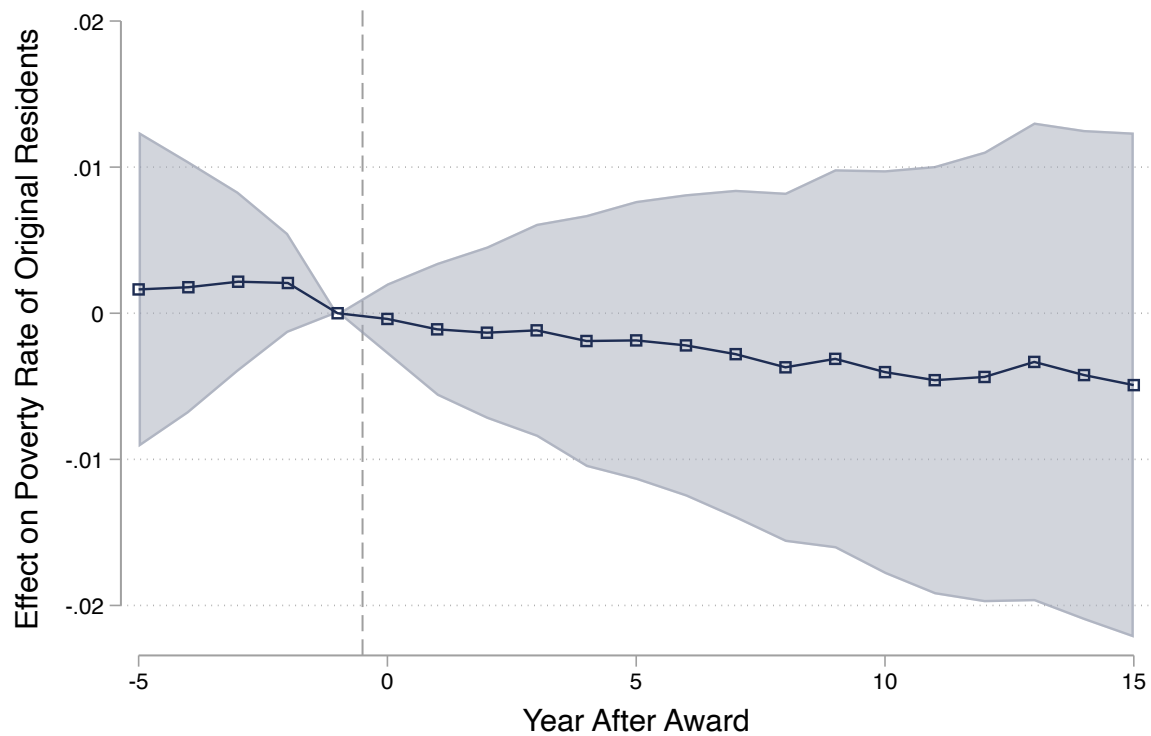
Figure B.20: Effect on Exposure to Poverty for Original Residents without Subsidized Housing, by Baseline Poverty Status



Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the poverty rate of the current neighborhood, and the same is the set of original residents without subsidized housing who are either poor or not poor in the year before the award. The shaded regions denote the 95 percent confidence intervals.

Source: Authors' calculations based on linked administrative data.

Figure B.21: Effect on Income of Original Residents

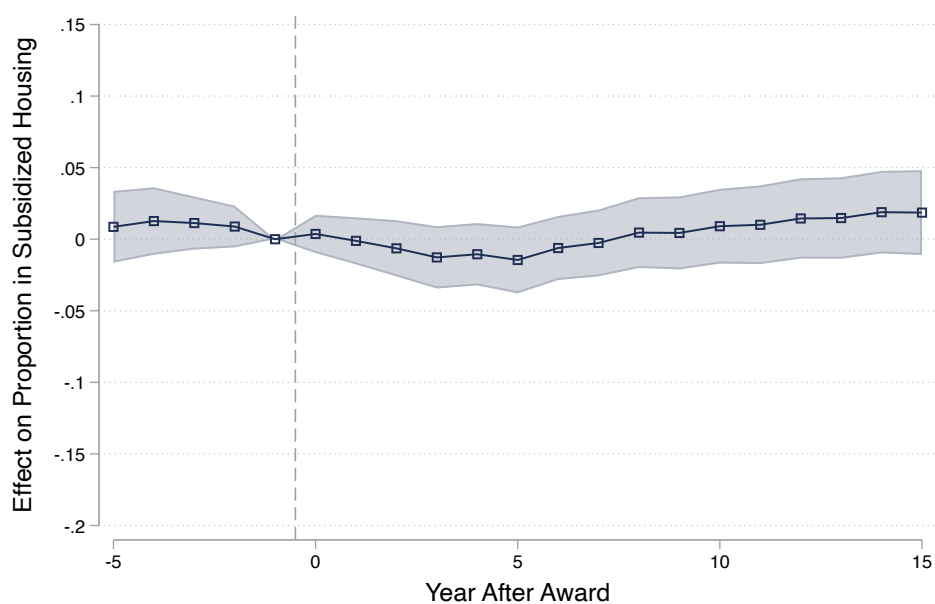


Notes: This figure plots estimates from the stacked difference-in-differences specification described in equation 10. The outcome is the proportion of the original residents who are poor. The shaded regions denote the 95 percent confidence intervals.

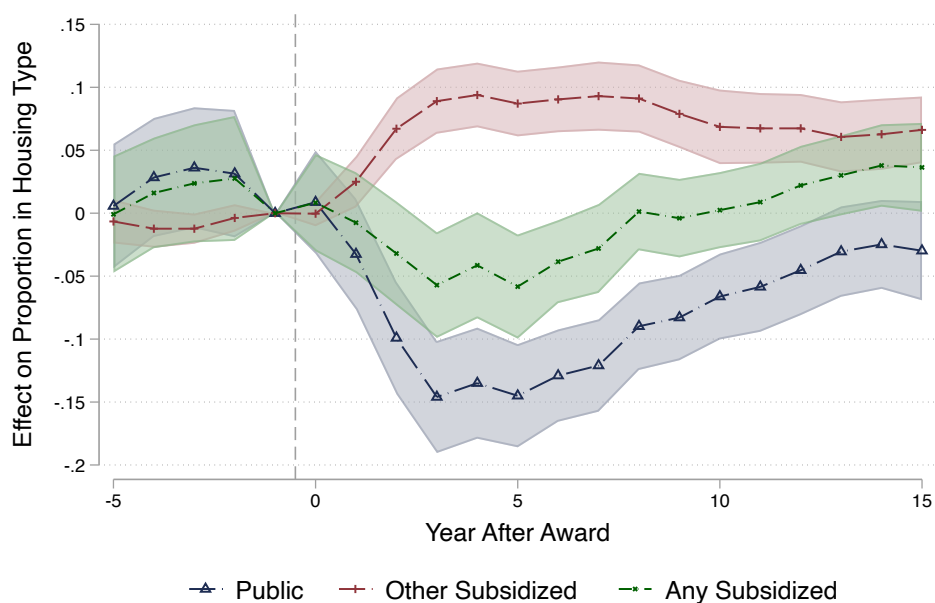
Source: Authors' calculations based on linked administrative data.

Figure B.22: Effect on Subsidy Status, Original Residents

(A) All Original Residents

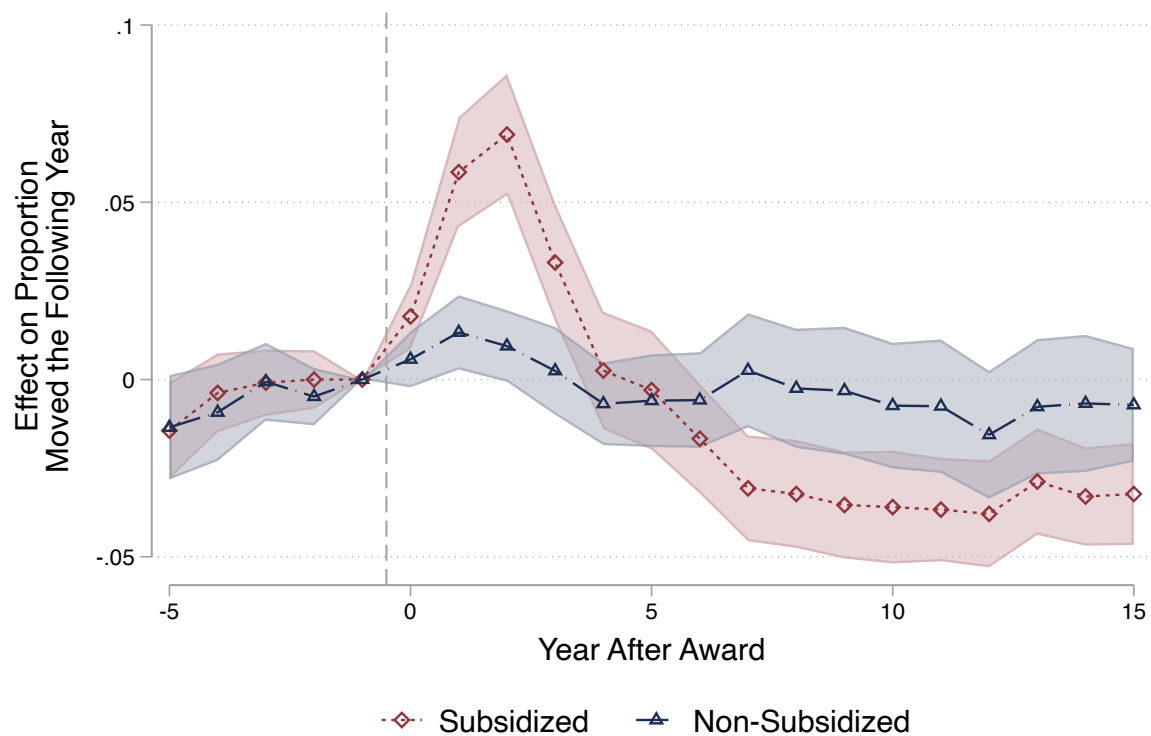


(B) Original Residents of Public Housing



Notes: This figures plots estimates from the stacked difference-in-differences specification described in equation 10. The sample in panel in includes all original residents and the outcome is the proportion of individuals in subsidized housing. The sample in Panel B includes all original residents of the public housing projects and the outcome is the proportion of the these residents who were living in public, other subsidized, or non-subsidized housing. The shaded regions denote the 95 percent confidence intervals. Source: Authors' calculations based on linked administrative data.

Figure B.23: Effect on Out-Migration



Notes: This figures plots estimates from the specification described in equation 10. The outcome variable is proportion of current residents who move to a different neighborhood in the following year. The shaded region denotes the 95 percent confidence intervals.

Source: Authors' calculations based on linked administrative data.

Appendix C Residential Sorting Model

Consider a city that consists of two neighborhoods, the central city (C) and the suburbs (B), which both have a fixed, homogeneous housing stock of equal size. There are two periods (t and $t + 1$). The population is fixed and equal to the size of the housing stock. There are two types of households: low-income (l) and high-income (h). All low-income and no high-income households have subsidized housing.⁴² The poverty rate of neighborhood j is P_{jt} (i.e., the share of households that are low-income) and it is defined in equilibrium, while the poverty rate of the city is fixed at P .⁴³

The Revitalization program sought to reduce the average neighborhood poverty rate for subsidized renters ($\bar{P}_t^l$), where $\bar{P}_t^l \equiv P_{Ct} \times \omega_{Ct} + P_{Bt} \times \omega_{Bt}$ and $\omega_{jt} = \frac{P_{jt}}{P_{Ct} + P_{Bt}}$, taking the city-level poverty rate as given. Figure C.1 depicts the relationship between segregation and exposure to poverty by plotting $\bar{P}_t^l$ against the difference between the neighborhood-level poverty rates, $P_{Ct} - P_{Bt}$. Distributing poor households evenly throughout the city minimizes exposure to neighborhood poverty for subsidized households.

Low-income households either live in public housing or receive housing vouchers and join high-income households in their search for housing in the private market.⁴⁴ In each neighborhood a share of units, H_{jt} , are reserved for public housing. Each household i in the private market chooses a neighborhood j to maximize their utility, given by:

$$\begin{aligned} u_{ijt}^h &= y^h - r_{jt} + A_{jt} - P_{jt} + \epsilon_{ijt} \\ u_{ijt}^l &= y^l - \phi r_{jt} + \delta A_{jt} - \alpha P_{jt} + \epsilon_{ijt}, \end{aligned} \tag{C.1}$$

where y is income; r_{jt} is the cost of housing; A_{jt} is the amenity value of neighborhood j ; P_{jt} is the neighborhood poverty rate; and ϵ_{ijt} represents household-level heterogeneity in neighborhood valuations, which is assumed to be independently, identically distributed extreme value. $0 < \phi < 1$ captures the fact that low-income households who are not in public housing have housing vouchers and therefore pay less than full market rent. While housing subsidies typically require households to contribute a fixed percentage of their income to rent, a positive value of ϕ could capture other dimensions of the cost of living (e.g., more expensive groceries, etc.). $\delta > 0$ and $\alpha > 0$ allow low-income households to value neighborhood amenities and neighborhood poverty differently than high-income households. A spatial equilibrium is a vector of prices, $\{r_{Ct}^*, r_{Bt}^*\}$, and allocations, $\{P_{Ct}^*, P_{Bt}^*\}$, such that no household in the private market could be made better off by moving neighborhoods.

To model the city before the program we consider a unique, interior, stable equilibrium where there is no public housing in the suburbs and the city center has a higher poverty rate than the suburbs.⁴⁵ There are two cases in which such an equilibrium exists, *Case 1*: $A_{Ct} > A_{Bt}$ and $\delta > \phi$; and *Case 2*: $A_{Ct} < A_{Bt}$ and $\delta < \phi$. We model the Revitalization

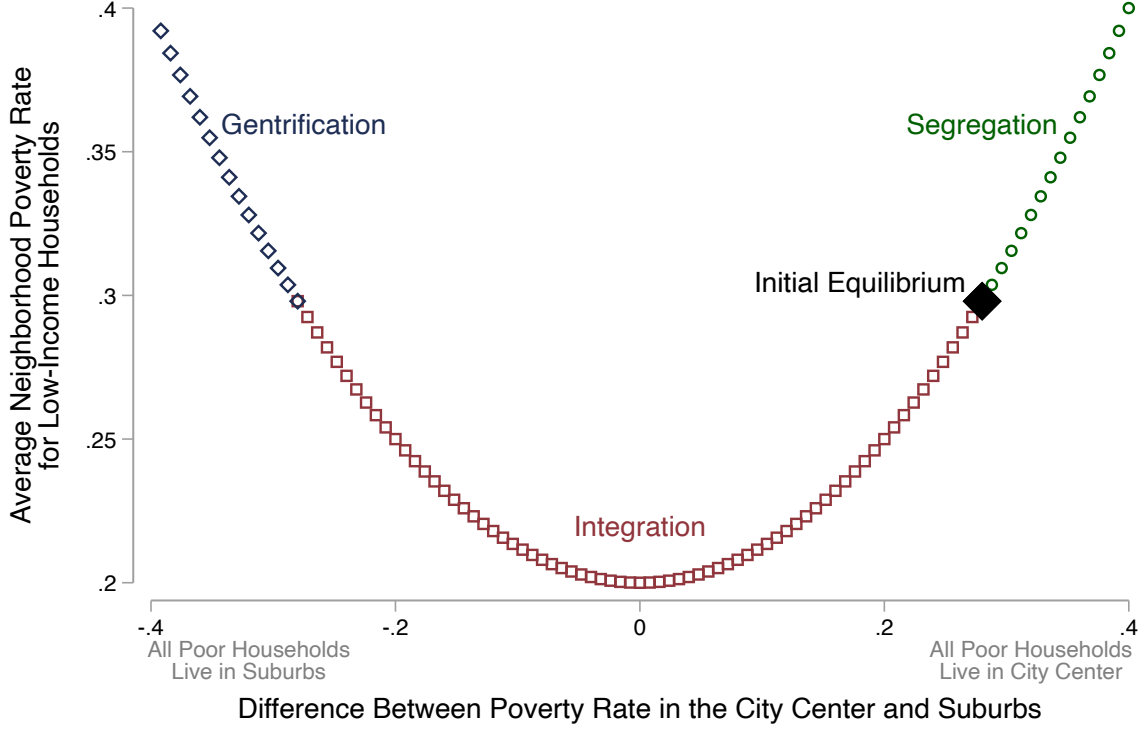
⁴²In practice, recent estimates suggest that just one in four income-eligible households receives federal rental assistance. See <https://www.cbpp.org/research/housing/more-housing-vouchers-most-important-step-to-help-more-people-afford-stable-homes>.

⁴³We drop the time subscript for variables that are time invariant.

⁴⁴We assume that public housing is rationed and sufficiently appealing such that low-income households would always choose to accept public housing over searching for housing in the private market.

⁴⁵Formally, the equilibrium satisfies $0 = H_B$, $0 < H_{Ct} < P_{Ct}^*$, and $0 < P_{Bt}^* < P_{Ct}^* < 2P$.

Figure C.1: Segregation and Exposure to Neighborhood Poverty



Notes: This figure plots $\bar{P}_t^l$ on the vertical axis against $P_{Ct} - P_{Bt}$ on the horizontal axis under the assumption that $P = 0.2$. The solid diamond depicts the initial equilibrium. The hollow diamond, square, and circle markers depict the three possible equilibria after the Revitalization intervention.

Source: Simulated data.

program as increasing neighborhood amenities ($A_{Ct} < A_{Ct+t}$) and reducing the stock of public housing ($H_{Ct} > H_{Ct+1}$), keeping fixed the the city-wide poverty rate. In equilibrium, neighborhood poverty rates are pinned down by: $(\delta - \phi)(A_B - A_{Ct}) = 2(\alpha - \phi)(P - P_{Ct}^*) + \log(\frac{P_{Bt}^*}{P_{Ct}^*}) - \phi \log(\frac{1-P_{Bt}^*}{1-P_{Ct}^*})$. Taking the derivative of P_{Ct}^* with respect to A_{Ct} yields, $\frac{dP_{Ct}^*}{dA_{Ct}} = \frac{\phi - \delta}{2} \left[\phi \left(\frac{(1-P_{Ct}^*)(1-P_{Bt}^*) - (1-P)}{(1-P_{Ct}^*)(1-P_{Bt}^*)} \right) - \left(\alpha + \frac{P}{P_{Ct}^* P_{Bt}^*} \right) \right]^{-1}$. The conditions of case 1 imply that $\frac{dP_{Ct}^*}{dA_{Ct}} > 0$ and the conditions of case 2 imply that $\frac{dP_{Ct}^*}{dA_{Ct}} < 0$. See Appendix C.1 for proofs.

The Revitalization program produces three possible equilibria. First, case 1 implies that $\bar{P}_t^l < \bar{P}_{t+1}^l$. The program attracts low-income households to the city center because they value the amenities and housing vouchers prevent them from paying the full cost. We refer to this case as the *Segregation* equilibrium, which is depicted by the green circles in Figure C.1. Second, case 2 in addition to $A_B \geq A_{Ct+1}$ or $H_{Ct+1} \geq 2p - P_{Ct}^*$ implies that $\bar{P}_t^l > \bar{P}_{t+1}^l$. An increase in amenities attracts high-income households to the city center because they value the amenities more. We refer to this case as the *Integration* equilibrium, which is depicted by the red squares in Figure C.1. Third, case 2 in addition to $A_B < A_{Ct+1}$ and $H_{Ct+1} < 2p - P_{Ct}^*$ implies that $\bar{P}_t^l < \bar{P}_{t+1}^l$. In this case the poverty rate in the suburbs after

the award is higher than the poverty rate in city center before the award ($P_{Bt+1}^* > P_{Ct}^*$) because the improvement in amenities attracts high-income households and the reduction in public housing allows the low-income households to be priced out. We refer to this scenario as the *Gentrification* equilibrium, which is depicted by the blue diamonds in Figure C.1. The program reduces poverty rates in the city center, but increases poverty rates in the suburbs enough to increase the average, city-wide neighborhood poverty rate for low-income households.

This model illustrates why the effect of the Revitalization program on neighborhood poverty for subsidized renters is theoretically ambiguous. The program could increase exposure to neighborhood poverty by attracting poor households to the revitalized sites (segregation equilibrium) or relocating too many poor households to the suburbs (gentrification equilibrium). Whether these unintended consequences occur depends on how low-income households value neighborhood amenities (δ) and the extent to which they pay for them (ϕ). The program will be successful only if it attracts high-income households into the revitalized neighborhoods while maintaining a sufficient amount of affordable units for low-income households (integration equilibrium).

C.1 Proofs

The logit choice probability that household of type k selects neighborhood j is

$$\mathbb{P}_{jt}^k = \frac{\exp\{\bar{u}_{ijt}^k\}}{\exp\{\bar{u}_{iBt}^k\} + \exp\{\bar{u}_{iCt}^k\}}, \quad (\text{C.2})$$

where $\bar{u}_{ijt}^k \equiv u_{ijt}^k - \epsilon_{ijt}$. In equilibrium, the choice probabilities, $\mathbb{P}_{jt}^k$ must be consistent with the poverty rates that enter the utility function, P_{jt} . Without loss of generality assume the size of the population in each neighborhood is 1. Then a spatial equilibrium requires that,

$$P_{jt} = \mathbb{P}_{jt}^l 2P = 1 - \mathbb{P}_{jt}^h 2(1 - P). \quad (\text{C.3})$$

Combining the identity $(P_{Ct} + P_{Bt})/2 = P$ with equations C.2 and C.3 yields,

$$\begin{aligned} \log(\mathbb{P}_{Bt}^l) - \log(\mathbb{P}_{Ct}^l) &= -\phi(r_{Bt} - r_{Ct}) + \delta(A_{Bt} - A_{Ct}) - \alpha(P_{Bt} - P_{Ct}) \\ \log\left(\frac{2P - P_{Ct}}{P_{Ct}}\right) &= -\phi(r_{Bt} - r_{Ct}) + \delta(A_{Bt} - A_{Ct}) - \alpha 2(P - P_{Ct}) \end{aligned} \quad (\text{C.4})$$

and

$$\begin{aligned} \log(\mathbb{P}_{Bt}^h) - \log(\mathbb{P}_{Ct}^h) &= -(r_{Bt} - r_{Ct}) + (A_{Bt} - A_{Ct}) - (P_{Bt} - P_{Ct}) \\ \log\left(\frac{2(1 - P) - (1 - P_{Ct})}{1 - P_{Ct}}\right) &= -(r_{Bt} - r_{Ct}) + (A_{Bt} - A_{Ct}) - 2(P - P_{Ct}). \end{aligned} \quad (\text{C.5})$$

Rearranging terms in equations C.4 and C.5, we can write the price differential between neighborhoods B and C based on the difference in choice probabilities for household type k ,

$f^k(P_{Ct}) \equiv r_{Bt} - r_{Ct}$, as a function of the poverty rate in C,

$$\begin{aligned} f^l(P_{Ct}) &= \frac{\delta}{\phi}(A_B - A_{Ct}) - 2\frac{\alpha}{\phi}(P - P_{Ct}) - \frac{1}{\phi}\log\left(\frac{2P - P_{Ct}}{P_{Ct}}\right) \\ f^h(P_{Ct}) &= (A_B - A_{Ct}) - 2(P - P_{Ct}) - \log\left(\frac{2(1 - P) - (1 - P_{Ct})}{1 - P_{Ct}}\right) \end{aligned} \quad (C.6)$$

Equation C.6 is a system of two equations and two unknowns, which pins down the equilibrium neighborhood poverty rate, P_{Ct} , and relative prices, $r_{Bt} - r_{Ct}$.⁴⁶

We consider a unique, interior, stable equilibrium in which there is public housing in C but not B and C has a higher poverty rate at time t . Let P_{jt}^* denote the equilibrium poverty rate in neighborhood j . Then the equilibrium satisfies the following conditions: $0 = H_B$, $0 < H_{Ct} < P_{Ct}^*$, and $0 < P_{Bt}^* < P_{Ct}^* < 2P$. Let $g(P_{Ct}) \equiv f^h(P_{Ct}) - f^l(P_{Ct})$, then the equilibrium level of price differentials, $r_{Bt} - r_{Ct}$, and the spatial distribution of poverty, P_{Ct} and P_{Bt} , is defined by the condition: $g(P_{Ct}^*) = 0$.

We prove the existence of a unique, interior, stable equilibrium by showing that two conditions are met: i) $\lim_{P_{Ct} \rightarrow 0} g(P_{Ct}) > 0$ and ii) $dg(P_{Ct})/dP_{Ct} < 0$. To illustrate the intuition behind why these conditions define an equilibrium, Figure C.2(a) plots $f^k(P_{Ct})$ and Figure C.2(b) plots $g(P_{Ct})$. If both conditions are met, then there is a unique poverty rate, P_{Ct}^* such that $g(P_{Ct}^*) = 0$. Furthermore, the fact that $g(P_{Ct})$ is downward sloping in Figure C.2(b) implies that the equilibrium is stable. To see why, consider a case in which poverty rates in C are below the equilibrium value and the price differential satisfies the constraint for the low-income households. In this scenario, C is too expensive for the high-income households and they will move to B, which will drive down poverty rates in C. If the price differential shifts to satisfy $f^l(P_{Ct})$, then poverty rates in C will fall until $g(P_{Ct}) = 0$. For i), note that $\lim_{P_{Ct} \rightarrow 0} \frac{1}{\phi}\log\left(\frac{2P - P_{Ct}}{P_{Ct}}\right) = \infty$ for $\phi > 0$, which implies that $\lim_{P_{Ct} \rightarrow 0} g(P_{Ct}) > 0$. For ii), note that

$$\frac{dg(P_{Ct})}{dP_{Ct}} = \frac{\phi - \alpha}{\phi} - \frac{2(1 - P)}{(1 - P_{Bt})(1 - P_{Ct})} - \frac{P}{\phi P_{Bt} P_{Ct}} < 0. \quad (C.7)$$

Under the assumption that $\alpha > 0$ and $\phi > 0$, the inequality in equation C.7 always holds because $\frac{\phi - \alpha}{\phi} \leq 1 < \frac{1}{1 - P} \leq \frac{(1 - P)}{(1 - P_{Bt})(1 - P_{Ct})}$ and $\frac{P}{\phi P_{Bt} P_{Ct}} > 0$.

Under what conditions will $P_{Bt}^* < P_{Ct}^*$? We can rewrite $g(P_{Ct}^*) = 0$ as,

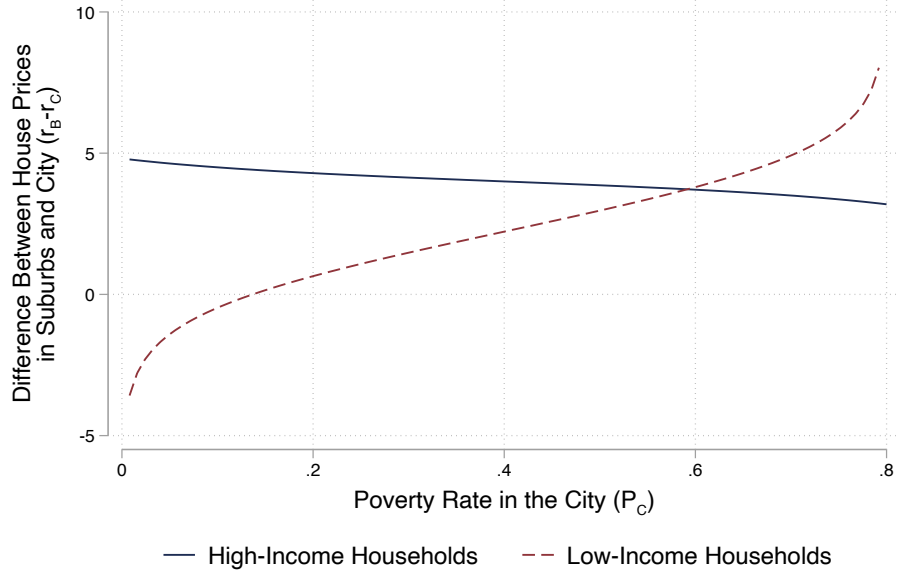
$$\frac{\delta - \phi}{\phi}(A_B - A_{Ct}) = -\frac{1}{\phi} \left[\alpha(P_{Ct}^* - P_{Bt}^*) + \log\left(\frac{P_{Ct}^*}{P_{Bt}^*}\right) \right] + \left[(1 - P_{Bt}^*) - (1 - P_{Ct}^*) + \log\left(\frac{1 - P_{Bt}^*}{1 - P_{Ct}^*}\right) \right]. \quad (C.8)$$

When $P_{Bt}^* < P_{Ct}^*$ the right hand side of equation C.8 is always negative because $\alpha(P_{Ct}^* - P_{Bt}^*) + \log\left(\frac{P_{Ct}^*}{P_{Bt}^*}\right) > 0$ and $(1 - P_{Bt}^*) - (1 - P_{Ct}^*) + \log\left(\frac{1 - P_{Bt}^*}{1 - P_{Ct}^*}\right) < 0$. There are two conditions under which the left hand side is also negative: case i) $A_B < A_{Ct}$ and $\phi < \delta$ and case ii) $A_B > A_{Ct}$ and $\phi > \delta$. These are the two cases in which there exists an equilibrium that satisfies the above conditions (i.e., the equilibrium is a unique, interior, stable equilibrium in which there is public housing in C but not B and C has a higher poverty rate at time t).

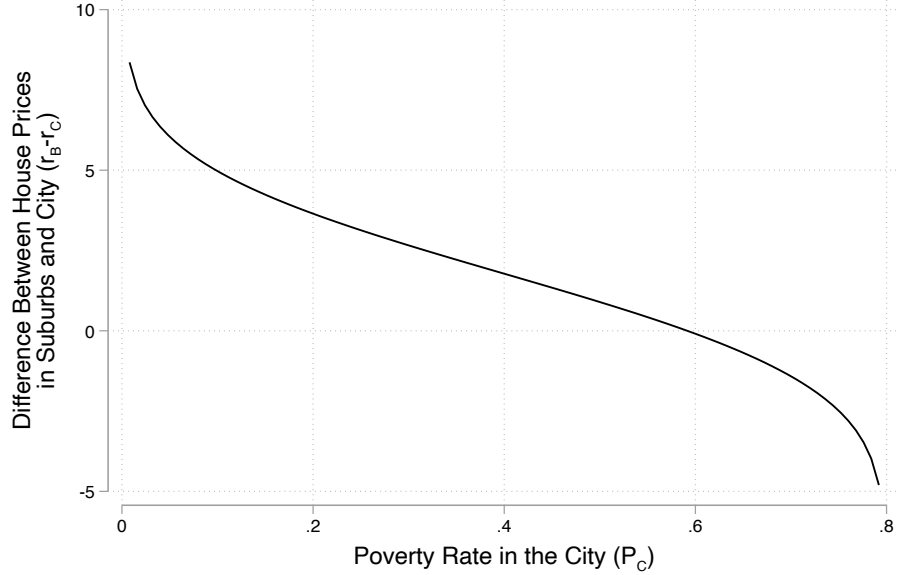
⁴⁶Because we do not model housing supply, we can only pin down relative prices in equilibrium.

Figure C.2: Equilibrium

(A) High- and Low-Income Households



(B) Difference



Notes: Panel A plots $f^l(P_{Ct})$ and $f^h(P_{Ct})$, which describes the relationship between relative prices, $r_{Bt} - r_{Ct}$, and poverty rates in the city, P_{Ct} . Panel B plots $g(P_{Ct})$, which is defined as $g(P_{Ct}) \equiv f^h(P_{Ct}) - f^l(P_{Ct})$.

Source: The figure is based on simulated data under the assumptions that $P = 0.4$, $\phi = 0.9$, $\delta = 0.5$, $\alpha = 0.8$, and $(A_B - A_{Ct}) = 4$.

Our main comparative static asks how poverty rates in the city will change if the amenity value increases. Using implicit differentiation with the condition $g(P_{Ct}^*) = 0$, yields,

$$\frac{dP_{Ct}^*}{dA_{Ct}} = \frac{\phi - \delta}{2} \left[\phi \left(\frac{(1 - P_{Ct}^*)(1 - P_{Bt}^*) - (1 - P)}{(1 - P_{Ct}^*)(1 - P_{Bt}^*)} \right) - \left(\alpha + \frac{P}{P_{Ct}^* P_{Bt}^*} \right) \right]^{-1}. \quad (\text{C.9})$$

Note that $\phi \left(\frac{(1 - P_{Ct}^*)(1 - P_{Bt}^*) - (1 - P)}{(1 - P_{Ct}^*)(1 - P_{Bt}^*)} \right) < 0$ because $(1 - P_{Ct}^*) < (1 - P) < (1 - P_{Bt}^*)$ and $\left(\alpha + \frac{P}{P_{Ct}^* P_{Bt}^*} \right) > 0$. Thus, $\frac{dP_{Ct}^*}{dA_{Ct}} > 0$ if $\phi < \delta$ and $\frac{dP_{Ct}^*}{dA_{Ct}} < 0$ if $\phi > \delta$.⁴⁷

⁴⁷If $\phi > \delta$ and the increase in A_{Ct} is sufficiently large such that $P_{Bt}^* > P_{Ct}^*$, then $\frac{dP_{Ct}^*}{dA_{Ct}} < 0$ still holds because $(1 - P_{Bt}^*) < (1 - P) < (1 - P_{Ct}^*)$.

Appendix D Identification Results

D.1 Original Residents

Starting with the estimator, we have,

$$\begin{aligned} & \mathbb{E}[\bar{P}_{t''}(N_{it''}) \mid D(N_{it'}) = 1, S_{it''} = 1] - \mathbb{E}[\bar{P}_{t''}(N_{it''}) \mid C(N_{it'}) = 1, S_{it''} = 1] \\ &= \mathbb{E}[\bar{P}_{t''}^1(N_{it''}^1) \mid D(N_{it'}) = 1, S_{it''}^1 = 1] - \mathbb{E}[\bar{P}_{t''}^0(N_{it''}^0) \mid C(N_{it'}) = 1, S_{it''}^1 = 1] \\ &= \mathbb{E}[\Delta_i \mid \mathcal{G}_1] \end{aligned} \quad (\text{D.1})$$

where the first equality holds by Assumption 2 and the second by Assumption 1.

D.2 New Residents

We start by decomposing the target estimand into the effect on neighborhoods and the sorting effect,

$$\begin{aligned} \mathbb{E}[\Delta_i \mid \mathcal{G}_2] &= \mathbb{E}[\bar{P}_{t''}^1(N_{it''}^1) - \bar{P}_{t''}^0(N_{it''}^1) \mid \mathcal{G}_2] + \mathbb{E}[\bar{P}_{t''}^0(N_{it''}^1) - \bar{P}_{t''}^0(N_{it''}^1) \mid \mathcal{G}_2] \\ &= \mathbb{E}[\bar{P}_{t''}^1(n) - \bar{P}_{t''}^0(n) \mid D(n) = 1] + \mathbb{E}[\bar{P}_{t''}^0(N_{it''}^1) - \bar{P}_{t''}^0(N_{it''}^0) \mid \mathcal{G}_2]. \end{aligned} \quad (\text{D.2})$$

Assumption 3 allows us to rewrite the estimator as,

$$\begin{aligned} & \mathbb{E}[\bar{P}_{t''}(N_{it'}) \mid D(N_{it'}) = 0, D(N_{it''}) = 1, S_{it''} = 1] \\ & - \mathbb{E}[\bar{P}_{t''}(N_{it'}) \mid C(N_{it'}) = 0, C(N_{it''}) = 1, S_{it''} = 1] \\ &= \theta(\mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) \mid D(N_{it'}) = 0, D(N_{it''}) = 1, S_{it''} = 1] \\ & \quad - \mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) \mid C(N_{it'}) = 0, C(N_{it''}) = 1, S_{it''} = 1]) \\ &= \theta(\mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) \mid D(N_{it'}) = 0, D(N_{it''}) = 1, S_{it''} = 1] \\ & \quad - \mathbb{E}[\bar{P}_{it''}^0(n) \mid C(n) = 1]) \end{aligned} \quad (\text{D.3})$$

Assumption 1 implies that, $\mathbb{E}[\bar{P}_{it''}^0(n) \mid C(n) = 1] = \mathbb{E}[\bar{P}_{it''}^0(n) \mid D(n) = 1]$, which yields,

$$\begin{aligned} & \theta(\mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) \mid D(N_{it'}) = 0, D(N_{it''}) = 1, S_{it''} = 1] - \mathbb{E}[\bar{P}_{it''}^0(n) \mid C(n) = 1]) \\ &= \theta(\mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) \mid D(N_{it'}) = 0, D(N_{it''}) = 1, S_{it''} = 1] - \mathbb{E}[\bar{P}_{it''}^0(n) \mid D(n) = 1]) \\ &= \theta(\mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) \mid D(N_{it'}) = 0, D(N_{it''}) = 1, S_{it''} = 1] \\ & \quad - \mathbb{E}[\bar{P}_{it''}^0(N_{it''}^1) \mid D(N_{it'}) = 0, D(N_{it''}) = 1, S_{it''} = 1]) \\ &= \theta \mathbb{E}[\bar{P}_{it''}^0(N_{it''}^0) - \bar{P}_{it''}^0(N_{it''}^1) \mid \mathcal{G}_2]. \end{aligned} \quad (\text{D.4})$$

D.3 Residents Who Would Have Moved in Absent the Award

We can rewrite the target estimand as,

$$\mathbb{E}[\Delta_i \mid \mathcal{G}_3] = \mathbb{E}[\bar{P}_{t''}^1(N_{it''}^1) \mid \mathcal{G}_3] - \mathbb{E}[\bar{P}_{t''}^0(N_{it''}^0) \mid \mathcal{G}_3]. \quad (\text{D.5})$$

For the first component, Assumption 4 implies that,

$$\begin{aligned} \mathbb{E}[\bar{P}_{t''}^1(N_{it''}^1) \mid \mathcal{G}_3] &= \mathbb{E}[\bar{P}_{t''}^1(N_{it''}^1) \mid D(N_{it'}) = 0, D(N_{it''}^1) = 0, D(N_{it''}^0) = 1, S_{it''}^1 = 1] \\ &= \mathbb{E}[\bar{P}_{t''}^0(N_{it''}^0) \mid D(N_{it'}) = 0, D(N_{it''}^0) = 1, S_{it''}^0 = 1]. \end{aligned} \quad (\text{D.6})$$

Assumption 1 then implies,

$$\begin{aligned}
& \mathbb{E}[\bar{P}_{t''}^0(N_{it'}) \mid D(N_{it'}) = 0, D(N_{it''}^0) = 1, S_{it''}^0 = 1] \\
&= \mathbb{E}[\bar{P}_{t''}^0(N_{it'}) \mid C(N_{it'}) = 0, C(N_{it''}^0) = 1, S_{it''}^0 = 1] \\
&= \mathbb{E}[\bar{P}_{t''}^0(N_{it'}) \mid C(N_{it'}) = 0, C(N_{it''}) = 1, S_{it''} = 1].
\end{aligned} \tag{D.7}$$

For the second component, Assumption 1, implies that,

$$\begin{aligned}
\mathbb{E}[\bar{P}_{t''}^0(N_{it''}^0) \mid \mathcal{G}_3] &= \mathbb{E}[\bar{P}_{t''}^0(N_{it''}^0) \mid D(N_{it'}) = 0, D(N_{it''}^1) = 0, D(N_{it''}^0) = 1, S_{it''} = 1] \\
&= \mathbb{E}[\bar{P}_{t''}^0(n) \mid D(n) = 1] \\
&= \mathbb{E}[\bar{P}_{t''}^0(n) \mid C(n) = 1] \\
&= \mathbb{E}[\bar{P}_{t''}^0(N_{it''}) \mid C(N_{it'}) = 0, C(N_{it''}) = 1, S_{it''} = 1].
\end{aligned} \tag{D.8}$$

Combing the expressions for the two components yields the identification result,

$$\mathbb{E}[\bar{P}_{t''}(N_{it'}) - \bar{P}_{t''}(N_{it''}) \mid C(N_{it'}) = 0, C(N_{it''}) = 1, S_{it''} = 1] = \mathbb{E}[\Delta_i \mid \mathcal{G}_3]. \tag{D.9}$$

D.4 Spillovers

Assumption 5 and the fact that poverty rates are bounded between 0 and 1 implies the following,

$$\begin{aligned}
\bar{P}_{t''}^1(n) - \bar{P}_{t''}^0(n) &= \lambda_n \mathbb{E}[P_{it''}^1 \mid N_{it''}^1 = n, D(N_{it''}^0) = 1] - \lambda_n \bar{P}_{t''}^0(n) \\
&\leq \lambda_n \\
&= \frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it''}^0) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}} \\
&= \frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it''}^0) = 1, D(N_{it'}) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}} \\
&\quad + \frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it''}^0) = 1, D(N_{it'}) = 0\}}{\sum_i \mathbb{1}\{N_{it''} = n\}} \\
&\leq \frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it'}) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}} + \frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it''}^0) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}} \\
&= \frac{\sum_i \mathbb{1}\{N_{it''}^1 = n, D(N_{it'}) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}} + \frac{\sum_i \mathbb{1}\{N_{it'} = n, D(N_{it''}^0) = 1\}}{\sum_i \mathbb{1}\{N_{it''} = n\}},
\end{aligned} \tag{D.10}$$

for $D^1(n) = 1$.

Appendix E Weights for Aggregate Effects

This appendix describes how we calculate the weights, $\omega_g(s)$ in equation 8. Note that we partition the sample into mutually exclusive and exhaustive groups, which implies that $\sum_g \omega_g(s) = 1$ and $\omega_g(s) \in [0, 1]$ for all g and $s \in [0, 1]$.

- We find that 27.1 percent of the original residents of the Revitalization neighborhoods resided in subsidized housing 15 years after the award and assume that there is not change in the total number of subsidized households over time. Thus, $\omega_1(s) = 0.271 \times s$.
- Table 1 shows that the program led to a 20 percent reduction in the number of subsidized units in the revitalized neighborhoods. Furthermore, Table 1 shows that 90 percent of the residents of revitalized neighborhoods were new residents (as opposed to original residents).⁴⁸ Thus, $\omega_2(s) = (1 - 0.2) \times 0.9 \times s$.
- Table 1 shows that the program led to a 20 percent reduction in the number of subsidized units in the revitalized neighborhoods. Thus, $\omega_3(s) = 0.2 \times s$.
- The share of subsidized renters who live in other neighborhoods and are not directly affected by the program (i.e., are not in groups 1 through 3) is equal to: $1 - [\omega_1(s) + \omega_2(s) + \omega_3(s)]$. These other households can be organized into a group that is and is not affected by the program due to spillover effects on their neighborhoods. Let $g = 4$ and $g = 5$ be the group that is and is not affected by spillover effects, respectively and let c be a constant such that $c \times s = \omega_1(s) + \omega_2(s) + \omega_3(s)$. We assume that $\omega_4(s) = 2 \times s \times (1 - c \times s)$, which implies that no one is affected by spillover effects when no or all units are revitalized ($\omega_4(0) = 0$ and $\omega_4(1) = 0$) and all units not directly affected by the program are affected by spillover effects when half the units are revitalized ($\omega_4(s) = 1 - \frac{1}{2}c = 1 - [\omega_1(s) + \omega_2(s) + \omega_3(s)]$). Section 5.2.3 argues that the displacement of the poor original residents, poor new residents, and non-poor new residents increase poverty rates for subsidized renters living in in other neighborhoods by 0.296, 0.194, and 0.270 percentage points, respectively. To provide an upper bound we assume the spillover effects from these three groups of displaced households all affect the same neighborhoods and therefore sum the effects from all three groups. Thus, $E[\Delta_i | G_i = 4] = 0.0076$, $\omega_4(s) = 2 \times s \times (1 - c \times s)$, $\Delta_5 = 0$, and $\omega_5(s) = 1 - [\omega_1(s) + \omega_2(s) + \omega_3(s) + \omega_4(s)]$.

⁴⁸We arrive at this number using the estimates in columns 7, 9, 11, and 13 from Table 1. We estimate the size of the population in the Revitalization neighborhood as the failed applicant average plus the coefficient estimate: $0.9 = (696 - 168 + 335 - 2)/(108 - 43 + 696 - 168 + 52 - 20 + 335 - 2)$.

Appendix F Cost Benefit Calculation

Did the benefits of the program justify the costs? We use our estimates to quantify the benefits and costs for the average PHA, which contained 8,904 units in 2018 and received two Revitalization grants for a total of \$47.8 million to revitalize 22 percent its subsidized units.

The aggregate effects in Section 5.3 imply that a PHA that revitalized 22 percent of the subsidized housing stock reduced neighborhood poverty rates among all subsidized renters by 1.7 percentage points, on average. Using MTO-driven experimental variation in housing prices to identify households' marginal utility of consumption along with structural estimates of households' WTP for neighborhood amenities, Galiani et al. (2015) present results showing that a similarly low-income sample of subsidized renters have an annual WTP for a one percentage point decrease in their neighborhood poverty rate of \$282.⁴⁹ Together, this suggests that subsidized renters in Revitalization PHAs would be willing to pay \$479 more for housing in neighborhoods with poverty rates that are 1.7 percentage point lower ($479 = 1.7 \times 282$).

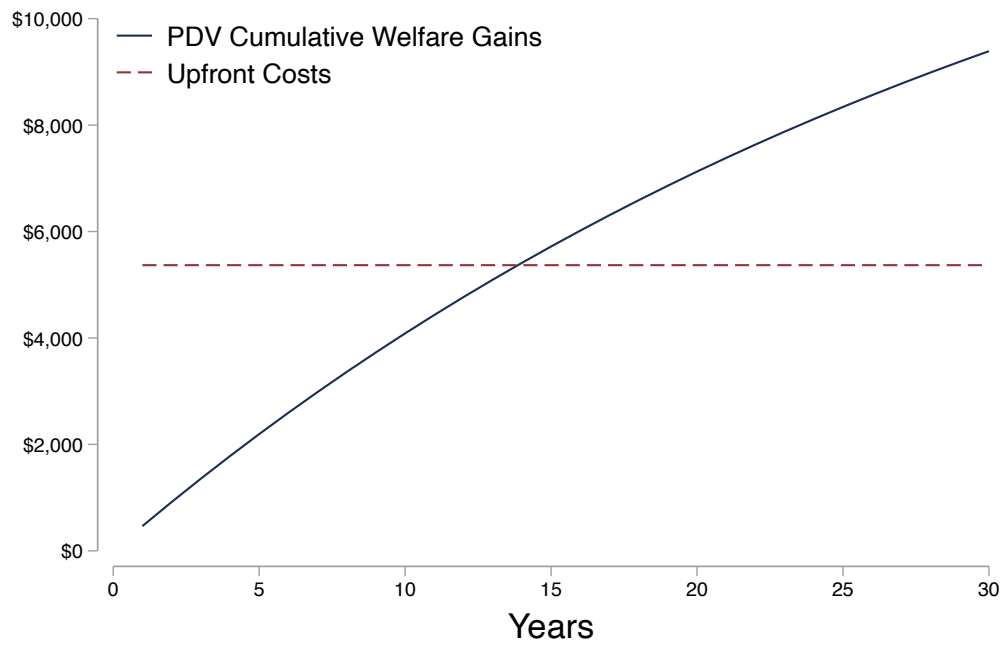
The upfront costs of the program are substantial: \$5,368 per subsidized unit in the PHA ($5,368 = 47,800,000/8,904$). However, the program produced durable reductions in neighborhood poverty for subsidized renters that lasted substantially longer than a single year—we observe poverty reductions that extend at least to 15 years post-Revitalization, the last year we consider due to data limitations. Figure F.1 plots the present discounted value (PDV) of the cumulative annual welfare gains per subsidized unit as a function of the number of years over which the benefits accrue, assuming a 3 percent annual discount rate. The figure illustrates that the PDV of welfare gains will exceed the upfront costs if the reductions in exposure to poverty persist for at least 14 years.⁵⁰ Thus, when considered over a sufficiently long horizon, the Revitalization program produced meaningful welfare gains for subsidized households. Furthermore, these net gains in subsidized renter welfare would remain even if the full HOPE VI program costs had been passed along to subsidized households through rent increases.

Our estimates of the costs and benefits of the Revitalization program are intentionally simple. In particular, they only consider the financial costs and only consider the benefits from a reduction in exposure to neighborhoods poverty for subsidized renters. While these are key inputs to any cost-benefit analysis, the program could have imposed other costs and produced other benefits not captured by our analysis. Importantly, such an analysis ought to also consider impacts on poor households who lack subsidized housing.

⁴⁹These WTP estimates correspond to what the model in Galiani et al. (2015) predicts WTP would be at the sample average value for all observable characteristics that determine household WTP for neighborhood amenities.

⁵⁰The PDV of 14 years of cumulative welfare gains from the reduction in exposure to neighborhood poverty is $5,411 = 479 \times (1 + 0.03)^{-14}$.

Figure F.1: Cumulative Welfare Gains vs Upfront Costs



Notes: This figures plots the present discounted value (PDV) of the cumulative welfare gains per subsidized unit as a function of the number of years over which these gains accrue. The red dashed line represents the upfront cost per subsidized units.